Math 246 notes #1: Introduction and order of magnitude calculations

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1 What is modeling?

- A model is a description of a situation being studied that captures all the *relevant* features. Examples: road map, geologic map.
- A *good* model makes correct predictions when applied to problems it was intended to solve. For example, if a road map is missing a major highway, then it is a poor model.
- In this class, we will focus on *mathematical* models and attempt to describe realistic situations using the language of mathematics.
- Mathematics is extremely useful for modeling because 1) it is a precise language, 2) it is concise and can be manipulated, 3) we can write down theorems, 4) we can carry out the resulting calculations with computers if needed.

2 Why should you care?

- If you're interested in the physical sciences, applied math, etc, all of that work revolves around building models. How do you describe thousands of interacting galaxies? What happens when two populations of predators and prey interact with each other?
- A computer science example: rendering human skin. How complicated does your code need to be in order to make skin look realistic? Can you get away with treating skin as a single, partially reflective surface? No: it turns out you need to model several skin layers, including subsurface reflectance. For details, check out `http://http.developer.nvidia.com/GPUGems3/gpugems3_ch14.html`
- Business, economics, consulting: suppose a company asks you how many salespeople to employ. How would you advise them? More salespeople means more sales, but also increased overhead costs. How do you analyze this tradeoff?

3 Properties of models

- Definition: *A mathematical model is an abstract, simplified, mathematical construct related to a part of reality and created for a particular purpose.*
- A model can be divided into three parts:
  1. Things whose effects can be neglected. A good model keeps things as simple as possible and uses only the information that is relevant.
  2. Things that affect the model but whose behavior the model is not designed to study. These are the *inputs* to the model (also called parameters, independent variables, or exogenous variables), and consist of constants, functions, etc.
  3. Things the model is designed to study the behavior of. These are the *outputs* of the model (also called dependent variables or endogenous variables), i.e. the things that the model seeks to explain.

4 The art of simplification: figuring out what effects can be neglected

- How ruthlessly can we simplify? If the wrong things are neglected, then the model is worthless. If too many inputs are used, then the model will be too complex and unwieldy.

- To develop intuition for simplification, we're going to start with *order of magnitude* calculations. One of the most famous order-of-magnitude calculations was Enrico Fermi's estimation of the atomic bomb energy yield at the 1945 Trinity test: "*About 40 seconds after the explosion the air blast reached me. I tried to estimate its strength by dropping from about six feet small pieces of paper before, during, and after the passage of the blast wave. Since, at the time, there was no wind I could observe very distinctly and actually measure the displacement of the pieces of paper that were in the process of falling while the blast was passing. The shift was about 2 1/2 meters, which, at the time, I estimated to correspond to the blast that would be produced by ten thousand tons of T.N.T.*" The actual answer: 20,000 tons of TNT, only a factor of two different! How did Fermi do that?

- General rules of thumb for simplification:

  1. An "order of magnitude" is a factor of 10. If your estimated quantity is within a factor of 10 of the truth, then your answer is good enough.
  2. When in doubt, establish lower and upper bounds for the quantity you're trying to estimate, and then take the geometric mean for your answer.
  3. In order to keep track of factors of 10, express numbers in scientific notation. Only keep track of factors of a few when enough of them combine to form 10. Ignore decimal points.
  4. Sanity check your answer. Does it intuitively make sense?

- Once you have an order-of-magnitude estimate, your answer will lie in one of three categories:

  1. Too big
  2. Too small
  3. In the middle

  Answers in the first two categories lead to obvious actions (in the case of modeling, use this to determine whether an input is essential or negligible). Answers in the third category require a bit more thought.

5 Example: armored cars

How much money can a fully loaded armored car carry?

Yes, that is the entire statement of the problem, and no, there is no formula. How are we going to tackle it? Let's **divide and conquer** and worry about one thing at a time: 1) how many bills can fit in the car, and 2) what are the bill denominations?

Let's start with the first question. If V is the volume of the car and v is the volume of a typical bill, then we can get the number of bills N from the ratio of the volumes: $N \sim V/v$. (The $\sim$ sign indicates that the quantities on either side are proportional to each other and that the dimensions match. We'll come back to the issue of dimensions later.) Now we have two simpler questions: how do we determine V and v?

To determine the volumes, we need to **lie skillfully**. We can pretend that the storage space in an armored car is a rectangular prism, and let's say that the length on each side is roughly 2 m. The volume is then

$$V \sim (2\mathrm{m})^3 \sim 10\mathrm{m}^3 = 10^7\mathrm{cm}^3.$$

Note that we've used the third rule of thumb for simplification: three factors of two multiplied together is essentially an order of magnitude, i.e., $2^3 \sim 10$.

To estimate the volume of a typical bill, we apply the technique of **guerrilla warfare** and use any piece of information that we can get. A good guess for the area of a bill is roughly 5 cm by 15 cm (you can check this with a ruler). What about the thickness? Bills are made of paper, and a typical bound notebook has roughly 100 sheets of paper and is roughly 1 cm thick. Therefore, we can guess that

the thickness of a bill is roughly 10^{-2} cm. Putting the numbers together, we now have the volume of the bill:

$$v \sim 5\text{cm} \times 15\text{cm} \times 10^{-2}\text{cm} \sim 1\text{cm}^3.$$

Now that we have the volumes of the armored car and of a typical bill, we can calculate the number of bills that fit in the car: $N \sim V/v \sim 10^7\text{cm}^3/1\text{cm}^3 = 10^7$. The last thing we need is the denominations of the bills, and then we can calculate the total amount of money. Cash machines typically dispense R100 notes, so let's use that for our calculation. Our final answer is then R100 $\times 10^7$ = R1,000,000,000.

Does this make sense? We need to **sanity check** our answer and see if it jives with our intuition. South Africa's GDP per capita is roughly R100,000, so our estimate of 1 billion rands corresponds to 10,000 people's worth of per-capita-GDP being carried by a single truck. That seems highly unlikely! We therefore need an alternate cross-check of our calculation.

The armored car can carry only finite mass, so we can ask ourselves: what is the weight of 10^7 bills? We know that bills are made of paper, which is made of wood, and wood barely floats in water. Therefore, the density of paper is roughly the same as the density of water: $\rho \sim 1\text{g}/\text{cm}^3$. A single bill thus weighs $1\text{cm}^3 \times 1\text{g}/\text{cm}^3 = 1\text{g}$, and 10^7 bills have a total mass of $10^7\text{g} = 10$ tons. This is huge. Does it make sense? Most mass limits on bridges are typically a few tons, and most trucks can carry one or two tons of cargo. So let's say that from mass arguments, our original answer of 10^7 bills was off by a factor of 10, and let's revise our answer to 10^6 bills, or R100,000,000. This answer turns out to be pretty close to the truth. One of the largest South African cash-in-transit heists netted R31-million from an SBV van (the film "31 Million Reasons"is loosely based on these events), so our revised estimate was only off by a factor of three!

In summary, here are some of the important take-away messages from this exercise that you should keep in mind as you do more of these calculations:

- In order-of-magnitude calculations, there's generally more than one way to solve the problem.
- Calculate with confidence: use the techniques of lying skillfully and guerrilla warfare to get as far as you can with estimates.
- There isn't a single correct answer, but you must be able to *justify your steps*.
- Always, ALWAYS sanity check your answer and ask yourself if you believe it.

6 A refresher in dimensional analysis

We did some unit manipulation in the previous example, but just in case you're a bit rusty on dimensional analysis, here's a quick refresher before we continue onward.

Measurements, parameters, etc. are expressed as *concrete numbers*, which are combinations of a numerical quantity and a dimensional unit. For example, speed is a combination of length and time: 80 kilometers per hour = 50 miles per hour = 22 meters per second = etc. In this class, we will generally use SI units. Some types of numbers, such as percentages, are dimensionless quantities. When taking a derivative of a quantity, the dimension of the differentiating variable appears in the denominator of the final answer. For example:

- Position, x, has dimensions of length.
- Speed, dx/dt, has dimensions of length per time.
- Acceleration, d^2x/dt^2, has dimensions of length per time-squared.

We can change the units of a given quantity by applying a *conversion factor*. For example, 1 kilometer = 0.62 mile, so if we wanted to convert 80 km/hr to mile/hr, we would do the following:

$$80\frac{\text{km}}{\text{hr}} \times \frac{0.62\text{ mile}}{1\text{ km}} = 50\frac{\text{mile}}{\text{hr}}.$$

The (0.62 mile)/(1 km) is the conversion factor, and it is equal to one because the numerator and denominator are equivalent. We choose km to be in the denominator of the conversion factor so that it cancels the km that is present in the numerator of 80 km/hr. For comparison, if we wanted to convert 80 mile/hr into km/hr, we would apply the reciprocal of the above conversion factor:

$$80\frac{\text{mile}}{\text{hr}} \times \frac{1\text{ km}}{0.62\text{ mile}} = 129\frac{\text{km}}{\text{hr}}.$$

You can change more than one unit by applying multiple conversion factors. For example, to convert 80 mile/hr to meter/s, we would do the following:

$$80\frac{\text{mile}}{\text{hr}} \times \frac{1\text{ km}}{0.62\text{ mile}} \times \frac{1000\text{ meter}}{1\text{ km}} \times \frac{1\text{ hr}}{3600\text{ s}} = 36\frac{\text{meter}}{\text{s}}.$$

We have used three conversion factors, (1 km)/(0.62 mile), (1000 meter)/(1 km), and (1 hr)/(3600 s), and we have arranged them so that the mile, km, and hr units all cancel, leaving behind meter/s in our final answer.

When we deal with equations in this class, you must check that the dimensions are equal on both sides—otherwise, it isn't an equation! Only *commensurable quantities*, i.e. quantities with the same dimensions, can be *compared, equated, added, or subtracted*.

7 Example: nose picking

How many people in the world are picking their nose right now?

Let's start by considering an individual person rather than the entire world's population. What fraction of time do *you* spend picking your nose? To answer this question, we can invoke our second rule of thumb for simplification: a person's typical nose-picking time is probably more than 1 second in a day (lower bound) and probably less than 100 seconds in a day (upper bound). So for our estimate, let's take the geometric mean: $\sqrt{1 \times 100} = 10$ seconds in a day, which corresponds to a time fraction or probability of 10 s / 10^5 s = 10^{-4}. (If you haven't done so already, convince yourself that there are, in fact, roughly 10^5 seconds in a day.)

To obtain the average number of people picking their nose right now, we multiply the world's population by the nose-picking probability (which is given by the time fraction calculated in the previous paragraph): (7×10^9 people) $\times$ (10^{-4}) = 7×10^5 people, i.e. just under *a million people* are picking their nose at this very moment.

8 Why do we use the geometric mean?

Let's revisit our second rule of thumb for simplification. It's often easier to guess lower and upper bounds than an exact estimate. Why do we take the geometric mean of the two bounds rather than the arithmetic mean? Let's take a specific example: how many clowns can fit into a Cooper Mini? The answer must be more than 1 and less than 100. If we take the arithmetic average, 50, this is a bad estimate because it's a factor of 50 greater than the lower bound and only a factor of 2 less than the upper bound. If we take the geometric mean instead, $\sqrt{1 \times 100} = 10$, then our estimate lies exactly a factor of 10 above the lower bound and a factor of 10 below the upper bound. In other words, our estimate is halfway between the lower and upper bounds on a logarithmic scale.

9 Example: trash

How much trash does South Africa generate yearly?

As with the previous example, let's start with an individual person and consider the amount of trash generated in a week. We can guess a lower bound of 1 kg and an upper bound of 50 kg of trash generated per week, so our estimate is the geometric mean: $\sqrt{1 \times 50} \sim 7$ kg of trash per person per week.

The population of South Africa is about 50 million $= 5 \times 10^7$ people, so the total trash generated per year is (7 kg per person per week) $\times$ (52 weeks per year) $\times$ (5×10^7 people) $\sim 2 \times 10^{10}$ kg. To

account for business and industry, let's multiply this number by a factor of two so that our final answer is 4×10^{10} kg or 4×10^7 tons per year.

It turns out that you can actually look up national waste management reports from the South African Waste Information Centre (`http://sawic.environment.gov.za`). The total waste disposal for 2015, spanning all provinces, is **43,217,127.3 tons**, which is astonishingly close to our estimate. We were admittedly a bit lucky in our guess, and the answers do fluctuate from year to year (for example, 2014 had 74 million tons), but we did get the correct order of magnitude.

How much landfill space is needed for a century's worth of trash?

If the trash generation rate is 4×10^7 tons per year, then the total mass generated after a century is 4×10^9 tons. Let's estimate the typical trash density to be 0.5 ton/m^3, which is a factor of two less than the density of water. The total volume of a century's worth of trash is then (4×10^9 tons) / (0.5 ton/m^3) $\sim 10^{10}$ m^3.

If we assume that a typical landfill can pile trash 100 m high, then the area needed to contain a century's worth of trash is 10^{10} m^3 / 100 m = 10^8 m^2 = 10^2 km^2. This is roughly equivalent to the combined area of Westville (30 km^2) plus Pinetown (80 km^2), or 10^{-4} (0.01%) of South Africa's total land area. In other words, we don't have to worry about running out of landfill space.

There are two important take-away messages from this example. First, when interpreting large (or small) numbers, you must put them into the *relevant context*. The total amount of trash generated per year, 4×10^7 tons, may sound like a daunting number. However, when we calculated the required landfill space, we found that it was negligible. The second important message: look at just how far we were able to get with simple estimation techniques and without relying on experts. Our conclusion that landfill space isn't a problem is quite a powerful statement, and we were able to arrive at that answer simply by guessing intelligently.

10 Example: cars, petrol, and batteries

What is the energy density of batteries?

We're going to invoke the technique of guerrilla warfare and obtain our estimate by using an example that's familiar from everyday life: a hand torch powered by a D-cell battery. We can guess that the light output of a typical torch is more than 1 W but less than 10 W, which gives us an estimate of 3 W. (Recall that a Watt is the SI measure of power and has units of Joules per second.) If we leave the torch on continuously, it'll probably last more than 1 hour, but less than 1 day. We can therefore estimate that the total stored energy is power $\times$ time, or (3 W) $\times$ (5 hours) = (3 W) $\times$ (2×10^4 seconds) = 6×10^4 J.

To obtain the energy density of a battery, we need to divide its energy by its mass. We can estimate that there are between 5 and 10 D-cell batteries per kilogram, so our estimated energy density is (6×10^4 J/battery) $\times$ (7 batteries/kg) = 4×10^5 J/kg.

Why do we drive petrol-powered cars?

Petrol has a typical energy density of 4×10^7 J/kg, which is 100 times greater than that of batteries! Let's consider a related question: what is the energy transfer rate of petrol when refueling your car? When we refuel, we typically transfer about 50 L of petrol, and it takes about 2 minutes = 10^2 seconds. The mass density of petrol is approximately the same as water, 1 kg/L. The energy transfer rate is therefore (4×10^7 J/L) $\times$ (50 L) / (10^2 s) = 10^7 W = 10 MegaWatts! For comparison, home power cords only carry about 10^3 W, so it's much more difficult to transfer energy to your car via electricity than petrol.

Can we "grow" our petrol by relying entirely on corn for ethanol?

Another way of phrasing this question is: is it feasible to rely on food production for generating car fuel? To answer this question, let's estimate our current food consumption and then ask how much we'd have to increase it in order to fuel our cars. Let's say that each person consumes 2500 Cal = 10^7 J per day. A car consumes between 1 L and 10 L of petrol per day, so let's call our estimate 3 L

per day. This amount of fuel corresponds to a daily energy consumption of (3 L) $\times$ (4×10^7 J/L) = 10^8 J. Cars therefore consume 10 times as much energy as humans, so if we wanted to fuel our cars with corn, we would need 10 times as much crop land to grow our ethanol. In other words, it doesn't make much sense to burn food in cars.

11 Example: plastic bottles

Should you recycle plastic drink bottles?

We need to estimate the value of a plastic bottle, and we can do this by considering its energy content. Plastic, like petrol, is composed of hydrocarbons, so we'll use the energy density of 4×10^7 J/kg that we quoted in the previous example. A single plastic bottle weighs more than 1 g but less than 100 g, so let's guess the mass to be 10 g. The energy content is therefore (4×10^7 J/kg) $\times$ (10 g) = 4×10^5 J. We can convert this to Watt-hours with (4×10^5 J) $\times$ (1 hr / 4×10^3 s) = 100 W-hr. The cost of 100 W-hr is 0.1 rand, or 10 cents. It's up to you to decide whether or not that amount of savings is worth the effort...

12 Example: Einstein and you

What is the probability that one atom or molecule in your body was once part of Einstein?

You are exchanging matter with your environment constantly through food consumption, breathing, etc. Let's consider breathing and take that to be the primary method of matter exchange. To tackle the above question, we need to calculate two quantities: the total number of air molecules in the earth's atmosphere, and the number of air molecules that have passed through your lungs.

We can obtain the mass of earth's atmosphere by considering the pressure. Atmospheric pressure is $\sim 10^5$ Pa in SI units, and recall that pressure is defined as force divided by area. The force, in this case, comes from the total weight of the atmosphere in earth's gravitational field. The force associated with weight is defined as the product of mass (kg) and earth's gravitational acceleration, $g = 9.8$ m/s^2. We can therefore write down the following expression for the total mass M of the atmosphere:

$$M \times (9.8\ \mathrm{m/s^2}) = (10^5\ \mathrm{Pa}) \times (4\pi \mathrm{r}^2),$$

where r is the radius of the earth (roughly 6000 km), and $4\pi r^2$ gives the surface area over which the weight of the atmosphere pushes down. Solving for M, we get an approximate mass of 5×10^{18} kg. To convert this mass to a number of molecules, we'll have to recall some basic chemistry: air weighs roughly 30 g/mole, and 1 mole is 6.02×10^{23} molecules. Therefore, (5×10^{21} g) / (30 g/mole) $\times$ (6×10^{23} molecules/mole) $\sim 10^{44}$ total molecules in the atmosphere.

What about the number of air molecules that have passed through your lungs during your lifetime? Let's guess that you exchange 0.5 L with every breath, take one breath every second, and that you're about 20 years old. This gives us a total of $\sim 10^8$ L of air. From chemistry, we know that one mole of air is roughly 25 L at ambient temperature and standard pressure. The number of molecules that have passed through your lungs is therefore (10^8 L) / (25 L/mole) $\times$ (6×10^{23} molecules/mole) $\sim 10^{30}$ molecules.

Putting the above numbers together, we can calculate that the probability of any particular air molecule having passed through your lungs is $10^{30}/10^{44} \sim 10^{-14}$. The same is true for Einstein. Therefore, the probability of any given air molecule having passed through both your lungs *and* Einstein's lungs is $(10^{-14})^2 = 10^{-28}$. To obtain the *average number of air molecules* that satisfy this condition, we multiply the probability by the total number of air molecules in the entire atmosphere: $10^{-28} \times 10^{44}$ molecules = 10^{16} molecules. In other words, on average, your lungs have exchanged 10^{16} air molecules that once passed through Einstein's lungs, and it is therefore almost completely certain that there's a little bit of Einstein in each of us!

13 Back to our first example: Fermi and the atomic bomb

Fermi estimated the energy yield of the atomic bomb by dropping a few small pieces of paper on the ground when the air blast reached him. He was standing 16 km from the blast site, and the pieces of paper moved by 2.5 m. What was the TNT equivalent of the bomb energy?

To tackle this calculation, we need to invoke a bit of elementary physics. The work done by the shock wave of the bomb is $W = P\Delta V$, where P is the pressure of the shock wave and ΔV is the change in volume. Work has units of energy, and in SI units, it is measured in Joules. We obtain ΔV by calculating the volume of a hemispherical shell with a radius of 16 km and thickness equal to the 2.5-m displacement of the paper: $\Delta V = 2\pi r^2 d = 6(1.6 \times 10^4 \mathrm{m})^2(2.5\mathrm{m}) = 4 \times 10^9$ m^3.

Our next task is to estimate the pressure, which is measured in force per area (N/m^2). We can guess a lower limit of 10^2 N/m^2, which is negligible (equivalent to the weight of 1 g spread out over 1 cm^2), and we can guess an upper limit of 10^4 N/m^2, which is enough to knock you down (equivalent to 1 ton spread out over 1 m^2). Our estimate is therefore $P = 10^3$ N/m^2.

Putting the above numbers together, we get $W = P\Delta V = (10^3 \mathrm{N/m^2})(4 \times 10^9 \mathrm{m^3}) = 4 \times 10^{12}$J $=$ 1 kiloton, the equivalent energy of one kiloton of TNT. Note that this energy corresponds to the work done by the shock wave alone, but the total energy of the bomb is spent in several different ways: light, nuclear radiation, moving the ground, etc. Therefore, to obtain our final estimate for the bomb's total energy yield, we multiply by a fudge factor of a few, and we'll say that our answer is 4 kilotons of TNT.

So in summary:

- We got 4 kilotons.
- Fermi got 10 kilotons.
- The actual blast was 20 kilotons.

Our answer isn't perfect, but it's not too bad! And we shouldn't feel disheartened that we didn't do as well as Fermi; after all, he was a Nobel Prize winner!

14 We don't always win...

We've looked at some of the surprising successes of order-of-magnitude calculations, but I'll end with an example where we don't quite win. The problem we'll examine here is called the *Fermi paradox*, and it originated, once again, with Enrico Fermi. The story goes that Fermi was having lunch with his colleagues in 1950, and several of them were expressing optimism that our Galaxy is full of intelligent civilizations. Fermi's response was, "then where is everybody?"

Let's look at some numbers and try to figure out whether or not our own Milky Way should be filled with intelligent beings.

- Speed of fastest outbound spacecraft we've built so far: ~ 15 km/s
- Milky Way size: ~100,000 light years = $\sim 10^{18}$ km
- Age of Milky Way: $\sim 10^{10}$ years = $\sim 3 \times 10^{17}$ s

With our fastest spacecraft (which we can consider to be modest technology from our still-young civilization), we calculate the Galaxy crossing time to be (10^{18} km) / (15 km/s) $\sim 10^{17}$ s. This number is a factor of a few smaller than the age of Milky Way, so in other words, we've had enough time to colonize the entire Galaxy. If our extraterrestrial friends got a head start on us and developed spacecraft that was just a factor of a few faster than ours, they certainly should have found us by now. (So what's the holdup?!)

There are, of course, some considerations that we've left out. One oversimplification is that we've assumed a straight-line journey in our initial calculation. If we actually wanted to colonize all of the

stars in our Galaxy, we'd have to visit all of them. There are about 100 billion stars, with an average separation of a few light years. The total travel distance to visit all stars is then around a trillion light years, or $\sim 10^{25}$ km. There is no way a single spaceship, even traveling at the speed of light, could even come close to covering this distance.

Let's consider a slightly more complicated colonization picture. Suppose our sentient, extraterrestrial friends have a desire to conquer and that they take the following strategy:

- Send a ship to the nearest star. If the typical distance is a few light years, where one light year is $\sim 10^{13}$ km, then the time it takes to arrive is roughly 10^{12} s = $\sim 10^5$ years. (Here I'm assuming that their space travel technology is as primitive as ours currently is.)
- Upon arrival at the nearby star, spend 1000 years acclimating, breeding, and settling in. Note that this is a long time compared to human history, but it's negligible compared to the 10^5 years of space travel time.
- After the 1000-year settling-in period, both the old world and new world would send out spacecraft to different stars, each taking 10^5 years to repeat the travel process, plus an additional 1000 years to settle in. After all four worlds are established, each would send out spacecraft again, and process repeats itself from there.

The scenario described above is *exponential growth*, and we'll see this model behavior come up again in this class. After 10^5 years, our extraterrestrial friends have two worlds. After 2×10^5 years they have four, after 3×10^5 years they have eight, and so on. The number of worlds N can thus be described by the equation

$$N = 2^{t/10^5 \text{ years}}.$$

We can now calculate the estimated time it takes to colonize all of the stars in the Milky way by setting N equal to 100 billion = 10^{11} and solving for t. If you work out the math, the answer is *a few million years*, which is almost nothing—a few tenths of a percent—compared to the age of the Milky Way. In other words, an enterprising, intelligent, extraterrestrial civilization should have trivially occupied every star in our Galaxy by now.

The fact that we have no evidence for alien overlords on every nearby star, despite our not-obviously-unreasonable assumptions above, is the statement of the Fermi paradox.

This is just one example of a back-of-the-envelope calculation that produces a result that doesn't match reality. I include it here as an illustration that calculations that fail are just as interesting as calculations that succeed, perhaps even more so. The Fermi paradox opens up a whole host of new questions. I'll leave it to you to read more details online if you're interested (I also suggest reading about the Drake Equation), and I'll just list a handful of possible resolutions here:

- Are we unique in our Galaxy?
- Are advanced civilizations short-lived? (One could speculate how long humans have before we self-destruct.)
- Is interstellar travel just too hard?
- Are the aliens just holding off on contacting us, or are we too primitive to communicate with them?

Math 246 notes #2: Scaling relations

1 Introduction

- In the previous section of this class, we discussed methods for obtaining rough estimates in various types of calculations. We will now talk about how to extend calculated quantities to other situations using arguments of proportionality. This technique is known as applying *scaling relations*.
- A simple example is a scale model of an object. These models have a length scale factor, N, which describes the ratio of a length in the model compared to the corresponding length in real life. For example, a map that shows 1 km in real-world distance as 1 cm on the drawing would have a length scale factor of $N = 10^{-5}$. The *surface area* changes by a factor of N^2, and the *volume* (for a 3D model) changes by a factor of N^3. In the case of our map example, the area of a city in the drawing would be $N^2 = 10^{-10}$ times the actual area of the city in real life.
- The importance of scaling relations can be illustrated by considering what scale models of objects are used for. For example, if you build a 1/10 scale model of a car and place it in a wind tunnel, how fast should the air flow be in order to simulate realistic drag forces of the full-size car traveling at, say, 100 km/hr? (You might naively guess that the wind speed should be divided by 10 for the scale model, but this turns out to be incorrect!)
- Scaling laws are often used to solve cutting-edge, complex problems in scientific research. If you search for "scaling" on the Nobel Prize website, you'll find dozens of references. (Try it yourself: `http://www.google.com/search?q=scaling+site:nobelprize.org`.)

2 Warmup exercises: basic geometry

Some of the most familiar scaling relations are those from basic geometry. Let's start by considering a square, 1 m on a side. If we double the length scale and enlarge the square so that it's now 2 m on a side, *all* linear quantities, e.g. the perimeter or the diagonal length, increase by a factor of two. The area of the square increases by a factor of $2^2 = 4$. Similarly, if we increase the length scale by a factor of 3, the area increases by a factor of $3^2 = 9$. In other words,

$$\text{(ratio of areas)} = \text{(ratio of lengths)}^2.$$

We can also write down the relationship between length and volume. If we consider a cube, rather than a square, increasing the length scale from 1 m to 2 m increases the corresponding volume by a factor of $2^3 = 8$. Similarly, increasing the length scale by a factor of 3 increases the volume by a factor of $3^3 = 27$. Our scaling relation is then

$$\text{(ratio of volumes)} = \text{(ratio of lengths)}^3.$$

It's important to note that the above relations hold for *uniform* scaling where the objects are *self-similar*. Furthermore, the relations are true for all self-similar objects regardless of shape (not just squres and cubes), and they apply even for non-integer scale factors. In the case of non-uniform scaling, you need to think a little more carefully. For example, a cylinder of length ℓ and radius r has a volume that scales as ℓr^2. Therefore, doubling only ℓ (and keeping r fixed) would double the volume, and doubling only r (and keeping ℓ fixed) would quadruple the volume. When the scaling is non-uniform, the volume does not follow the naive length-cubed scaling relation.

Let's briefly consider the appropriate terminology and language when describing scaling. A statement such as "triangle A is twice as big as triangle B" is ambiguous because it is unclear whether you're referring to lengths or areas. A practical example is that a two-liter bottle of soda is "twice

as big" as a one-liter bottle, but the height and diameter are only 1.26 times as big. You must be *precise* when you make comparisons. For example, the precise way of phrasing the above scaling relations is "area scales as the square of the length, and volume scales as the cube of the length." In math, we express these relations as $A \propto L^2$ and $V \propto L^3$, where L, A, V are length, area, and volume, respectively. The $\propto$ symbol denotes proportionality (but not necessarily dimensional homogeneity, as we will see in later sections).

Simple example #1: how does volume scale with surface area?

Because V scales as L^3 and A scales as L^2, we can combine those relations to establish that $V \propto A^{3/2}$.

Simple example #2: we have two spheres made of the same material, but one has a diameter that's twice as large as the other. How much does the larger sphere weigh in comparison to the smaller sphere?

If the spheres are made of identical material, then the density (mass per unit volume) is the same in both cases. We can thus state that the mass m scales as volume V, or $m \propto V$. The volume, in turn, scales as the cube of the length scale L. Therefore, $m \propto L^3$, and the larger sphere weighs $2^3 = 8$ times as much as the smaller sphere. (Note that this proportionality relation is an example of one that is not dimensionally homogeneous; that is, the units of mass differ from the units of length cubed.)

3 Gravity on the moon

Let's move on from geometry and consider an example that involves gravity. The gravitational force between two objects is

$$F = G\frac{Mm}{r^2},$$

where G is the gravitational constant, M and m are the masses, and r is the separation distance of the objects. Let's use this equation to answer the question: how much gravity would you feel on the moon relative to what you experience on earth? In other words, we are interested in calculating

$$\frac{F_{\text{moon}}}{F_{\text{earth}}} = \left(G\frac{M_{\text{moon}}\, m}{r_{\text{moon}}^2}\right) \Big/ \left(G\frac{M_{\text{earth}}\, m}{r_{\text{earth}}^2}\right),$$

where m denotes your mass in this particular case. Rather than evaluate this expression in full gory detail, let's write it in terms of a scaling relation so that we can extrapolate our answer to any size.

The mass of any uniform object scales as its volume, which in turn scales as r^3. If we assume that the moon and the earth have roughly the same densities (not quite true, and we'll come back to this), then we can write the approximate scaling relation $M_{\text{moon}}/M_{\text{earth}} \propto r_{\text{moon}}^3/r_{\text{earth}}^3$, where the densities have cancelled each other. The ratio of gravitational forces then scales as

$$\frac{F_{\text{moon}}}{F_{\text{earth}}} \propto \left(\frac{M_{\text{moon}}}{r_{\text{moon}}^2}\right)\left(\frac{r_{\text{earth}}^2}{M_{\text{earth}}}\right) \propto \left(\frac{r_{\text{moon}}^3}{r_{\text{moon}}^2}\right)\left(\frac{r_{\text{earth}}^2}{r_{\text{earth}}^3}\right) = \frac{r_{\text{moon}}}{r_{\text{earth}}}.$$

Thus, the ratio of moon-to-earth radii should also give us the ratio of gravitational forces. The moon's radius is 1700 km and the earth's radius is 6400 km, so our answer is that the gravity on the moon is roughly $1/4$ the gravity we feel on earth. This answer turns out to be close but not exactly right: the actual answer is $1/6$, and the discrepancy comes from the fact that the average density of the moon is about 60% that of the earth. If we account for the density factors ρ in our scaling relation, we get a better answer:

$$\frac{F_{\text{moon}}}{F_{\text{earth}}} \propto \frac{\rho_{\text{moon}}\, r_{\text{moon}}}{\rho_{\text{earth}}\, r_{\text{earth}}} = 0.6 \times \frac{1}{4} \approx \frac{1}{6}.$$

4 Size effects in animals

There are many rich and interesting scaling relations that arise when we consider size effects in animals. Why do heavier animals have stockier bodies and thicker legs? For example, an impala is not a scale model of an elephant. Why is the largest bird much smaller than a large mammal? Why can fleas and humans jump about the same height even though a human weighs vastly more than a flea? Why can insects survive a fall from 50 m while a human falling from that distance would splatter? (You'll answer this last question on your tutorial.)

4.1 Trunk proportions

Let's start by trying to make sense of how the proportions of an animal's trunk are affected by its size. For this exercise, we'll focus on four-legged creatures. Animals are complicated objects, and we could easily get lost in constructing a correspondingly complicated model. Let's try to keep the picture as simple as possible, and we'll start off with some very crude models.

We'll start by considering the dimensions of an animal's body (trunk) and figuring out how they depend on weight. For our crude model, we can pretend that the animal's trunk is a flexible beam that's supported on both ends by legs. You've probably walked over suspended wooden planks before, and you know that the sagging depends on both the shape of the plank and your weight. To proceed with our calculation, we need to invoke a result from elasticity theory. If a beam of length l, vertical thickness t, and cross-sectional area A is subjected to a uniform load (force) F while its end points are held fixed, the maximum deflection δ satisfies

$$\delta \propto \frac{Fl^3}{t^2 A}.$$

The force F arises from the weight of the trunk under gravity, and this force scales as the mass m of the trunk. We can thus write

$$F \propto m \propto lA$$

since the mass scales as the volume of the trunk, and the volume is given by the product lA. Let's now define the *relative sagging*, δ/l, and express it as

$$\frac{\delta}{l} \propto \frac{(lA)l^3}{t^2 A} \times \frac{1}{l} = \frac{l^3}{t^2}.$$

We can argue that δ/l should be more or less constant for animals and that it is a *scale-invariant quantity*. The argument for this can be constructed by considering limiting cases: 1) if δ/l is large, the deflection is large and the animal is likely cripplingly deformed; 2) if δ/l is small, then the animal has excess body material that isn't necessary for providing support (that is, the animal is overengineered, and such inefficiencies are disfavored by evolution). Thus, if we take our crude model to the next step where we assume δ/l is a constant, then we can obtain the following scaling relation:

$$t \propto l^{3/2}.$$

In other words, **our first prediction is that quadruped trunk thickness scales as trunk length to the 3/2 power.** So, how well does our crude model perform? Figure 1 shows our model prediction compared to measurements, and the agreement is astonishing. (We almost don't deserve to have done so well with so little work!)

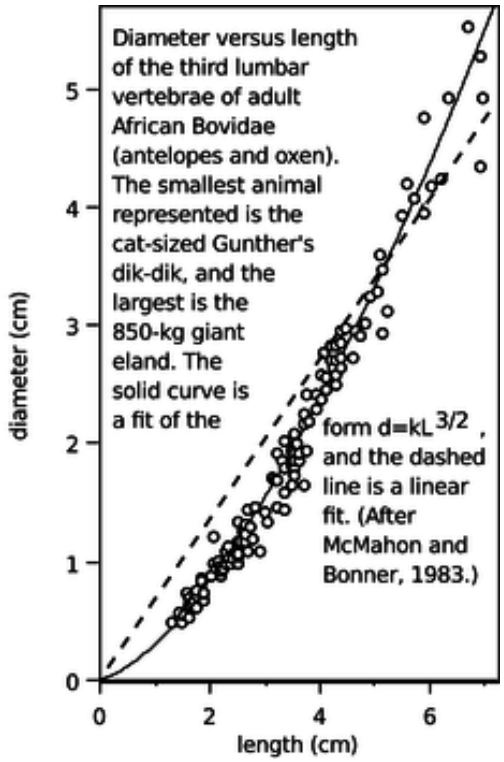


Figure 1: Plot of vertebrae diameter versus length, which illustrates the scaling of trunk thickness with trunk length. Data points are shown for African Bovidae, which range from the slender dik-dik to the giant eland. The solid curve is a fit to $t \propto l^{3/2}$ and is an excellent match with the observational data.

4.2 Limb proportions

The next question we'll try to answer is: how does limb size vary with body weight? To address this problem, we'll use a model that's even cruder than our previous model. Leg bones must be strong enough so that they don't break when an animal moves. We need to invoke one more result from elasticity theory, which is that the ability of a beam to withstand a force F is proportional to its

cross-sectional area A. Our beam, in this case, is a leg bone. From Newton's second law, we know that force is equal to the product of mass and acceleration. We have a sense of how mass scales, but how does acceleration vary? Let's consider two broad classes of animals, "slow" and "fast." For slow animals, the acceleration is dominated by gravitational acceleration, which is a constant (g = 9.8 m/s^2). For fast animals, acceleration depends on the rate of muscle contraction, which has about the same maximum value in all species. We can thus pretend that acceleration can be treated as a constant, and we obtain the following relation:

$$A \propto F \propto m.$$

Since the cross-sectional area of the leg bone scales as the square of the bone diameter d, we can write

$$d \propto m^{1/2}.$$

In other words, **our crude model predicts that limb thickness scales as the square root of body mass**. It is very important to note that if we had assumed that all animal parts remain in proportion as size varies, we would have obtained the result $d \propto m^{1/3}$. Our model thus predicts that leg thickness must increase *faster* than what proportionality would dictate; that is, legs of larger animals are relatively thicker than the legs of smaller animals. And this is why elephants are chunkier than impalas.

4.3 How high can animals jump?

To jump a height h, an animal must expend energy equal to mgh, which is the expression for work done against the force of gravity. How much stored energy does an animal have, and how does it scale? We can assume that the amount of available energy is proportional to the volume of muscle V_m, so putting the above statements together, we get

$$mgh \propto V_m.$$

But V_m is proportional to the total mass, so our scaling relation reduces to

$$h = \text{constant},$$

which is a rather surprising result! Table 1 illustrates that our answer turns out to be close to reality. Although there's a factor of 3 spread in the tabulated jump heights, this is a small variation considering that the masses span 8 orders of magnitude!

	Flea	Click beetle	Locust	Human
Mass (g)	5×10^{-4}	4×10^{-2}	3	7×10^{4}
Jump height (cm)	20	30	60	60

Table 1: Maximum jump heights for animals of varying mass

4.4 Metabolic rates

We've already seen that as the size of an animal varies, its surface area scales as $m^{2/3}$. This scaling relation between surface area and mass (volume) has important consequences for mammals, which must maintain a constant body temperature in order to avoid dying from becoming too cold. Mammals lose heat through their skin, so we can guess that the rate of heat loss must be proportional to their surface area. In order to generate body heat, animals need to consume food, which is then metabolized to produce energy. Oxygen is is a necessary ingredient for metabolic processes, so oxygen consumption is a good proxy for the rate of heat production. Measurements of guinea pigs of different sizes support our hypothesis that oxygen consumption scales as $m^{2/3}$. In other words, although smaller animals metabolize less oxygen and produce less heat in absolute terms, the amount of food and oxygen they must consume is greater in proportion to their own mass. For example, the Etruscan pygmy shrew is one of the smallest mammals, weighing about 2 g as an adult, and it must eat nearly continuously in order to survive.

On the opposite end of the size spectrum, large animals face the problem of overheating. The amount of food consumption can't drop to infinitesimal amounts, as cells have minimum energy requirements in order to run their internal machinery. Elephants are an example where evolution has implemented a design change—large ears—in order to increase surface area and improve cooling. Elephant ears are just one example of a change in shape necessitated by a change in scale.

5 The cost of packaging

You've probably noticed that when you buy some goods in bulk, e.g. flour, detergent, sugar, the larger packages often cost less per unit weight. The reason for the cost savings is often attributed to savings in the cost of packaging and handling. In this example, we will test this hypothesis by constructing a simple model and studying the subsequent scaling relations. We are fundamentally interested in understanding how the cost of a product (output) depends on the size of that packaged product (we'll describe this with several input variables shortly). Let's consider the wholesale cost—that is, the total cost to the producer—and we can assume that the cost to the consumer is just some uniform scale factor on top of that.

We can come up with the following list of inputs that affect the wholesale cost:

- a = Cost of producing the product. We can assume that this cost is proportional to the amount of product, so we can write $a \propto w$, where w is the weight.
- b = Cost of packaging the product. This cost depends on how long it takes to fill the package, close the package, and load it into a container for shipping. The first of these is proportional to the weight w, but the last two are probably roughly constant for packages within a reasonable range. We will thus write $b \approx fw + g$, where f and g are positive constants.
- c = Cost of shipping the product. Shipping costs generally depend on both weight and volume, but since volume scales as weight for filled packages, we can write $c \propto w$.
- d = Cost of the packaging material. This cost is a bit more complicated, and we can imagine several different scenarios. The amount of packaging may scale with the package's surface area S, but it may also scale with the package volume (e.g. might need thicker material to support a heavier volume). If the range of packaged volumes doesn't cover too large a spread, one could also imagine roughly constant-sized packaging material. We'll thus account for all three scenarios and write $d \approx hw + kS + \ell$, where h, k, ℓ are positive constants.

Now that we have our inputs, we can combine them to form the total cost per unit weight:

$$\frac{\text{Cost}}{w} = \frac{a+b+c+d}{w} = n + \frac{p}{w^{1/3}} + \frac{q}{w},$$

where n, p, q are positive constants formed from combinations of the constants introduced in the list above. Here we have assumed that the various packages are geometrically similar, so $S \propto w^{2/3}$. From the above expression, we can see that the cost per unit weight decreases as the weight increases, so our simple model correctly predicts our observation that goods tend to be cheaper when purchased in bulk.

We'd like to test our model further and see if it makes other interesting predictions. We could use data to solve for the constants n, p, q and then attempt predicting the per-unit-weight cost for packages of various sizes. However, since our model is quite crude, we're unlikely to get very far with this method. Let's instead take a broader approach and investigate how the *rate of change* of cost-per-weight varies:

$$r \equiv -\frac{d(\text{cost/w})}{dw} = \frac{p}{3w^{4/3}} + \frac{q}{w^2}.$$

The above expression is a decreasing function of w, so the increase in rate of savings per unit weight is less when the package is larger. Another way of translating this math into English is: "when

purchasing prepackaged goods, doubling the size of the purchased package tends to result in greater savings per unit weight when the packages are small than when they are large." If we multiply the previous expression by w, we can also calculate the rate of total savings:

$$rw = \frac{p}{3w^{1/3}} + \frac{q}{w}.$$

This is also a decreasing function of w.

The main take-away message in this example is that we started with an extremely crude model, and yet we were not only able to replicate the predicted cost trends of bulk packaged goods, but we were also able to make a couple of other sensible predictions. These sorts of *qualitative* predictions tend to be quite robust, and they provide an excellent way of testing whether or not a model is reasonable.

6 The speed of racing shells

Competitive rowers use small, thin boats (shells) in races, and these shells are built to certain specifications and can hold one, two, four, or eight rowers. For the eight-person crew, there is a lightweight and heavyweight category in sporting competitions. Empirically, the best racing times vary depending on the number of rowers in the shell, with small-number teams systematically taking slightly longer to finish a race than a large-number team. We would like to write down a simple model of the racing shell so that we can devise a method for adjusting the shape in order to pit different teams against each other on an equal basis in practice sessions.

Let's first consider what forces are at play when the racing shells are in motion. The rowers provide power that propel the shell forward, but the shell is subject to drag forces from the water. The balance of these forces determines the speed of the shell and its time in the race. If we invoke a result from hydrodynamics, we can write that the drag force on the shell is due to skin friction, and this force is proportional to Sv^2, where S is the wetted surface area and v is the velocity. The power P (energy per unit time) required to maintain a certain velocity is equal to the product of the velocity and the drag force, so $P \propto Sv^3$, or $v \propto (P/S)^{1/3}$.

Our next assumption is that all rowers have the same weight and the same constant power output during the race. Therefore, if the combined power of the rowers is constant and balances the required power $P \propto Sv^3$, then v is also a constant. The time t it takes to complete the race is proportional to v^{-1}, so we can write

$$t \propto \left(\frac{S}{P}\right)^{1/3}.$$

Let's use the above scaling relation to help us understand the differences between lightweight and heavyweight eight-person crews. Data collected from racing events show that the finish times of heavyweight crews (average rower = 86 kg) are consistently 5% better than those of lightweight crews (average rower = 73 kg). We'll start by writing the ratio of finish times:

$$\frac{t_L}{t_H} = \left(\frac{S_L}{S_H}\right)^{1/3} \left(\frac{P_H}{P_L}\right)^{1/3},$$

where the subscripts H and L denote heavyweight and lightweight, respectively. In order to explain the 5% difference in finish times, we must say something about the difference in wetted surface areas and power output of the rowers. The only information we have is the ratio of average rower body mass: m_H/m_L = (86 kg)/(73 kg) = 1.18.

We argued previously that the stored energy of a human, animal, etc. depends on factors such as muscle volume and lung volume, and these quantities scale with total volume or, equivalently, weight. Therefore, we can guess that $P_H/P_L = m_H/m_L = 1.18$, and our expression for ratio of finish times reduces to

$$\frac{t_L}{t_H} = (1.18)^{1/3} \left(\frac{S_L}{S_H}\right)^{1/3} = 1.06 \left(\frac{S_L}{S_H}\right)^{1/3}.$$

If we make the rough assumption that $S_L \approx S_H$, then the above expression comes close to the 5% observed difference in finish times. In reality, S_H is slightly larger than S_L, and this difference would bring our 6% calculated difference closer to the actual answer. Although our answer isn't perfect, it's in the right range. We'll come back to S a little later.

If we want to redesign the shells in order to make the finish times more equal, then we need to adjust the wetted surface areas of the shells. Using the relation $t \propto S^{1/3}$ for fixed power output, we can write

$$\frac{S_r}{S_o} = \left(\frac{t_r}{t_o}\right)^3,$$

where the subscripts r and o denote the redesigned and original shells, respectively. The lightweight crews will have finish times equal to those of the heavyweight crews if $t_r/t_o = 0.95$, so the ratio of wetted surface areas must be set to $S_r/S_o = 0.95^3 = 0.86$. In other words, we can either decrease the wetted surface area of a loaded lightweight shell by 14%, or we can increase the wetted surface area of a loaded heavyweight shell by 16% (1/0.86 = 1.16).

Now that we have a rough understanding of heavyweight versus lightweight eight-person teams, let's extend our model to describe the other team sizes. We'll express the finish time in terms of number of team members, which is the input variable that we ultimately want to vary. From our assumption that power output is the same for all members in one team, we can write

$$t \propto \left(\frac{S}{P}\right)^{1/3} \propto \left(\frac{S}{N}\right)^{1/3},$$

where N is the number of team members. In order to calculate S, we need some more information about the relative sizes of the racing shells for different team sizes. We'll assume that the shells are geometrically similar, and their loaded weights are proportional to N. We'll additionally assume that the submerged parts of the loaded shells are geometrically similar. These are both extremely crude assumptions, but it's the best we can do given the lack of information. The volume of water displaced by a shell is proportional to its total weight, which in turn depends on the number or rowers. Since we've assumed geometric similarity, the volume also scales as the cube of the length scale ℓ. Putting these relations together, we obtain

$$N \propto \ell^3.$$

Taking the geometric similarity assumption one step further, we can approximate the submerged (wetted) surface area as being proportional to the ℓ^2. Therefore, $S \propto \ell^2 \propto N^{2/3}$, and we can re-express the finish time as

$$t \propto \left(\frac{S}{N}\right)^{1/3} \propto \left(\frac{N^{2/3}}{N}\right)^{1/3} = N^{-1/9}.$$

In other words, **our simple model predicts that the finish times are proportional to the number of rowers raised to the power -1/9**. It turns out that this model agrees fairly well with measurements presented in an article by T. A. McMahon (1971). So, in order to equalize racing conditions for teams of differing size, we can combine the scaling relations $t \propto N^{-1/9}$ and $S_r/S_o = (t_r/t_o)^3$ to redesign a shell's wetted surface area depending on the number of rowers.

7 One last animal example: the law of urination

A couple of years ago, an interesting biological scaling relation achieved ultimate internet fame. The result, the "law of urination," states that animals empty their bladders over a nearly constant duration of 21 ± 13 seconds, and the result holds true for animals spanning 3 to 5000 kg. (Test subjects included dogs, goats, cows, and elephants.) In other words, the time it takes to pee is a nearly scale-invariant quantity, changing by only a factor of few over three orders of magnitude in mass. Let's try to explain this with a model.

The first ingredient we need is a geometric model of the bladder itself. We can crudely model the bladder as a volume V of some shape that's connected to the urethra, which we model as a tube of some length L and diameter D. We can make the guess that the entire bladder system scales in constant proportion across animals, and it turns out that this guess is supported reasonably well by observation. The mass M of an animal scales as its total volume, so $V \propto M$ (actual data suggest $V \propto M^{0.97}$, so this is pretty close). Because urethra length and diameter are both linear quantities, we can approximate $L \propto D \propto M^{1/3}$ (actual exponents are 0.41 and 0.38 for L and D, respectively).

The second model ingredient we need is a description of the flow of fluid exiting the bladder. For this part, we need to invoke some simple fluid dynamics and write down the pressure balance:

$$P_{\text{bladder}} + P_{\text{gravity}} = P_{\text{inertia}},$$

where P_{bladder} is the driving pressure supplied by the bladder, P_{gravity} is the gravitational pressure associated with a fluid column of height L, and P_{inertia} is a pressure term that describes the kinetic energy of the fluid. (There are also effects that arise from viscosity and surface tension at the outlet, but we will restrict our model to large animals where these effects can be neglected.) Fluid dynamics tells us that the pressure terms can be expressed as

$$P_{\text{bladder}} + \rho g L = \rho u^2/2,$$

where ρ is the fluid density, g is gravitational acceleration (constant), and u is the fluid velocity. Solving for the fluid velocity, we get $u = \sqrt{2P_{\text{bladder}}/\rho + 2gL}$.

In order to get the total *flow rate*, Q, we need to multiply u by the cross-sectional area of the urethra (this gives us a volume per unit time); therefore, $Q \propto D^2 u$. The total time T required to empty the bladder is then

$$T = \frac{V}{Q} \propto \frac{V}{D^2\sqrt{2P_{\text{bladder}}/\rho + 2gL}}.$$

If we substitute the mass scalings that we wrote earlier, we obtain the result

$$T \propto \frac{M}{M^{2/3}\sqrt{2P_{\text{bladder}}/\rho + M^{1/3}}} = M^{1/3}(2P_{\text{bladder}}/\rho + M^{1/3})^{-1/2}.$$

When the mass of the animal is large so that the P_{bladder} term is small in comparison, the scaling relation reduces to $T \propto M^{1/6}$. In other words, the time it takes to empty a bladder is not entirely independent of mass, but the scaling relation is *extremely* flat. For example, over the 3–5000 kg mass range quoted at the beginning of this section, our model predicts that the bladder emptying time changes by a factor of $(5000/3)^{1/6} \approx 3$, and this small level of variation is more or less consistent with observations.

For further details (and some choice diagrams), I recommend reading the original paper, *Law of Urination: all mammals empty their bladders over the same duration*, by P. J. Yang et al., which won a prestigious Ig Nobel prize in 2015[1]. The paper also contains an interesting discussion of smaller animals, where viscosity and surface tension effects can't be ignored.

[1] `http://www.improbable.com/ig/winners/#ig2015`

Math 246 notes #3: Dimensional analysis

1 What and why?

- Dimensional analysis is a trick for getting an answer without actually having to solve anything—the ultimate lazy person's approach.
- The basic idea: units (meters, seconds, kilograms, etc.) are artificial. *Meaningful, valid laws that govern reality must have the same form regardless of what units you choose.* Only dimensionless quantities, i.e. numbers, do not depend on units, so we should look for those quantities.
- More often than not, the dimensionless quantities are formed from unique combinations of parameters. This uniqueness constrains the problem that we are trying to solve and often gives us the answer we were looking for—without us having to actually do any work!

2 Fundamental dimensions and dimensions of common quantities

Generally speaking, the basic physical dimensions that we use are mass, length, and time. Note that this is not an exhaustive list—as an example, there are other basic dimensions required to describe electricity and magnetism—but it's a good starting point for many problems. We'll express these quantities as $[M], [L], [T]$, where the square brackets indicate that these quantities are dimensions and not variables. Most physical quantities that we'll deal with can be reduced to some combination of these three dimensions. For example, velocity v is expressed as meters per second, so its dimension is length per time, or $[v] = [L]/[T]$. (Note that the square brackets are used to indicate the dimensions of v; I will continue using this notation for the rest of these notes.) Similarly, acceleration a is meters per second-squared, so its dimension is $[a] = [L]/[T]^2$.

For reference, here is a list of dimensions for some physical quantities that we'll encounter frequently in this class:

- Frequency: $[T]^{-1}$ (number of cycles, which is dimensionless, per unit time)
- Mass density: $[M][L]^{-3}$ (mass per unit volume)
- Force: $[M][L][T]^{-2}$ (mass times acceleration)
- Pressure: $[M][L]^{-1}[T]^{-2}$ (force divided by area)
- Energy: $[M][L]^2[T]^{-2}$ (force times distance)
- Power: $[M][L]^2[T]^{-3}$ (energy per unit time)

3 Warmup exercise: dimension of G

As a reminder, all equations that we consider in this class are *dimensionally homogeneous*, i.e. the dimensions are the same on both sides of the equal sign. Let's use this concept to determine the dimension of the gravitational constant G, using the law of gravity

$$F = \frac{Gm_1m_2}{r^2}.$$

Here F is the gravitational force between two objects with masses m_1 and m_2, separated by a distance r. The dimension of force is $[F] = [M][L][T]^{-2}$, and the dimension of m_1m_2/r^2 is $[M]^2[L]^{-2}$. In order to make the law of gravity dimensionally homogeneous, G must have a dimension given by

$$[G] = \frac{[M][L][T]^{-2}}{[M]^2[L]^{-2}} = [M]^{-1}[L]^3[T]^{-2}.$$

It is important to note that *a constant is not necessarily dimensionless*, as we have just seen here with G.

4 Newton's second law

In the previous example, we used a known physical relation (law of gravity) in order to determine the dimension of one of the constituent quantities (G, the gravitational constant). We will now look at an example where we *reverse* this process. We will start with a handful of relevant variables with known dimensions, and we will form dimensionless quantities in order to *synthesize* physically meaningful relationships.

Let's suppose that we live in Newton's time and that we're trying to find the relationship between force (F), mass (m), and acceleration (a). From experiments, we know that there's *some* relationship between these three quantities, but we don't know the exact mathematical form. The most general relation between the three variables is

$$h(F, m, a) = 0,$$

where h is a function that we must find. The main guiding principle from dimensional analysis is that the general form of h can be written using only dimensionless quantities. If we succeed in finding this form, then the prize we win is a law of nature.

In order to form dimensionless quantities, we first need to write down the dimensions of our input variables:

- $[F] = [M][L][T]^{-2}$
- $[m] = [M]$
- $[a] = [L][T]^{-2}$

The following variable combinations are examples of dimensionless groups where all of the units cancel each other:

$$\frac{F}{ma}, \quad \left(\frac{F}{ma}\right)^2, \quad \sin\left(\frac{F}{ma}\right), \quad \frac{ma}{F}.$$

Note that all of these combinations are fundamentally functions of F/ma, so we can rewrite our general relation as

$$h(F, m, a) = f(F/ma) = 0,$$

where f is a dimensionless function that takes a dimensionless number as its input and produces a dimensionless number as its output. At this point, we only need to identify the zeros of f (we don't need to solve for f itself), and then we're done. In general, a function can have many zeros, so how do we choose which one is correct? Unfortunately this is where the buck stops with dimensional analysis—we simply don't know. We therefore assign a name, Π, to our unknown zero, and our final result is then $F/ma = \Pi$, or

$$F = \Pi ma.$$

This equation is a famous result in physics, Newton's second law. The exact form of Newton's second law has $\Pi = 1$ so that $F = ma$. Although we were not able to determine the value of Π using dimensional analysis, we did obtain the correct form of the relation up to a proportionality constant. For most of this text, we'll punt and assume (or hope) that Π is close to unity.

5 The Buckingham Pi theorem

The Π used in the previous section is in honor of the **Buckingham Pi theorem**, which lies at the heart of dimensional analysis:

The solution to any physical problem involving n variables $v_1, v_2, ..., v_n$ depending on r independent dimensions has the form $f(\pi_1, \pi_2, ..., \pi_{n-r}) = 0$. Here f is some function, and $\pi_1, \pi_2, ...$ are dimensionless products (and quotients) formed from the variables and constants appearing in the original equation. Not all dimensionless combinations need to appear in the list $\pi_1, \pi_2, ...$; only a set from which all others can be formed by multiplication and division is needed (i.e. the π quantities must be independent).

A full proof of this theorem is beyond the scope of this class, but those of you who have taken linear algebra may recognize that the Buckingham Pi theorem is a clever application of the rank–nullity theorem.

Let's go back to the law of gravity and reconstruct it using the Buckingham Pi theorem. Consider a product of the form

$$\pi = G^a m_1^b m_2^c r^d F^e,$$

where the exponents a, b, c, d, e are arbitrary constants that we have yet to determine. The dimension of this product is

$$([M]^{-1}[L]^3[T]^{-2})^a [M]^b [M]^c [L]^d ([M][L][T]^{-2})^e = [M]^{b+c+e-a}[L]^{3a+d+e}[T]^{-2(a+e)}.$$

In order for π to be dimensionless, the constants must satisfy

$$b + c + e - a = 0, \qquad 3a + d + e = 0, \qquad a + e = 0.$$

According to the Buckingham Pi theorem, we have five variables (G, m_1, m_2, r, F) and three dimensions ($[M], [L], [T]$), so we expect our solution to depend on $5 - 3 = 2$ dimensionless combinations. We have three equations to solve and five unknowns, so we must arbitrarily choose values for two of the constants in order to solve for the rest. For this example, let's decide to choose a and b arbitrarily, and we'll rewrite the expressions for the constants as $c = 2a - b, d = -2a$, and $e = -a$ (so that a and b are the independent quantities). Let's choose the two cases $(a, b) = (1, 0)$ and $(a, b) = (0, 1)$, which give us the solutions $(c, d, e) = (2, -2, -1)$ and $(c, d, e) = (-1, 0, 0)$, respectively. Substituting these two sets of constants into our original expression for π gives us the dimensionless combinations

$$\pi_1 = \frac{G m_2^2}{r^2 F}, \qquad \pi_2 = \frac{m_1}{m_2}.$$

At this point, it's always useful to *sanity check* your π groups and verify that they are, indeed, dimensionless. As an example, π_2 is obviously dimensionless (and therefore correct) because it is simply the ratio of two masses, so the units cancel. Our final answer, according to Buckingham's theorem, is a function of π_1 and π_2 so that $f(\pi_1, \pi_2) = 0$. Finding the solution isn't obvious because we have two π variables (as opposed to only one when we synthesized Newton's second law in Section 4). We'll come back to this point later, but for now, I'll give you the final answer, which is that the law of gravity satisfies Buckingham's theorem with $f = \pi_1 \pi_2 - 1 = 0$.

6 Period of a pendulum

Let's consider an ideal pendulum, a classic system that's studied in physics. An ideal pendulum consists of a mass m concentrated at a distance ℓ from the pivot, and there is no friction or air resistance. From observation, a pendulum has a characteristic period τ, the time it takes for one complete back-and-forth swing. We'd like to solve for τ using only dimensional analysis.

We need to start by making a list of all quantities that might possibly affect τ. The pendulum is characterized by m and ℓ, so we clearly need to include these in our list. The period may depend on the initial angle, so we also include θ_0. Finally, the pendulum only moves because gravity is present, so we include the gravitational acceleration constant, $g = 9.8\ \mathrm{m/s^2}$. The dimensions of the quantities we've listed are:

- $[\theta_0] =$ dimensionless (because radians are ratios of arc length to radius)
- $[m] = [M]$
- $[\ell] = [L]$
- $[\tau] = [T]$
- $[g] = [L][T]^{-2}$

There are five variables and three basic dimensions, so Buckingham's theorem tells us we can form two independent dimensionless groups. Although we could explicitly write down a generic expression for a dimensionless π product with arbitrary exponents as we did in Section 5 for the law of gravity, the quantities here are simple enough that we can identify dimensionless combinations just by inspection. Since guesswork seems like it'll be an easier approach, let's start looking through the list. We get our first dimensionless group for free because the initial angle is dimensionless, so $\pi_1 = \theta_0$. For our second group, note that m is the only quantity with a dimension of mass, so it *cannot* appear in the group because there's nothing else to cancel the mass dimension. We're therefore left with ℓ, τ, g, and the only way to make the dimensions cancel is to form the quantity $\pi_2 = g\tau^2/\ell$.

Now that we have our two dimensionless groups, we must solve $f(\pi_1, \pi_2) = 0$ in order to obtain our final result. Recall that when we reconstructed Newton's second law, $F = ma$, using dimensional analysis, we had only one dimensionless group and could therefore write $f(\pi_1) = 0$ and its solution, $\pi_1 = \Pi$, where Π was an unknown constant corresponding to the zero of f. Our current situation is a bit more complicated because we now have two dimensionless groups instead of one. We can simplify f by transforming it to a family of one-parameter functions, f_{π_1}, that are indexed by π_1. The general form of the function then becomes

$$f(\pi_1, \pi_2) = 0 \quad \rightarrow \quad f_{\pi_1}(\pi_2) = 0.$$

At this point, we make one final transformation:

$$f_{\pi_1}(\pi_2) = 0 \quad \rightarrow \quad \pi_2 = h(\pi_1),$$

where $h(\pi_1)$ is the zero of f_{π_1}. (This last transformation is valid when f_{π_1} has only one physically valid zero, which we assume for simplicity.)

We can now rewrite our transformed equation in terms of the dimensionless groups. Using $\pi_1 = \theta_0$ and $\pi_2 = g\tau^2/\ell$, we obtain

$$\pi_2 = h(\pi_1) \quad \rightarrow \quad \frac{g\tau^2}{\ell} = h(\theta_0).$$

Our original goal was to solve for the period, so the above relation gives us

$$\tau = k(\theta_0)\sqrt{\frac{\ell}{g}},$$

where $k(\theta_0) \equiv \sqrt{h(\theta_0)}$ is some undetermined function of the initial angle. This is as far as dimensional analysis takes us, and we need some other method (like observational measurements) to determine $k(\theta_0)$. However, look at just how far we were able to get by only manipulating dimensions: our final expression for τ is indeed the correct one, modulo the unknown $k(\theta_0)$, and we obtained it without having to solve any differential equations or physics! Despite not knowing the form of $k(\theta_0)$, we now have a valid *scaling relation* that can be applied elsewhere.

For completeness, the actual form of $k(\theta_0)$ is an elliptic integral, and in the limit that θ_0 is small, the integral evaluates to a nearly constant 2π.

7 Trinity revisited

We already discussed Fermi's remarkable estimation of the atomic bomb energy yield at the Trinity test. There is an equally remarkable sequel to this story: several years after Trinity, a popular magazine published several still photos of the explosion, showing the steady expansion of the hemispherical shock front (Figure 1). These stills included both a size scale and time stamps. A Soviet physicist, Leonid Sedov, analyzed these images using some basic dimensional analysis techniques, and he was able to obtain his own estimate of the blast energy. (The energy yield was still classified information at the time.)

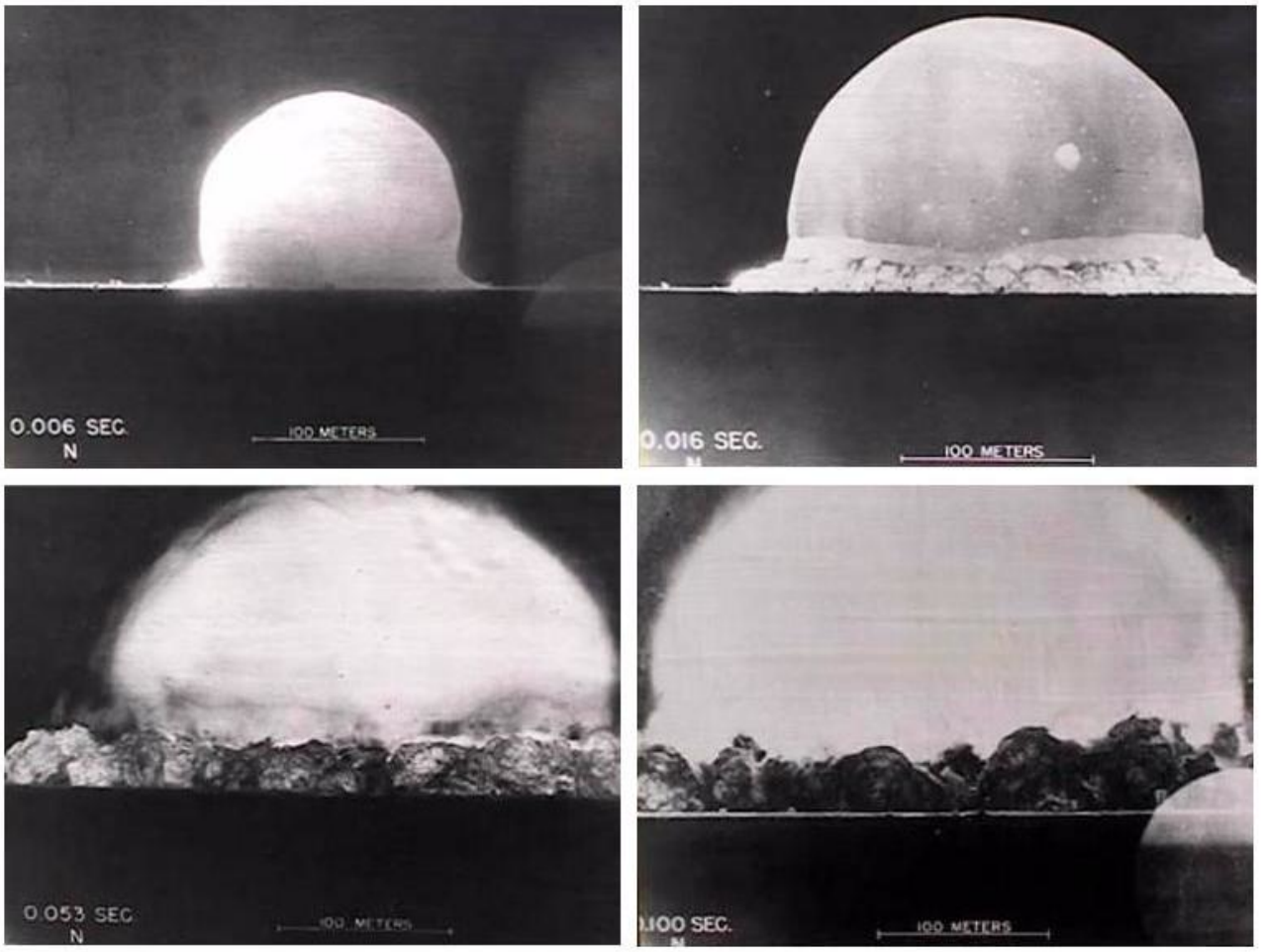


Figure 1: Still photos of the Trinity test. Note that the images include the size scale and time stamps.

We start with two assumptions: 1) the energy E was released in a small space, and 2) the shock wave was spherical. The still photos give us the approximate radius R of the fireball at several discrete time steps t. The radius depends on E, t, and the density ρ of the surrounding medium. Our dimensional analysis of the relevant quantities gives us the following:

- $[R] = [L]$
- $[E] = [M][L]^2[T]^{-2}$
- $[t] = [T]$
- $[\rho] = [M][L]^{-3}$

We have four variables and three dimensions, so we expect to form one dimensionless quantity

$$[R]^a[E]^b[t]^c[\rho]^d,$$

where a, b, c, d are exponents that we have yet to determine. Writing the variables in terms of their dimensions, we obtain

$$[L]^a\ ([M][L]^2[T]^{-2})^b\ [T]^c\ ([M][L]^{-3})^d = [M]^{b+d}[L]^{a+2b-3d}[T]^{c-2b},$$

so the exponents must satisfy

$$b + d = 0, \qquad a + 2b - 3d = 0, \qquad c - 2b = 0.$$

We have the freedom to choose one of the arbitrary constants, so let's pick $a = 1$. We can then solve for the remaining exponents: $b = -1/5$, $c = -2/5$, and $d = 1/5$. Our dimensionless group is thus

$$\pi_1 = R\, E^{-1/5}\, t^{-2/5}\, \rho^{1/5}.$$

By Buckingham's theorem, this group satisfies $f(\pi_1) = 0$, or $\pi_1 = \Pi$, where Π is our unknown zero. Our solution is therefore

$$R\, E^{-1/5}\, t^{-2/5}\, \rho^{1/5} = \Pi.$$

At this point, we're going to punt (as we did in Section 4) and pretend that $\Pi = 1$. We can now rewrite the equation and obtain our final answer for energy as a function of radius:

$$E = R^5 \rho / t^2.$$

Now that we have an expression for energy in terms of radius and time, we can go back to the still photos and plug in some approximate numbers. At $t = 0.006$ seconds, the radius R of the shock wave was approximately 80 m. Recall that the density of air is $\rho = 1.2$ kg/m^3. Plugging these numbers into our expression gives us

$$E = (80\text{ m})^5 (1.2\text{ kg/m}^3)/(0.006\text{ s})^2 \sim 10^{14}\text{ J}.$$

One kiloton equivalent of TNT is roughly 4×10^{12} J, so 10^{14} J ~ 25 kilotons of TNT. Recall that the actual answer was 20 kilotons (and Fermi's estimate was 10 kilotons), so Sedov's calculation yielded a remarkably accurate result! As a side note, Sedov's solution is generally applicable for any sort of spherical explosion originating from a point, and as an example, the equation is often used in astronomy for modeling supernova explosions.

8 Roasting a turkey

As advertised in the course survival guide, we're going to use the power of dimensional analysis to figure out how long to roast a turkey in the oven. If you've ever cooked meat before, you know that the purpose of cooking is to apply heat until the meat has reached some internal temperature that's safe for consumption. Let's consider what quantities are relevant for this calculation:

- We must obviously consider the cooking time t, since this is the quantity we wish to determine.
- The size of the turkey matters since larger birds take longer to cook. Let's assume that all turkeys are geometrically similar and can be characterized by a single length scale ℓ.
- The temperatures of the meat and the oven will impact the cooking time. We can list two relevant temperature *differences*: (1) ΔT_m, the temperature difference between the raw meat and the oven (you know that frozen meat takes longer to cook than thawed meat), and (2) ΔT_c, the temperature difference between the cooked meat and the oven.
- We need one last quantity that describes how easily heat flows through the meat. We'll introduce the *thermal conductivity*, κ, which is defined as the amount of energy crossing one unit of cross-sectional area per second divided by the temperature gradient perpendicular to that area. Or equivalently, in mathspeak:

$$\kappa = \frac{\text{energy}/(\text{area} \times \text{time})}{\text{temperature}/\text{length}}.$$

We want to form the dimensionless group

$$(\Delta T_m)^a (\Delta T_c)^b \kappa^c \ell^d t^e$$

and solve for the unknown exponents a, b, c, d, e. Let's consider the dimensions of each of the input variables. Temperature is a measure of energy per volume, so

$$[\Delta T_m] = [\Delta T_c] = [M][L]^{-1}[T]^{-2}.$$

From the above definition of thermal conductivity, we obtain

$$[\kappa] = \frac{([M][L]^2[T]^{-2})([L]^{-2}[T]^{-1})}{([M][L]^{-1}[T]^{-2})([L]^{-1})} = [L]^2[T]^{-1}.$$

Therefore, if we expand the dimensions within our dimensionless group, we obtain

$$([M][L]^{-1}[T]^{-2})^a([M][L]^{-1}[T]^{-2})^b([L]^2[T]^{-1})^c[L]^d[T]^e = [M]^{a+b}[L]^{-a-b+2c+d}[T]^{-2a-2b-c+e}.$$

The exponents must satisfy $a + b = 0$, $a + b - 2c - d = 0$, and $2a + 2b + c - e = 0$. Since we have five physical variables and three dimensions, we can form two dimensionless groups. If we choose $(a, e) = (1, 0)$ and $(0, 1)$ we can identify our two dimensionless groups as $\pi_1 = \Delta T_m/\Delta T_c$ and $\pi_2 = \kappa t/\ell^2$. Following the arguments presented in Section 6, we can write the final answer $f(\pi_1, \pi_2) = 0$ and transform it into $\pi_2 = h(\pi_1)$, which gives us

$$\kappa t/\ell^2 = h(\Delta T_m/\Delta T_c),$$

where h is some undetermined dimensionless function. Our expression for the total cooking time is therefore

$$t = h(\Delta T_m/\Delta T_c)\ \ell^2/\kappa.$$

What are the implications of this formula for turkey roasting in practice? Generally speaking, we can assume that the quantities ΔT_m and ΔT_c are roughly the same for all turkeys—that is, the oven temperature, thawed turkey temperature, and minimum cooked meat temperature do not vary. Therefore, $h(\Delta T_m/\Delta T_c)$ can be treated like a constant. Since κ is also roughly constant, we get $t \propto \ell^2 \propto m^{2/3}$, i.e. the roasting times of heavier turkeys should scale as mass m to the $2/3$ power.

9 Fluid mechanics

One area of applied math and physics where dimensional analysis particularly shines is the study of fluid mechanics. Fluid mechanics is notoriously complex and usually requires navigating a morass of partial differential equations. As an example, the equations governing the flow of an incompressible Newtonian fluid are

$$\frac{\partial \mathbf{v}}{\partial t} + (\mathbf{v} \cdot \nabla)\mathbf{v} = -\frac{1}{\rho}\nabla p + \eta \nabla^2 \mathbf{v},$$
$$\nabla \cdot \mathbf{v} = 0,$$

where $\mathbf{v}$ is the fluid velocity field, ρ is the fluid density, η is the kinematic viscosity, and p is the pressure. If we expand the above expressions, we end up with six coupled, nonlinear, partial differential equations. To quote a famous internet meme, ain't nobody got time for that.

9.1 Relevant quantities within fluid mechanics

To illustrate dimensional analysis within the arena of fluid mechanics, we will pick a specific problem to solve: calculating the *terminal velocity* of a falling object, i.e. the velocity that the object reaches after falling for a sufficiently long period of time. There are two relevant, competing forces that we need to consider. Gravity causes the object to fall, but the object is slowed by the drag force of the surrounding medium. We already know how gravity acts, so let's focus on finding the form of the drag force, F_d.

The quantities that affect F_d can be divided into two broad categories: characteristics of the falling object, and characteristics of the surrounding medium/fluid. Let's start with the former. The fluid interacts with only the *surface* of the object, so there are only two relevant characteristics of the object—its size, which we describe with a radius R, and its speed v. Moving on to the surrounding medium, imagine different fluids that you've interacted with before, e.g. water, air, molasses. All of these have different densities (ρ) and different degrees of "gumminess." This gumminess (or thickness) of a liquid is quantified by *viscosity*. For our purposes, we will consider the kinematic[1] viscosity η, which has dimension $[L]^2[T]^{-1}$. So, to summarize, we have the following quantities to work with:

[1]There is also a *dynamic viscosity*, μ, that arises in fluid mechanics, and it's related to the kinematic viscosity through the relation $\mu = \rho\eta$.

- Drag force: $[F_d] = [M][L][T]^{-2}$
- Object radius: $[R] = [L]$
- Object velocity: $[v] = [L][T]^{-1}$
- Density: $[\rho] = [M][L]^{-3}$
- Kinematic viscosity: $[\eta] = [L]^2[T]^{-1}$

Buckingham Pi tells us that we expect to form two independent dimensionless groups from this list. Rather than writing out the formalism this time, let's just take a careful look at what we have. The dimensions of velocity and viscosity are very close to each other, and by adding the radius into the mix, we can make all the units cancel:

$$\pi_1 = \frac{vR}{\eta}.$$

This dimensionless group has a special significance in fluid mechanics and is known as the *Reynolds number*, $Re \equiv vR/\eta$. It is essentially a dimensionless measure of the flow speed: low values of Re indicate slow, viscous flow (like honey oozing out of a jar), and high values indicate turbulent flow (like a jet flying at 1000 km/hr). For our second dimensionless group, we must include F_d and ρ since they were absent from our first group. With a little work, we can write down the result

$$\pi_2 = \frac{F_d}{\rho R^2 v^2}.$$

The most general relation is $f(\pi_1, \pi_2) = 0$ or, equivalently, $\pi_2 = h(\pi_1)$, and we expect it to yield a physically meaningful result. If we substitute the original variables, we obtain the following expression for the drag force:

$$F_d = \rho R^2 v^2 h(vR/\eta).$$

Alas, dimensional analysis cannot take us beyond this point, but we can push further using our tactic of dividing and conquering. We will consider two extreme cases: viscous, low-speed flow ($Re \ll 1$) and turbulent, high-speed flow ($Re \gg 1$).

9.2 Viscous limit

In slow, viscous flows, the drag force arises from viscous forces, which are proportional to η. We'll therefore make the educated guess that $F_d \propto \eta$ is this particular limiting case. If we choose $h(x) \sim 1/x$, then we obtain the correct proportionality in our expression for the drag force:

$$F_d = \Pi \eta \rho v R,$$

where Π is a dimensionless constant. Although dimensional analysis cannot tell us the exact value of Π, this expression is, in fact, the correct form in the viscous limit. (For completeness, the value of Π is 6π for a sphere, and this number was first calculated by Stokes.) Qualitatively, the result makes sense. Larger or faster objects feel more drag, and a denser or more viscous fluid produces more drag.

We can now return to the question that we originally asked: what is the terminal velocity? Now that we have an expression for the drag force, we can balance it against the force of gravity, $F_g = mg$, where m is the mass of the object and g is gravitational acceleration. If we assume that the object has a uniform density ρ_o, we can rewrite the gravitational force as $F_g \sim \rho_o g R^3$. Balancing with the drag force gives us $\eta \rho v R \sim \rho_o g R^3$ (where we have neglected constants), and solving for v gives us the expression for terminal velocity:

$$v \sim \frac{gR^2}{\eta} \frac{\rho_o}{\rho}.$$

Again, let's sanity check the answer wherever we can. Gravity accelerates objects, so g should be in the numerator. An object should fall faster if it is denser, so ρ_o should also be in the numerator. Viscosity slows objects, so η should be in the denominator. Although we didn't check the exponents or the location of R, the other tests help boost our confidence in the answer. (In reality our answer is nearly correct, but there's an additional correction that's needed if we want to account for the effects of buoyancy.)

9.3 Turbulent limit

Let's now consider our other extreme case, where the flow is high-speed and turbulent. Imagine, for example, a person who is thrown out of an airplane. At these sorts of high speeds (high values of Re), the viscosity actually becomes irrelevant because it doesn't influence the shearing and whirling of turbulent flows. This isn't much information to go on, but let's see how far we get simply by eliminating η. Removing η removes π_1 from our list, and we are left with π_2 as our only dimensionless group. The general relation that follows is then $f(\pi_2) = 0$, or $\pi_2 = \Pi$ for some undetermined constant Π (which we will ignore and assume is of roughly order unity). Substituting our original variables gives us

$$F_d \sim \rho R^2 v^2.$$

This drag force is about right for a sphere, and for other complex shapes, we modify the expression so that it contains the cross-sectional area A instead of radius R. To make this substitution, we use the fact that $A \propto R^2$:

$$F_d \sim C_d \rho v^2 A/2,$$

where we have defined the *drag coefficient* C_d, a dimensionless measure of the drag force that depends on the shape of the object (i.e. how streamlined it is).

To find the terminal velocity, we balance the drag force with gravity once again:

$$\rho R^2 v^2 \sim \rho_o g R^3.$$

Solving for the terminal velocity, we obtain

$$v \sim \sqrt{gR\rho_o/\rho}.$$

As usual, let's sanity check our answer. The terminal velocity is smaller in a denser fluid, so it makes sense that ρ is in the denominator. If earth's gravity somehow increased, we would expect the terminal velocity to also increase, so g should be in the numerator. A denser object experiences more gravity and a higher terminal velocity, so ρ_o should also be in the numerator. The argument for R is a bit more subtle: as R increases, the gravitational force scales as R^3, but the drag force scales only as R^2. Gravity therefore wins at larger values of R, and higher gravity means greater speeds, so R should also be in the numerator. As a reminder, η does not feature at all here because viscosity does not affect drag at high speeds.

Let's use the result of our hard work and actually calculate the terminal velocity of the unlucky person who was thrown out of an airplane. We'll assume that the person tucks himself into a spherical fetal position, with radius $R \sim 40$ cm. A person is mostly made of water, so $\rho_o \sim 1$ g/cm^3. The density of air is about $\rho \sim 10^{-3}$ g/cm^{-3}. Gravitational acceleration is $g \sim 1000$ cm/s^2. Putting the numbers together, we get

$$v \sim [(1000 \text{ cm/s}^2) \times (40 \text{ cm}) \times (1 \text{ g/cm}^3) \,/\, (10^{-3} \text{ g/cm}^3)]^{1/2} \sim 6 \times 10^3 \text{ cm/s},$$

which is roughly 200 km/hr. One final point we should confirm is that we are, in fact, in the turbulent limit in this case (we have assumed that $Re \gg 1$). Let's check this assumption. We have $Re = vR/\eta$, and the only piece of information I haven't given you yet is the kinematic viscosity of air, which is $\eta \sim 0.2$ cm^2/s. The Reynolds number for the falling person is then

$$Re \sim (6 \times 10^3 \text{ cm/s}) \times (40 \text{ cm})/(0.2 \text{ cm}^2/\text{s}) \sim 10^6.$$

The value is much greater than unity, so the flow is indeed highly turbulent.

10 Water waves

A classic application of dimensional analysis is constructing the speed of waves in water. For these calculations, we assume that water is an ideal fluid with no viscosity or compression, and we consider two broad categories of waves.

10.1 "Gravity waves": long wavelength, deep water

For our first case, we consider something like a storm wave at sea. The body of water is large and deep, and the wavelength is long enough so that surface tension effects don't matter (but the wavelength is still small compared to the depth of the water). At these large scales, we expect gravity to be the dominant force that governs the wave movement. We can then guess that the relevant physical quantities needed in this case are gravitational acceleration g, wavelength λ, water density ρ, and, of course, the speed v of the traveling wave. The relation that we want to construct is

$$v \propto \rho^a \lambda^b g^c,$$

and we need to determine the exponents a, b, c. The dimensions of the above quantities are

$$[L][T]^{-1} = ([M][L]^{-3})^a [L]^b ([L][T]^{-2})^c = [M]^a [L]^{-3a+b+c} [T]^{-2c}.$$

In order to make the equation dimensionally homogeneous, the exponents must satisfy $a = 0$, $-3a + b + c = 1$, and $2c = 1$. Solving these equations, we obtain $a = 0$, $b = 1/2$, $c = 1/2$, which gives us

$$v \propto \sqrt{\lambda g}$$

as our final answer. One interesting feature of this solution is that the wave speed increases with wavelength, so large waves like tsunamis can travel with impressively fast speeds.

10.2 "Capillary waves": short wavelength, deep water

For this case, we'll consider short wavelengths in deep water; an example of this is a ripple on a pond. At these small scales, we expect the effects of gravity to be subdominant to those of surface tension (which has units of force per unit length). We can therefore guess that the relevant physical quantities are surface tension s, wavelength λ, water density ρ, and the wave speed v. The proportionality relation we want to solve is

$$v \propto \rho^a \lambda^b s^c,$$

which has dimensions

$$[L][T]^{-1} = ([M][L]^{-3})^a [L]^b ([M][T]^{-2})^c = [M]^{a+c} [L]^{-3a+b} [T]^{-2c}.$$

The exponents must satisfy the conditions $a + c = 0$, $b - 3a = 1$, and $2c = 1$, so the values are $a = -1/2$, $b = -1/2$, $c = 1/2$. Our final answer is thus

$$v \propto \sqrt{\frac{s}{\rho\lambda}}.$$

In contrast to gravity waves, capillary waves move *slower* when the wavelengths are large.

11 Summary

We've shown that dimensional analysis is a powerful method for synthesizing physically meaningful results, and the main concept at the heart of this method is that *every valid physical equation can be written in a form without units*. To find these equations, follow the steps in this rough guideline:

1. Write down a list of the *relevant* physical variables. You need to use your intuition and wits in this step. Let's say that you end up with n variables.

2. Determine the dimension of each variable, and count the number of independent dimensions. We'll call this number r. Most of the examples that we've encountered have $r = 3$ for mass, length, and time.

3. Find the $n - r$ *independent* dimensionless combinations of the variables either by inspection or by explicitly enforcing constraints on the exponents. These dimensionless combinations are the Pi variables that are used in the Buckingham Pi theorem.

4. Write down the most general relation using the dimensionless groups, and then simplify the relation if possible. Even though you'll most likely end up with undetermined constants or dimensionless functions, you'll still have a valid scaling relation—in other words, you'll have almost the entire answer without doing any honest work!

Math 246 notes #4: Graphical methods

1 Why are we about to spend a week drawing pictures?

- Humans are well adapted for taking in an entire picture quickly and then deducing consequences using geometric intuition. In other words, your brain is a fabulous image processor.
- Graphs are an effective tool for gaining rapid, qualitative insight, but you must be aware of their limitations (as is true with every technique in modeling).
- Graphs are most useful for conveying qualitative relationships or approximate data which involve only a few variables. A graphical approach to a problem is most likely to be useful when not much information is available or when it is given in a rather imprecise form.
- Even when constructing a *quantitative*, analytic model, it is often useful to begin with a *qualitative* graphical model to help build intuition and to serve as a sanity check.

2 Warmup exercises: sketching graphs from words

We'll start with some simple warmup exercises in sketching qualitative graphs from written descriptions. As you think about how to construct the graphs, consider the following specific questions and try to be as precise as possible. Should the graph increase or decrease (what is the slope)? What is the concavity, i.e. does the graph increase/decrease with increasing or decreasing *rate*? Are there intercepts on the horizontal and vertical axes?

2.1 Example: picking strawberries

A strawberry farm employs some number of people to harvest the strawberries. Draw a graph to illustrate how the total harvest time depends on the number of people working.

Let's work through our list of questions to figure out how this graph should look. The total picking time clearly drops with the number of people employed, so the graph should decrease. What about the concavity? If we double the number of people, we expect the total picking time to halve. Similarly, tripling the number of people means reducing the picking time by a factor of three, and so on. The graph should therefore have positive concavity. There is no vertical intercept because in the limit that zero people are employed, the time to harvest is infinite. As more and more people are employed, the total harvest time drops closer to zero, but it never actually reaches that value. There is therefore no horizontal intercept either.

2.2 Example: ferris wheel

A ferris wheel rotates at 1 r.p.m, and all the seats on the wheel are different colors. When you begin observing the ferris wheel, you see a red seat at the very top of the wheel, and a blue seat on the side of the wheel (at the "nine o'clock" position) that is rising. On the same pair of axes, draw two graphs showing how the heights of the red and blue seats vary.

There is a minimum and maximum height for all the seats, so our graphs will be bounded by fixed lower and upper values. Within this range, the seats will move back and forth within the extremes. Should this back-and-forth movement be captured by straight or curved lines? We know that as a car passes through the very top or bottom of the wheel, the height does not change very quickly. In contrast, the height of the car changes rapidly when it passes through the sides of the wheels. We therefore expect the graph to have some curvature (or "flattening") near the upper and lower extremes, and to be somewhat straight in between. Finally, what about red versus blue? We know that the red seat starts at the top of the wheel, so its vertical intercept is the maximum height value. The blue seat starts halfway up, so its vertical intercept should be half the maximum height value, and it should begin with an increasing slope. Apart from the differences in starting positions, the curves for the red and blue seats should be identical, but they will have a relative horizontal offset.

3 Arms races

Using the basic techniques outlined in the previous section, we now move on to a more complicated example. Here we will analyze the factors that affect arms races between countries.

3.1 Mini history lesson: ICBM stockpiling

At various points in world history, countries have stockpiled intercontinental ballistic missiles (ICBMs) in order to avoid "nuclear blackmail." The idea is that if there is a sneak attack, a country wants to have enough surviving missiles to retaliate and ensure sufficient damage on the attacker. During the arms race between the United States and the Soviet Union, US General Maxwell D. Taylor is quoted as saying,

"*The strategic forces, having the single capability of inflicting massive destruction, should have the single task of deterring the Soviet Union from resorting to any form of strategic warfare. To maximize their deterrent effectiveness they must be able to survive a massive first strike and still be able to destroy sufficient enemy targets to eliminate the Soviet Union as a viable government, society, and economy, responsive to the national leaders who determine peace or war.*"

Taylor's description assumes a worst-case scenario in which the Soviet Union strikes first and destroys some fraction of the US's missile supply. The natural follow-up question is then: how many missiles is enough? Taylor suggests the following:

"*The number of weapons we shall need will be those required to destroy the specific targets within this system of which few will be hardened silos calling for the accuracy and short flight time of ICBMs. As a safety factor, we should add extra weapons to compensate for losses that may be suffered in a first strike and for uncertainties in weapon performance. The total weapons requirement should be substantially less than the numbers available to us in our present arsenal.*"

The most important point to note from this statement is the list of factors that influence the size of the arsenal. The number of missiles must be sufficiently large to destroy enemy targets, inflict enough damage, and compensate for any losses; the last factor implicitly depends on the size and effectiveness of the enemy forces. We would like to use some simple graphical methods to gain some qualitative insight on the dynamics of missile stockpiling. Specifically, what factors might cause countries to escalate arms races, and how many weapons is considered "enough?" Furthermore, what is the impact of introducing antiballistic missiles, missiles with multiple warheads, reinforced/hardened missile silos, missile-firing submarines, etc? The answers to each of these could have serious consequences, so we'd better get those answers right.

3.2 Graphical model for arms races

Let's say we have two countries, Country A and Country B, that are competing in a nuclear arms race. Each country adopts the following strategies:

- Survive a massive first strike and inflict sufficiently severe damage on the enemy by targeting population and industrial centers.
- Assume that the enemy intention is to conduct a massive first strike to destroy the home-based missile supply.

In other words, each country assumes the first point when determining its own missile supply and assumes that the second point is true for the opposing country.

Let's define x and y as the number of missiles possessed by Country A and Country B, respectively. Each country builds its missile stockpile in response to the number possessed by the opposing country. In other words, we can name a function $y = f(x)$ that determines the number of missiles Country B stockpiles in response to the number owned by Country A. Similarly, there is a function $x = g(y)$ that determines the number of missiles that Country A amasses in response to stockpile in Country B. Country A feels safe if and only if $x > g(y)$, and Country B feels safe if and only if $y > f(x)$.

What might these functions look like? Let's start with $y = f(x)$ and write down a list of factors that influence its shape:

- Even if x were zero, Country B would still want a nonzero number of weapons to destroy the population and industrial centers of Country A. We can therefore conclude that there is a nonzero intercept y_0 at $x = 0$.
- As Country A increases its missile supply, its goal is to use those weapons to destroy the stockpile that exists in Country B. Country B must therefore increase its missile supply in anticipation of sustaining losses. In other words, $f(x)$ is an *increasing* function.
- Let's assume for now that a single missile fired is capable of destroying at most one missile in enemy territory. However, there may be inefficiencies, and a country's missile force may not be 100% effective. This effectiveness impacts the *slope* of $f(x)$. If Country A's missiles were 100% effective, then Country B would need to match missiles one-for-one, i.e. $f(x)$ must have a slope of unity. This is the maximum slope needed, and any inefficiencies in Country A would result in a reduced slope.

Now that we have a gross sense of what $f(x)$ looks like, let's refine the picture. We will take the possible space for $f(x)$ and divide and conquer: by slicing the space into various subregions, we can make our estimate of $f(x)$ more precise. Specifically, our subregions will be defined by the lines $y = x$ (maximum slope), $y = x/2$, $y = x/3$, and so on. Within each of these subregions, we will analyze the behavior of $f(x)$.

Case 1: $y > x$**.** If Country A attacks in this situation, it fires all its x missiles. The surviving number of missiles in Country B is therefore $y - x$ at minimum, but if Country A is not completely effective, then some fraction s (where $0 < s < 1$) of the fired missiles x would not successfully destroy their targets. The total number of surviving missiles is thus $y - x + sx$. From our previous discussion, we know that the number of surviving missiles in Country B is also y_0, the number needed to inflict sufficiently severe damage on Country A. We therefore obtain the relation $y_0 = y - x + sx$, or

$$y = y_0 + (1 - s)x.$$

This equation gives the minimum number of missiles that Country B requires in order to be confident that y_0 missiles will survive an attack by Country A.

Case 2: $y = x$**.** We assume the same scenario as in Case 1: Country A fires all its x missiles, and some fraction s are ineffective at hitting their targets. The number of surviving missiles in Country B is therefore $sx = sy = y_0$, which gives us the total number of missiles required:

$$y = y_0/s.$$

Case 3: $y < x$ **and** $y > x/2$**.** In this case, note that Country A possesses *more total missiles* than Country B. If Country A fires all of its missiles, we'll assume that the "extra" missiles are used to double up on their targets (as protection against inefficiencies). In other words, there are $x - y$ missiles in Country B that are targeted twice, and $y - (x - y) = 2y - x$ number of missiles that are targeted only once. Of the missiles that are targeted once, a percentage $s(2y - x)$ survives the attack. What about the missiles that are targeted twice? After the first attack, there are $s(x - y)$ that remain, and after the second attack, there are $s[s(x - y)] = s^2(x - y)$ that survive. Therefore, Country B must have $y_0 = s^2(x - y) + s(2y - x)$ surviving missiles, or, solving for y,

$$y = \frac{y_0 + x(s - s^2)}{2s - s^2}.$$

Case 4: $y = x/2$**.** In this case, Country A possesses exactly twice as many missiles as Country B. Each of Country B's missiles is attacked twice, and the surviving number is $s^2 y$. The minimum number of required missiles is therefore

$$y = y_0/s^2.$$

Now that we've pinned the behavior of $y = f(x)$ in the above specific cases, let's piece the information together. We'll approximate $f(x)$ with a continuous function, which is a reasonable assumption in the limit that the number of missiles is large. Let's summarize the results of all four case studies and go through the results one at a time, in order of increasing x:

- In Case 1, with $y > x$, we obtained $y = y_0 + (1 - s)x$. As $x \to y$, the result approaches $y = y_0/s$, which is what we obtained in Case 2.
- In Case 3, the result $y = [y_0 + x(s - s^2)]/(2s - s^2)$ also reduces to $y = y_0/s$ (Case 2 result) in the limit that x approaches y. At the other extreme, when $x \to 2y$, the result reduces to $y = y_0/s^2$, the result from Case 4.

In other words, the line segments from Cases 1 and 3 form a continuous curve that join each other at the boundaries defined in Cases 2 and 4. Furthermore, the slope in Case 1 is $1-s$, which is *greater* than the slope in Case 2, $(1 - s)/(2 - s)$. By mentally extrapolating these results to other subregions (e.g. $y < x/2$ and $y > x/3$), you can convince yourself that $f(x)$ is a *piecewise linear curve with decreasing slopes*. Taking this result to the next level, we can replace the piecewise linear approximation by a single, continuous, smooth curve. From our case studies, we know that this curve must pass through the points $(0, y_0), (x, x), (x, x/s), ...$ and so on. If we generalize our analyses from Cases 2 and 4, we find that one curve that satisfies our requirements is

$$y = \frac{y_0}{s^{x/y}}.$$

Since Country A follows the same strategy as Country B when stockpiling its missiles, we can conclude that $x = g(y)$ follows the same general form as $f(x)$. The two curves must intersect, as shown in Figure 1. Because $y = f(x)$ represents the minimum number of missiles required by Country B, the region *above* the curve represents the missile levels that are satisfactory to Country B. Similarly, the region to the *right* of $x = g(y)$ represents the missile levels that are satisfactory to Country A. The dark gray wedge where these regions intersect represents missile levels where both countries are happy. The exact intersection point of $f(x)$ and $g(y)$ represents the *minimum* level at which both sides are satisfied.

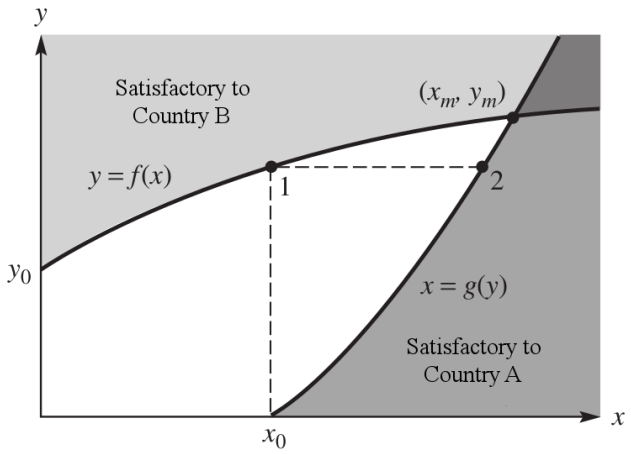


Figure 1: Model curves for number of missiles required by Country A ($x = g(y)$) and Country B ($y = f(x)$). Figure adapted from Giordano et al., 2014.

3.3 Graphical behavior of the candidate model function

Let's take a closer look at the candidate model function that we constructed, $y = y_0/s^{x/y}$. The graph of this function depends on three factors: the minimum number of required missiles y_0, the survivability percentage s, and the ratio x/y of missiles exchanged. If y_0 increases, then the curve $y = f(x)$ shifts upward and also has a larger slope at each point than before. If s increases, then the curve rotates downward toward the horizontal, and the slope decreases at each point along the curve. Finally, if x/y increases, then Country A is able to target Country B's missiles multiple times. Country B therefore requires more missiles to survive the attack, and the curve must increase in slope and rotate upward.

The function $x = g(y)$ describing Country A's missile supply depends similarly on minimum required missiles, survivability percentage, and ratio of missiles exchanged, but note that the values of these quantities are generally different from the corresponding quantities for Country B. In the following sections, we will introduce some new external factors to our model, and those factors will,

in turn, influence the behavior of both $f(x)$ and $g(y)$. We are interested in the how the intersection point shifts as both functions are subjected to change. In summary, our two model functions are:

- Country A's missile supply: $x = x_0/s_A^{y/x}$
- Country B's missile supply: $y = y_0/s_B^{x/y}$

Note that the survivability percentages s_A and s_B are now explicitly named for each country and can take on different values.

3.4 Model test case #1: increased civil defense budget

For our first thought experiment, let's suppose that Country A decides to double its annual budget for *civil defense*. In other words, more resources are given toward fortifying population centers, but the budget for *military offense* remains unchanged. We'll begin by analyzing the effect on Country B's missile supply, $y = y_0/s_B^{x/y}$. If Country A's defense is strengthened, then Country B requires a larger minimum number of missiles to inflict sufficiently severe damage. In other words, y_0 must increase. Let's discuss the other factors that may or may not be impacted:

- The survivability percentage s_B of Country B's missiles is unaffected because the effectiveness of Country A's missiles hasn't changed, and Country B has not actively improved the protection of their missiles.
- The exchange ratio x/y for Country B is unaffected because neither country has changed its total number of missiles.

Because only y_0 changes, the net effect is that $y = f(x)$ shifts vertically upward with increasing slope at every x.

For Country A, the function $x = g(y)$ remains the same because x_0, s_A, and the exchange ratio y/x are all unchanged. However, because $y = f(x)$ has shifted higher because of the increase in y_0, the *intersection point between the two functions also shifts higher*. In other words, although Country A's action of increasing civil defense is a passive move at first glance, the net result is that *both* countries require an increased number of missiles.

3.5 Model test case #2: mobile launching pads

In this scenario, we will assume that Country A puts its missiles on mobile launching pads, which can be easily relocated. What is the impact on Country B? Let's consider each factor in turn. The factor y_0 is unchanged because it depends only on the population and industrial centers of Country A that must be destroyed. The survivability s_B is also unchanged because the new mobility of Country A's missiles has nothing to do with their accuracy. The exchange ratio x/y is unaffected because neither side has actively changed their missile supply numbers. Therefore, the curve $y = f(x)$ remains the same in this scenario.

Now let's consider the curve for Country A. There is no change in x_0 because Country B's civil defense is unchanged. The exchange ratio y/x is also unchanged for the same reasons described above. However, Country A's survivability percentage s_A is *increased* because if its missile force is mobile, then Country B has a reduced chance of accurately targeting those missiles. Overall, the net effect is that $x = g(y)$ flattens toward the y axis and becomes more vertical.

If $f(x)$ is unchanged and $g(y)$ flattens toward the y axis, then the intersection point between the two functions moves down and to the left. In other words, the impact of Country A placing its missiles on mobile launching pads is that the required number of missiles *decreases* for both countries.

3.6 Model test case #3: multiple warheads

In this scenario, we will assume that both Countries A and B develop missiles that contain multiple warheads (multiple independently targetable reentry vehicles, or MIRVs). When we developed

our model, we assumed that a single missile was equivalent to a single warhead, but we will now have to change our model in order to account for MIRVs. Let's continue to count missiles rather than warheads, and we'll assume that each missile is armed with N smaller missiles, each with its own warhead.

Because civil defense for the population and industrial centers is unchanged for both countries, we assume that the same number of *warheads* is required to inflict sufficiently severe damage. The number of required *missiles* is thus reduced by N, so the factors x_0 and y_0 are also reduced by N.

Now let's consider survivability factors. If a warhead released independently from an in-flight missile has unchanged guidance technology, then we can assume that its targeting accuracy is also unchanged. The survivability factors s_A and s_B therefore remain the same.

Finally, let's consider exchange ratios. When a single missile is launched, carrying N warheads, each warhead can be used to target independent, separate missiles in enemy territory. In other words, in a first strike, a single launched missile can destroy up to N missiles. The exchange ratios thus increase significantly. For Country B, the curve $y = f(x)$ must rise more sharply toward the vertical, and the curve $x = g(y)$ for Country A must steepen away from the y axis.

In summary, we have two competing effects: reducing the intercepts x_0 and y_0 causes the curve slopes to flatten, but the increased exchange ratios cause the curve slopes to steepen. Without knowing detailed numbers, we cannot determine which effect wins, and we therefore don't know whether the intersection point shifts up or down. This analysis demonstrates a limitation of graphical models. In order to push this particular analysis further, we need more precise information.

3.7 Model test case #4: multiple warheads revisited

In the previous test case, we found that we were unable to predict the effect of multiple warheads on the number of *missiles* required by each country. Let's try the problem again, but this time we will count the total number of warheads (using our old variables x and y) rather than counting the total number of missiles.

The minimum number of warheads needed to inflict sufficient damage on each opposing country remains the same, so x_0 and y_0 are unchanged. Because each warhead is located on a missile, its chance for survival is the same as that of the missile, so s_A and s_B are also unchanged. What about the exchange ratio factors? Let's start with Country B. A single warhead released from an incoming missile from Country A has the potential of destroying N warheads of Country B simultaneously, because N warheads are clustered on a single missile. The exchange ratio for Country B is therefore increased. The same argument holds for Country A, so its exchange ratio also goes up.

The effect of increased exchange ratios is that both curves steepen dramatically: $y = f(x)$ rotates toward the y axis, and $x = g(y)$ rotates toward the x axis. The intersection point thus shifts up and to the right, i.e. both countries require more warheads if MIRVs are introduced.

3.8 Beyond arms races

The graphical arms race model that we've introduced here is a simple one. Nevertheless, it helps shed light on several of the problems involved, and it allows us to have a more intelligent debate about the impacts of various external factors. The arms race model can be extended to other situations as well; as an example, there are parallels in biology. There is an unusual predator–prey relationship between the garter snake *Thamnophis sirtalis* and the Oregon newt *Taricha granulosa*: the newt defends itself by producing a highly potent toxin (tetrodotoxin), and the snake has developed partial resistance to this toxin. However, the snake becomes temporarily immobilized after consumption, and it is susceptible to other predators during this time. In this odd relationship, the prey (newt) develops a defense by becoming poisonous, and the predator (snake) develops a resistance to the prey's toxicity. We will discuss standard predator–prey relationships in detail later in this course, but the snake–newt relationship is an example where the behavior differs because the two species follow a co-evolutionary biological arms race.

4 Nonrenewable resources and the energy crisis

We depend heavily on nonrenewable energy sources such as petroleum and natural gas. The steady supply of these sources is often uncertain, as supply security is affected by political unrest, oil embargoes, etc. Given these uncertainties, there is high motivation to reduce oil consumption on both long and short time scales, and the latter is particularly important if there is a crisis situation. A variety of solutions have been proposed to address these long- and short-term needs. One particular example that we'll study here is the introduction of petrol surcharge taxes. The idea is that petrol companies pass the tax to the consumer by increasing the price per liter, and the consumer, in turn, reduces consumption because of the higher price. We will construct a graphical model to qualitatively address the following questions:

1. What is the effect of the surcharge tax on short- and long-term consumer demand?
2. Who actually pays the tax, the consumer or the oil companies?
3. Does the tax contribute to inflation?

In other words, our goal is to determine the effect of a surcharge tax on both the market price of gasoline and consumer demand. A graphical analysis is appropriate because we are interested in a qualitative understanding of the principal factors, and because precise data are difficult to obtain.

4.1 Graphical model for nonrenewable resources

Let's start by taking a look at an oil company, which produces a single product within a large competitive industry. The goal of the company is to maximize profits, and the profits depend on the number of units produced. If the industry is sufficiently large (as is the case with oil), a single company's production has no appreciable effect on the market price. We will therefore assume that the oil price is constant, and the only relevant quantity the company must consider is the *difference* between the price and the company's costs of production.

The production costs of a company can be split into two categories. There are *fixed costs*, which are independent of production levels; these include rent and utilities, management costs, etc. Companies also have *variable costs*, which depend on the quantity produced. These include the cost of raw materials, labor, etc. When the fixed costs are divided by the quantity produced, the cost-per-unit is clearly higher when production quantities are low. As production quantities increase, the cost-per-unit tends to decrease because 1) the fixed costs are spread between a larger quantity of products, and 2) economies of scale reduce some of the variable cost rates (recall, for example, the analysis of packaging costs). However, as the production level continues to increase, it eventually strains the capabilities of the firm (e.g. more machinery required, overtime payments), and the per-unit-cost begins to increase again. The changes in cost-per-unit are directly related to the company's profit, which can be described by a function, $P(q)$. This function depends on the quantity of product, and we expect that there is an optimal production level, q^*, that maximizes profit over the range of production levels of interest.

We can break the total profit $P(q)$ into the difference between the total revenue $R(q)$ and total cost $C(q)$:

$$P(q) = R(q) - C(q).$$

In order for a local maximum to exist in $P(q)$, its derivative must vanish at q^*. We therefore obtain the relation

$$R'(q^*) = C'(q^*)$$

at the optimal production level, i.e., the slope of the total revenue curve equals the slope of the total cost curve. The derivative $R'(q)$ is interpreted as the *marginal revenue* of the product, and the derivative $C'(q)$ is the *marginal cost*. In addition to the first derivative $P'(q)$ vanishing, the second derivative must be negative if $P(q^*)$ is a maximum. We therefore obtain a second relation

$$P''(q^*) = R''(q^*) - C''(q^*) < 0,$$

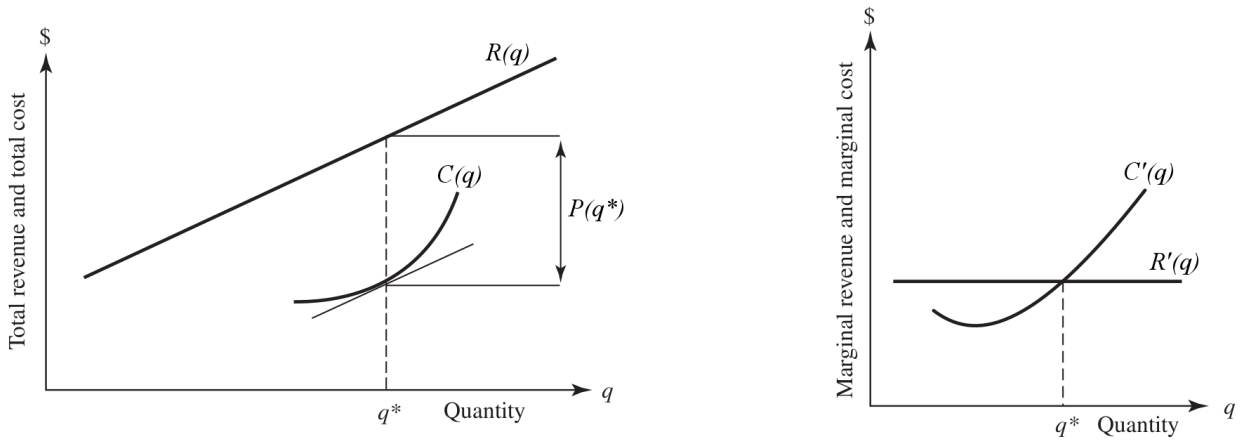


Figure 2: *Left:* The total profit $P(q)$ depends on the difference between the total revenue $R(q)$ and total cost $C(q)$. We expect $C(q)$ to have a local minimum at some optimal value q^*. *Right:* The marginal cost and marginal revenue curves are the first derivatives of $C(q)$ and $R(q)$, respectively. These marginal curves intersect at the optimal value q^*. Figures adapted from Giordano et al., 2014.

which gives us

$$R''(q^*) < C''(q^*).$$

Putting these two relations together, we obtain the result that the marginal revenue and marginal cost curves must intersect at q^*, with the marginal cost curve rising more rapidly than the marginal revenue. These results are illustrated in Figure 2. Note that the flatness of $R'(q)$ reflects the constant market price of the product.

4.2 Effects of taxation

Let's suppose that a company is currently operating in an optimal manner and maximizing its profits; that is, given a market price $R'(q)$, the company is producing q^* units. Now let's say that the government imposes a tax on each unit sold, and the company must pay this tax. The impact is that the marginal cost curve $C'(q)$ shifts upward by the amount of tax. In order to compensate for the new losses from tax, the entire industry raises the market price by adding the amount of tax to the price of each unit. The impact is that the market price (or marginal revenue) curve also shifts upward by the amount of tax. Since both $C'(q)$ and $R'(q)$ are increased by the same constant amount, the intersection point still remains at q^*. Our model therefore predicts that production levels are unaffected by the tax, but the company charges a higher price and contributes to inflation. Note also that it is the consumer who pays the full amount of the tax in the form of a price increase.

4.3 Supply and demand

In the previous section, we assumed that the industry was able to raise the price of the product with no consequences. However, this assumption may not be true—if the price of a product increases to the point of becoming unreasonable, then customers lose interest in purchasing it. To address this issue, we need to start by extending our model of a single company to construct a model of the industry as a whole.

Each company in the industry has different intersections between the marginal cost curve and various marginal revenue curves (the heights of which are set by the market price). For a given market price, we can calculate the optimal production quantity for each company, and we can sum these numbers together to yield the total quantity produced by the industry. The relationship between market price and total quantity produced is called the *supply curve*, and it represents the amount of product that the industry is willing to supply at various price levels. Qualitatively, as the market price increases, the industry is willing to produce greater quantities.

Now let's consider the consumer's point of view. The total quantity demanded by consumers as a whole is a function of market price, and the demand decreases as the market price increases. It is traditional to plot price versus quantity (rather than quantity versus price, which is perhaps more intuitive); the consumer *demand curve* therefore decreases as quantities increase. Plotting the supply and demand curves together yields an intersection point given by (q^*, p^*), as shown in Figure 3. This intersection represents an equilibrium state in which no excess supply exists at that particular price, and both consumers and suppliers are satisfied.

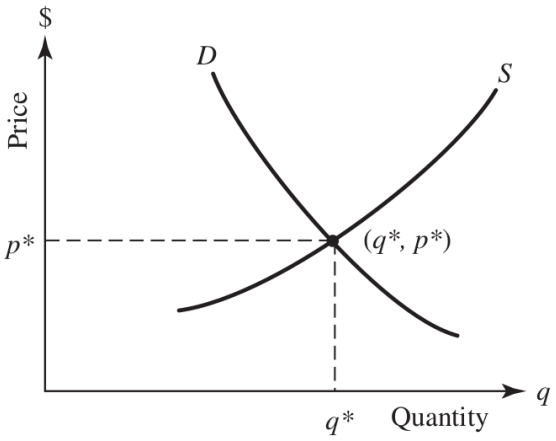


Figure 3: Supply and demand curves intersect at a point that corresponds to a market price and market quantity that satisfy both consumers and suppliers. Figure from Giordano et al., 2014.

Since it is impossible to know the exact demand curve for a product, it is highly unlikely for an industry to produce the exact quantity q^*. It is therefore important to understand the dynamics of what happens when the industry supplies an amount of product other than q^*. Let's say that the industry begins by supplying an amount $q_1 > q^*$, and let's trace the resulting actions:

- If the market is flooded with a larger-than-optimal quantity of product, then the customers are willing to pay only if the price *drops* to p_1, the value of the demand curve at q_1. Because the demand curve is a decreasing function, $p_1 < p^*$.
- If the market price drops to p_1, then the industry is willing to supply only q_2 units, where $q_2 < q^*$. The industry thus cuts production to this level.
- With only q_2 units on the market, there is not enough supply to satisfy the consumers. The units can therefore be sold at a higher price (greater than p^*) to meet consumer demand.

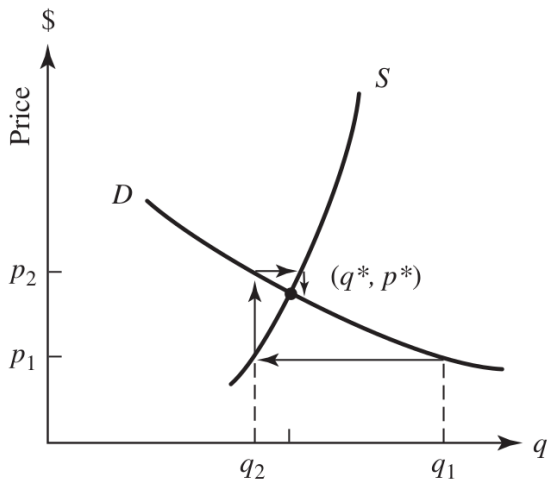


Figure 4: Supply and demand curves intersect at a point that corresponds to a market price and market quantity that satisfy both consumers and suppliers. Figure from Giordano et al., 2014.

These iterative changes in production quantity and market price continue until the process converges to (q^*, p^*), as shown in Figure 4. Because the quantity/price path follows a spiral pattern inward, this concept is called the *cobweb model* in economics. Note that we have made a couple of implicit assumptions in order to have the process converge as it does in the example figure. First, we assume that there is a time lag in the response of both the industry and the consumers to the market price/quantity. Second, the supply curve is steeper than the demand curve. (If you draw a graph where the demand curve is steeper than the supply curve, you can convince yourself that you don't necessarily reach a stable equilibrium.) In practice, the demand curve can be steep for certain industries where customers are not able to easily switch to an alternative product if the price increases—water and electricity are some examples. The supply curve is steep when the industry is strained and cannot supply more product without adding more capacity (in the form of machinery, employees), which in turn sharply increases the price.

4.4 Taxation and the energy crisis

Now that we have a general picture for supply and demand dynamics, let's introduce a tax that is imposed by the government upon each unit sold by the industry. Because the industry must pay the government extra, the supply curve shifts upward by the amount of tax (the extra cost). If there is no shift in the demand curve, then the intersection between the demand and supply curves shifts upward and to the left, as shown in Figure 5. The new intersection point, (q_1, p_1), represents the decrease in demand and the increase in price. Note that the price increase, $p_1 - p^*$, is *less* than the tax; in other words, the consumers and the industry share the tax and both pay some fraction. The exact details of the split depend on the slopes of the supply and demand curves.

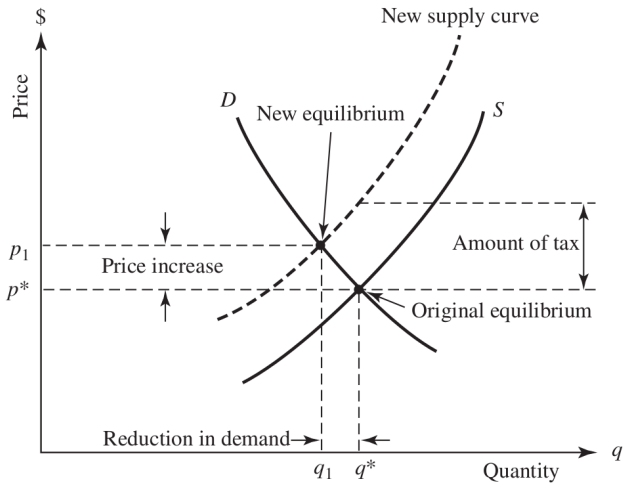


Figure 5: The effect of tax upon the equilibrium demand and price. Figure from Giordano et al., 2014.

We'll now apply our taxation model to the energy crisis. Many people depend critically on petrol for transportation and have essentially no other fuel-efficient options. Let's say an oil shortage suddenly hits—the consequence is that people are essentially forced to pay higher prices because they have no alternate options. As discussed in the previous section, these circumstances result in a very steep demand curve over a fairly wide range of q. The steepness of this curve changes slightly in the limiting cases:

- If the supply decreases to the point where the price skyrockets to unreasonable levels, then the cost of commuting eventually outweighs the money earned from going to work. The demand thus tapers off and becomes flatter at low q and high p.
- If the supply increases to the point where the market is flooded with petrol, then consumers have the option of using the extra petrol for e.g. leisure activities for very little additional cost. The demand curve therefore also becomes flatter at high q and low p. Another way of thinking about this situation is that if a consumer uses extra petrol for leisure and the price increases slightly, the consumer can easily reduce petrol usage by a large margin simply by cutting out leisure activities.

The flattening of the demand curve at its extremes is illustrated in Figure 6.

How is the supply curve impacted by the oil shortage? If the shortage hits suddenly, then the industry must find alternate oil sources, which are most likely more expensive. Because of the increased difficulty for the industry to supply oil, the supply curve shifts upward and also becomes more steep. This change, coupled with a steep demand curve, results in a dramatic price increase with only a small reduction in demand.

Now let's say that the government attempts to reduce the demand for oil and decides to impose a tax. As discussed earlier, the added tax shifts the supply curve upward. Because the demand curve is steep, this added tax does little to reduce the demand, and the situation is analogous to the oil shortage scenario previously described. Furthermore, the steepness of the demand curve implies that the new equilibrium point will be difficult to achieve, and it is likely that there will be large fluctuations in price. Our graphical model additionally predicts that it is the consumer, not the industry, that bears most of the tax burden.

Let's continue with the oil shortage thought experiment and say that the crisis continues for some time but then eventually passes (for example, if the original shortage was caused by a trade embargo

that is now lifted). During the long-term crisis, the industry was forced to develop new oil sources in order to supplement its supply. After the crisis passes, the industry has access to both its old and new oil sources, so it is able to supply a greater quantity. We therefore predict that the supply curve shifts downward and becomes more horizontal than before.

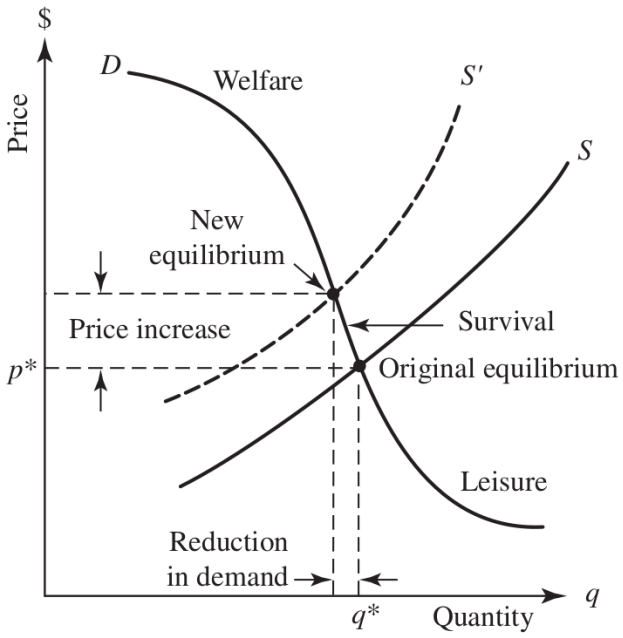


Figure 6: The demand curve for oil/petrol is steep over a wide range of q, but it tends to flatten at the extremes. Figure from Giordano et al., 2014.

Long-term consumer demand also evolves as a result of the oil shortage. Because the price increases during the crisis, consumers are forced to adopt alternate transportation solutions: for example, they may form carpools, use mass transportation systems, or switch to more fuel-efficient cars. These changes tend to flatten the demand curve and shift it downward. Because both the supply and demand curves shift downward, our graphical model predicts that there is a clear reduction in equilibrium price, but it is difficult to say whether or not there is a significant reduction in demand. (This uncertainty is another example of the limitation of graphical models, and we would need more quantitative information in order to push the prediction further.)

We will consider one final stage in our thought experiment: let's suppose that after the oil crisis has passed, the government is still dissatisfied with consumer demand and imposes a tax to reduce demand further. As discussed before, the supply curve shifts upward by the amount of tax. However, because the demand curve has evolved during the oil crisis and has become flatter, our graphical model makes three qualitative predictions:

- The reduction in demand is more significant than in our original case (before the oil shortage).
- The price increase for the consumer is small *relative* to the overall added tax. The implication is that the industry bears a larger fraction of the tax burden.
- The new equilibrium position is more easily obtained, and we do not expect to see large price fluctuations.

5 Summary

We've seen that in certain modeling situations with incomplete or imprecise information, we can still make some informed predictions by employing graphical models. Once you have identified the relevant input and output variables, it is possible to construct a qualitative graph by considering slopes, curvatures, and intercepts. This model can be tested by introducing external factors and by studying how the graph changes shape in response. If there are multiple graphs (corresponding to competing processes), the intersections can give us information about equilibrium points and their stability. Although graphical methods have their limitations because they are fundamentally qualitative, they allow us to make some predictions that are backed by logical reasoning, and they provide general guidelines for more quantitative analyses.

Math 246 notes #5: Optimization

1 Motivation

- Optimization of a particular process is a problem that frequently arises; as an example, we've already mentioned industries that must optimize their production quantities in order to maximize profits. Because optimization is such a ubiquitous problem to solve, it's worth developing the machinery to optimize within the context of mathematical models.
- The topic of optimization is actually the backbone for an entire field of study, operations research, which aims to model the activities of organizations and to find the best solutions for how those activities should be conducted.
- One example of optimization in practice is job scheduling on supercomputers. A supercomputer has many nodes for running programs that are submitted by multiple users, and each user requests the resources needed for his/her job (number of nodes, run time, etc). The scheduling software must decide the order in which to run the jobs in order to maximize efficiency. The solution to this problem turns out to be similar to solving an N-dimensional Tetris game.

2 Basic calculus refresher

Optimization can involve a wide range mathematical techniques. However, in many cases, the answers to "best solution" problems can be obtained, or at least approximated, by using derivatives to maximize or minimize functions. Here's a quick refresher on that process.

If a function $f(x)$ has a minimum or maximum value at $x = x_0$, then the first derivative is zero when evaluated at that critical point:

$$\left.\frac{df}{dx}\right|_{x_0} = 0.$$

In other words, the function is "flat" at this point and does not change appreciably if x is slightly greater or less than x_0. We can write this in mathspeak as $f(x_0 + dx) - f(x_0) = 0$, where dx is a small change in x.

To determine whether the critical point x_0 corresponds to a maximum or a minimum, we must look at the second derivative of the function:

- Positive second derivative, $d^2f/dx^2|_{x_0} > 0$: positive concavity, x_0 is a local minimum.
- Negative second derivative, $d^2f/dx^2|_{x_0} < 0$: negative concavity, x_0 is a local maximum.

We can also consider the implications for the behavior of the first derivative on either side of x_0. If x_0 is a minimum, then $f(x)$ rises on either side of that point. On the left side, $x = x_0 - dx$, the first derivative df/dx is negative. On the right side, $x = x_0 + dx$, the first derivative is positive. By similar arguments, if x_0 is a maximum, then $df/dx > 0$ when $x = x_0 - dx$, and $df/dx < 0$ when $x = x_0 + dx$. We can therefore also use the behavior of the first derivative to help determine whether x_0 is a minimum or maximum.

3 Snell's law

There is a phenomenon known as *refraction* in optics, where the path of light is bent as it passes through different media. A common example of this effect in action is that objects appear to be bent when partially submerged in water. Our understanding of refraction is based on Fermat's principle, which states that when light passes through one or more homogeneous media, it follows the path that requires the least amount of travel time. Thus, when light travels through a single, homogeneous

medium like air, it takes a straight path. However, when light crosses a boundary between e.g. air and water, its path is deflected. We would like to calculate the amount of deflection by performing the optimization exercise of minimizing light travel time. For simplicity, we will consider a water–air boundary, although the result can be generalized beyond this specific case.

Let's assume that the speed of light in water and air is v_1 and v_2, respectively. The path of light starts from point A beneath the water surface, as shown in Figure 1, and it arrives at point B above the water surface. The total travel time between points A and B is

$$T = \frac{d_1}{v_1} + \frac{d_2}{v_2} = \frac{\sqrt{a^2 + x^2}}{v_1} + \frac{\sqrt{b^2 + y^2}}{v_2}.$$

If we call the total horizontal distance $c = x + y$, then we can rewrite the total travel time as

$$T = \frac{\sqrt{a^2 + x^2}}{v_1} + \frac{\sqrt{b^2 + (c - x)^2}}{v_2}.$$

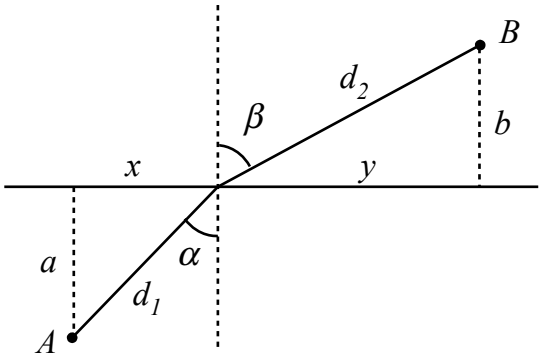


Figure 1: The path of light is bent as it passes from point A, which lies below the water surface, to point B (in air).

We are interested in determining the value of x, i.e. the location of the boundary crossing point, that minimizes the travel time. We therefore impose the constraint

$$\frac{dT}{dx} = \frac{x}{v_1\sqrt{a^2 + x^2}} + \frac{x - c}{v_2\sqrt{b^2 + (c - x)^2}} = 0,$$

and rearranging this equation gives us

$$\frac{v_1}{v_2} = \frac{x/\sqrt{a^2 + x^2}}{y/\sqrt{a^2 + y^2}} = \frac{\sin\alpha}{\sin\beta}.$$

This final relationship is known as *Snell's law*. If v_1 and v_2 are known, then we can compute the angle by which light is deflected when it passes from one medium to another. As a side note, the ratio v_1/v_2 is known as the index of refraction, and it is a material property of the medium of interest.

There is one final check we should perform, which is that our solution actually corresponds to a minimum in the travel time rather than a maximum. So let's check the second derivative:

$$\frac{d^2T}{dx^2} = \frac{a^2}{v_1(a^2 + x^2)^{3/2}} + \frac{b^2}{v_2(b^2 + (c - x)^2)^{3/2}}.$$

Since v_1 and v_2 are positive quantities, the second derivative is always positive. Therefore, our above expression for Snell's law does indeed correspond to a minimum in T.

4 Honeycomb geometry

This example will feature the handiwork of one of nature's craftiest engineers: the honeybee. Bees build wax comb to house their young and honey supply, and the comb has a remarkable hexagonal cell structure. We would like to understand the details of the cell construction.

Figure 2 shows the geometry of a single cell within honeycomb. The open face of the cell is a regular hexagon. The cell walls themselves form a hexagonal prism, and the closed face of the cell comprises three rhombic areas that form a pyramid shape. (The reason for this structure is that honeycomb cells are constructed back-to-back, and the layers are staggered so that the bottom of one cell meets the bottoms of three other cells on the reverse side.) Our optimization problem is: how can we maximize storage volume while using the least amount of wax, i.e. minimizing the surface area? The answer will depend on the pitch angle θ of the pyramidal face.

We need to start by writing down an expression for the total surface area. Let's begin with one of the rhombic areas, outlined in red in Figure 2. Each side of the hexagon has length s, and this is the same distance between the center and each of the vertices; therefore the length MN is equal to s. We can use this information to obtain the length VB, which is $s/\sin\theta$. To obtain the other dimension of the rhombus, note that the points MCN form an equilateral triangle with sides s and altitude $s\sqrt{3}/2$. The length AC is therefore $s\sqrt{3}$. With the lengths VB and AC in hand, we can calculate the area of a single rhombus as $(s^2\sqrt{3}/2)\csc\theta$. Because there are three rhombi, the total surface area of the closed face is $(3s^2\sqrt{3}/2)\csc\theta$.

Let's move on to the cell walls now. Using the fact that the length MN is equal to s, we can obtain the length VM as $(s/2)\cot\theta$. This distance is the same as the length NB. The area of a single cell wall is therefore $sh-(s/2)\cot\theta\cdot s/2 = sh-(s^2/4)\cot\theta$. Multiplying by six gives us $6sh-(3s^2/2)\cot\theta$.

Putting the rhombus and wall areas together, we obtain the total surface area $f(\theta) = 6sh - (3s^2/2)\cot\theta + (3s^2\sqrt{3}/2)\csc\theta$, where $0 < \theta < \pi/2$. Now let's take the derivative with respect to θ in order to find the optimal angle:

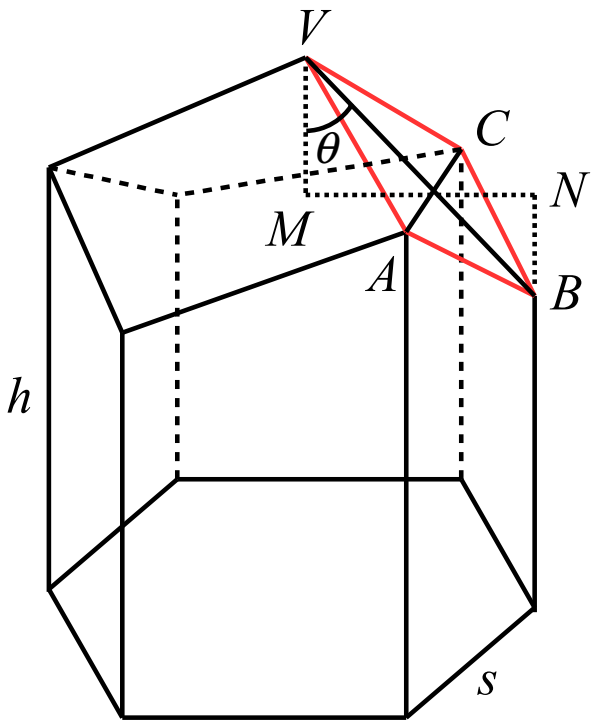


Figure 2: A single cell in a honeycomb has a hexagonal open face (depicted at the bottom in this figure). The closed face of the cell (at the top) comprises three rhombic areas, one of which is highlighted in red, that join to form a pyramid.

$$\frac{df}{d\theta} = \frac{3s^2}{2}\csc^2\theta - \frac{3s^2\sqrt{3}}{2}\csc\theta\cot\theta = \frac{3s^2}{2}(\csc\theta)(\csc\theta - \sqrt{3}\cot\theta).$$

Setting $df/d\theta = 0$ gives us $\csc\theta - \sqrt{3}\cot\theta = 0$, or $\cos\theta = 1/\sqrt{3}$. Solving for θ gives us an angle of 0.955 radians, or $54.7°$. Note that our solution for θ does not depend on s or h.

Our final step is to convince ourselves that the critical value of θ actually corresponds to a minimum, rather than a maximum. Instead of taking the second derivative, let's look at the behavior of the first derivative on either side of the critical point. The first derivative can be rewritten as $df/d\theta = (3s^2/2)\csc^2\theta(1-\sqrt{3}\cos\theta)$. Note that $\csc^2\theta$ is a positive quantity, so let's focus on the behavior of $1-\sqrt{3}\cos\theta$. In the range $0 < \theta < \pi/2$, $\cos\theta$ is a *decreasing* function. Therefore, when θ is slightly larger than $54.7°$, $1-\sqrt{3}\cos\theta$ is positive. Similarly, when θ is slightly smaller than $54.7°$, $1-\sqrt{3}\cos\theta$ is negative. The first derivative thus changes signs: $f(\theta)$ has a positive (negative) slope when θ is greater (less) than the critical value, so the critical value does indeed correspond to a minimum in total surface area.

Remarkably, bees construct their honeycomb cells with a pitch angle that is shockingly close to our derived optimal value, $\theta = 54.7°$. Measurements of honeycomb geometry suggest that the actual pitch angles seldom differ from our calculated value by more than two degrees! As a side note, there are other optimization problems in nature that yield the same geometry and resulting angle. One example is soap bubbles that are packed into a foam; the intersecting soap films always meet each other at tetrahedral angles. The reason for this geometry is that soap films are under surface tension, and they pull on each other in such a way as to minimize the total surface area between all connected points.

5 Geometry of blood vessels

Blood vessel systems within large animals are highly complex, and there is probably some sort of optimization that has been imposed by evolution. Is it possible to explain blood vessel geometry with an optimization model? We need to start by identifying the quantities that are either minimized or maximized in the optimization process: we will generically call this the "cost," and we will work out the details of this cost one step at a time.

We'll start by considering the branching of blood vessels, and we'll make the naive assumption that each vessel splits into two same-sized vessels that carry equal amounts of blood. We'll also assume that all three vessels lie in a plane because this configuration minimizes the amount of vessel wall material needed (and Mother Nature is an efficient engineer). From our assumptions, the two smaller branches must be symmetric: each has a radius r, flow rate f, and each makes the same angle θ with the larger vessel. We need corresponding quantities for the larger vessel, so we'll call its radius R, and the flow rate $F = 2f$ is equal to the sum of the flow rates of the smaller vessels.

We have yet to identify the exact nature of the cost in this case. Nevertheless, let's forge ahead and give the cost a function name and describe it as a *cost per unit length*, $C(r, f)$, which depends on radius and flow rate. Our goal is then to minimize this cost, and we obtain constraints on the blood vessel radii by enforcing the following partial derivative conditions:

$$\frac{\partial C(R,F)}{\partial R} = 0 \quad \text{and} \quad \frac{\partial C(r,f)}{\partial r} = 0.$$

Solving the above expressions yields optimal values for R and r. In addition to the radii, we also want to choose θ in a way that minimizes the total cost associated with the three vessels in the branch. If the large vessel has length L and the small vessels have lengths ℓ_1, ℓ_2, then the total cost is

$$C_{\text{tot}} = C(R,F)L + C(r,f)\ell_1 + C(r,f)\ell_2.$$

If we keep the start and end points of the vessel system fixed and consider a small change in $L \to L+\Delta L$, the result is that ℓ_1 and ℓ_2 decrease by a small amount that's approximately equal to $\Delta L \cos\theta$. We can express these small changes as

$$\frac{\Delta C_{\text{tot}}}{\Delta L} = \frac{C_{\text{tot}}(L+\Delta L) - C_{\text{tot}}(L)}{\Delta L} = C(R,F) - 2C(r,f)\cos\theta.$$

Because the total cost is a quantity that's minimized, it must satisfy $\Delta C_{\text{tot}}/\Delta L = 0$ at the minimum. In other words, there is essentially no change in C_{tot} as L is varied slightly. We therefore obtain the condition

$$C(R,F) - 2C(r,f)\cos\theta = 0$$

at an extreme point, which we can use to solve for the optimal angle $\theta = \theta_0$. Because the branching angle lies in the range $0 < \theta < \pi/2$, the function $\cos\theta$ is a *decreasing* function in this quadrant. We can use this information to convince ourselves that θ_0 minimizes, rather than maximizes, the cost:

- When θ is slightly *larger* than θ_0 (or equivalently, L is slightly larger than the optimal value), the value $\cos\theta$ is *smaller* than $\cos\theta_0$. The quantity $\Delta C_{\text{tot}}/\Delta L = C(R,F) - 2C(r,f)\cos\theta$ therefore becomes *positive*.
- When θ is slightly *smaller* than θ_0 (or equivalently, L is slightly smaller than the optimal value), the value $\cos\theta$ is *larger* than $\cos\theta_0$. The quantity $\Delta C_{\text{tot}}/\Delta L = C(R,F) - 2C(r,f)\cos\theta$ therefore becomes *negative*.

In other words, the first derivative of the cost function $\Delta C_{\text{tot}}/\Delta L$ is positive (negative) when L is slightly higher (less) than the optimal value. We therefore conclude that C_{tot} is, indeed, minimized.

Now let's try to write down a specific form for the cost function C. There are several factors that that might be important when considering blood transport through vessels:

- The basic mechanism we're dealing with is fluid flowing through pipes, and pushing the fluid through requires energy (work). We will invoke a result from fluid mechanics that the work-per-unit-length required to push fluid through a pipe is kf^2/r^4, where k is a constant that describes the fluid properties.
- Blood vessels are composed of cells that eventually wear out and die, so there is some cost associated with blood vessel maintenance and replenishing lost cells. The amount of maintenance depends on several factors, including the total volume of cells in the blood vessel, the inner surface area (where we expect the most wear and tear), or some combination of both. If we assume geometric similarity, the cost *per length* for each of these factors scales as r^2 and r, respectively. The geometric similarity assumption may break down in some cases (it's possible to imagine a range of blood vessel sizes where the wall thickness might be constant), so we will generically describe the maintenance cost as Kr^a, where K is a proportionality constant and a is an exponent that lies in the range $1 \leq a \leq 2$.

Combining the above factors together, we obtain a total cost per unit length of

$$C = \frac{kf^2}{r^4} + Kr^a.$$

Now that we have an explicit form of the cost function, we can apply the constraints that we developed earlier. We start by optimizing the radius:

$$\frac{\partial C}{\partial r} = -\frac{4kf^2}{r^5} + aKr^{a-1} = 0.$$

Rearranging the above expression, we obtain

$$\frac{f^2}{r^{a+4}} = \frac{aK}{4k}.$$

We can substitute this relation into our original expression for the cost per unit length:

$$C(r, f) = k\left(\frac{aK}{4k}\right) r^a + Kr^a \equiv \lambda r^a,$$

where λ is a combination of constants. Note that this expression applies for the two smaller branch vessels, but we can easily obtain the corresponding expression for the larger vessel by recalling that $F = 2f$. The cost function is $C(R, F) = kF^2/R^4 + KR^a = 4kf^2/R^4 + KR^a$, and enforcing $\partial C/\partial R = 0$ gives us the result

$$\frac{4f^2}{R^{a+4}} = \frac{aK}{4k},$$

and

$$C(R, F) = k\left(\frac{aK}{4k}\right) R^a + KR^a \equiv \lambda R^a.$$

In summary, we now have $C(r, f) = \lambda r^a$, $C(R, F) = \lambda R^a$, and combining $\partial C/\partial r$ with $\partial C/\partial R$ gives us $(R/r)^{a+4} = 4$.

Finally, we can use our previous equation $C(R, F) - 2C(r, f)\cos\theta = 0$ to solve for the optimal branching angle:

$$\cos\theta = \frac{C(R, F)}{2C(r, f)} = \frac{(R/r)^a}{2}.$$

We can further simplify this expression by using the result $(R/r)^{a+4} = 4$, which can be rewritten as $R/r = 2^{2/(a+4)}$. Substituting, we obtain our final result

$$\cos\theta = 2^{(a-4)/(a+4)}.$$

Let's put our hard work to good use and make a prediction! Because $1 \leq a \leq 2$, we obtain $2^{-1/3} \leq \cos\theta \leq 2^{-3/5}$, or $37^\circ \leq \theta \leq 49^\circ$. Blood vessel branching angles are difficult to measure in practice, but what little data exist in the literature suggest that our prediction is more or less correct (in the special case where the two smaller branches are the same size).

We can also ask the question of how many branchings there are between the the largest (aorta) and smallest (capillary) blood vessels within an animal. If we are given the aorta and capillary radii, we can establish the number of branchings, N, with the following relation:

$$\frac{R_{\text{aorta}}}{r_{\text{capillary}}} = \prod^{N} 2^{2/(a+4)} = 2^{2N/(a+4)}.$$

Empirically, a human's aorta is about 3 cm in diameter, and a typical capillary has a diameter of about 7 microns = 7×10^{-4} cm, so the ratio of radii is about $4000 \simeq 2^{12}$. Therefore, $12 \simeq 2N/(a+4)$, and since a lies in the range $1 \leq a \leq 2$, we obtain N of between 30 and 36. This number of branchings may not seem like much, but let's use it to calculate one more quantity: the total number of capillaries in a human. Because each branching doubles the number of blood vessels, the total number of vessels after N branchings is 2^N. If we plug in our typical values of 30–36 for N, we obtain a total of 10^9–10^{11} capillaries for a human. The actual answer turns out to be about 10^{10}, so our simple model didn't do too badly!

6 Maintaining inventories

Suppose that you are a management consultant and that you are asked for advice on production and warehousing policies. A company is able to choose how often it performs production runs of a particular product and, as a consequence, how much inventory of finished goods to maintain at any given time. There are many competing factors to consider: the cost of storage, setup expenses for a production run, etc. Before we get mired in the details of what factors to consider, let's first establish our ultimate goal: *we want to minimize the cost per unit time to the company, subject to the constraint that all orders be filled*. Furthermore, we will reduce the dependence of the cost per unit time to involve only one quantity that the company can control: the time between production runs. In other words, we are reducing our problem to a single-variable function that we hope to optimize.

Let's make a list of variables and assumptions that we should consider:

- When the production line is operating, we'll say that the finished goods are produced at a constant rate k per unit time.
- For each production run, there is a cost c associated with the setup. In practice, this cost can include things like fixed costs, additional material and salaries, etc.
- The storage costs of the finished product are s per item per unit time, independent of the quantity stored.
- We approximate that the orders arrive continuously at a constant rate r per unit time.
- We will introduce two time variables: T = the length of time between one production run and the next, and t = the length of a production run. (The time between t and T is the "dead time" when there is no production, and there is only consumption of the warehoused product.) Our goal is to find the optimal value of T.

Now let's start putting some of these variables together into meaningful relationships. Ideally, a company would like to balance the total production rate with the demands of the consumers: $kt = rT$, so that goods produced equal the goods sold during a cycle. During the production run (time from 0 to t), the rate of inventory accumulation in the warehouse is the difference between production and consumption rates, $k - r$. After the production run ends but before the next one begins (time from t to T), the warehoused inventory decreases with a rate r. In summary, the inventory versus time rises

with a slope $k - r$ from 0 to t, and then falls from t to T with slope $-r$. The integral of the inventory versus time curve has a triangular shape with area $A = (k - r)tT/2$, and it has units of (# items) $\times$ (time). The total storage cost during the production period is thus sA.

We now have enough information to write down the total cost per unit time, which is combination of the setup costs and the storage costs:

$$C = \frac{c + sA}{T} = \frac{1}{T}(c + s(k - r)tT/2) = \frac{c}{T} + \frac{srT(k - r)}{2k}.$$

(In the last step, we used the result $kt = rT$, or $t = rT/k$.) Now that we have the cost function in terms of our single independent variable T, we can optimize by differentiating:

$$\frac{dC}{dT} = -\frac{c}{T^2} + \frac{sr(k - r)}{2k} = 0.$$

Solving the above equation gives us the optimal values for T and t:

$$T = \sqrt{\frac{2ck}{sr(k - r)}} \quad , \quad t = \sqrt{\frac{2cr}{sk(k - r)}}.$$

We can convince ourselves that our solution corresponds to a minimum in cost, rather than a maximum, by examining the behavior of dC/dT when T deviates slightly from the critical value. If T is slightly increased (decreased) from the critical value, then $dC/dT > 0$ ($dC/dT < 0$). Therefore, our solution is indeed a minimum. Although this model is a crude one, it illustrates a simple starting point for more complicated inventory management calculations. For example, the model can be extended to include the impact of storage costs for raw materials, the possibility of back-ordering, etc. These extended models are often encountered in business management.

7 Summary

Optimization models are important in many different areas of research and industry, and we've touched on a simple class of these models that require only simple differentiation to determine a maximum or minimum value. As you've seen elsewhere in this class, the difficulty is not so much in application of this mathematical technique, but it is in the *setup* of the problem. An effective mathematical optimization model requires careful consideration of what inputs are relevant and what function(s) to optimize. Once you do the hard work of the setup, the actual optimization calculation should be smooth sailing.

Math 246 notes #6: Modeling change

1 Motivation

- We've spent a fair amount of time discussing techniques for calculating static properties and relationships within various modeling scenarios. We'll now turn our attention to the next level of complexity: modeling *change* in a dynamic system.
- Our mathematical weapon of choice for modeling change will be differential equations. In essence, our goal for each modeling scenario is to write a precise mathematical statement that corresponds to the words "*change = future value – present value*." Once we have the appropriate math written, we can solve for the evolution of our modeled system.
- Differential equation models feature prominently in many areas of study. Population biology is a classic example, and we will spend some time discussing these calculations in depth. Much of physics is governed entirely by differential equations, with Newton's second law being a prime example. There are further applications in economics and sociology as well.

2 Population growth

We'll begin our differential equation adventures by considering a model of population growth. We'll start off with a simple picture, and we'll refine it as we proceed. The basic question we'd like to answer is: how does a population grow numerically over some number of generations?

2.1 Unbounded growth

Let's start by identifying the relevant variables in our problem:

- We are interested in how the population evolves with time, so the time t must be one of our independent variables.
- Each individual member of the population has some net reproductive rate r, which represents the average number of offspring produced per unit time minus the number of deaths per unit time. This is an input parameter.
- We additionally need to know how many population members exist at the beginning. We'll call this N_0, and it is also an input parameter.
- The only dependent variable (or output) that we are concerned with is the total population at a given time, which we denote $N(t)$.

Now that we've identified the input and output variables, our task is to combine them into a sensible relationship. In this process, we will assume that r is a constant and that there are no time lag effects (so that only the present value of N and its derivatives are relevant in determining the future values of N). Let's consider what happens to the population size as time increases by some small amount: $\Delta N = N(t + \Delta t) - N(t)$. During this time increment Δt, each individual member of the population creates an average of $r\Delta t$ new population members. The total number of new members is thus $rN\Delta t$. This number of new members is equal to the incremental population increase, so we obtain the relationship $\Delta N = rN\Delta t$. If we divide both sides by Δt and take the limit in which $\Delta N/\Delta t \to dN/dt$, we arrive at our first differential equation governing population growth:

$$\frac{dN}{dt} = rN.$$

Our next step is to solve the above equation, which means that we need to talk about some basic techniques for tackling differential equations. The zeroth strategy when analyzing a differential equation is always: *try to guess the answer*. Our population growth equation states that the derivative of some function N is equal to the function itself (modulo a multiplicative constant). The only function with this property is an exponential, so we can correctly guess the solution that $N \propto e^{rt}$.

2.2 Separation of variables

Although guessing and careful inspection is always worth a try, we are seldom that lucky. Fortunately, our population growth equation falls into the category of *separable* differential equations, which take the following generic form:

$$\frac{dy}{dx} = \frac{f(x)}{g(y)}.$$

In other words, the derivative can be expressed as a ratio of two functions, each of which depends only on the independent or dependent variable. The general technique for solving this equation begins by treating the derivative as a fraction (ratio of two differential elements) and *separating* the independent and dependent variables onto opposite sides of the equation:

$$g(y)dy = f(x)dx.$$

We then integrate both sides to obtain the final solution:

$$\int g(y)dy = \int f(x)dx$$

$$G(y) = F(x) + C,$$

where $G(y)$ and $F(x)$ are the respective antiderivatives of $g(y)$ and $f(x)$, and C is an arbitrary constant. The exact value of the constant is determined by the *boundary conditions*, i.e. constraints on the differential equation at specific value(s) of the independent and dependent variables. There are a couple of obvious but important points to note about the process of separation of variables: (1) this procedure works only for differential equations that involve two variables, and (2) the differential elements dx and dy must be in the *numerator* on opposite sides of the equation.

Let's apply this technique to our population growth equation, $dN/dt = rN$. We can separate the independent and dependent variables, t and N, so that our equation reads as

$$\frac{dN}{N} = rdt.$$

Integrating both sides gives us $\ln N = rt + C$, where C is an arbitrary constant. Solving for N, we obtain $N = e^{rt+C} = C_1 e^{rt}$, where $C_1 \equiv e^C$ is the constant that's been renamed for convenience. To solve for the value of C_1, we need to use the initial condition $N(0) = N_0$. Plugging these values into the equation gives us $C_1 = N_0$, so our final solution to the differential equation is

$$N = N_0 e^{rt}.$$

2.3 Limited growth

Does our final solution make sense? As always, we need to *sanity check* the limiting cases. Unless r is zero (no births, no deaths) the population eventually either dies out completely ($r < 0$) or grows to infinity ($r > 0$) when t is very large. Our model therefore fails to capture realistic behavior in this limiting case, which means that we have to try harder. Clearly the growth rate of a population must depend on the size of the population because of effects such as exhaustion of the food supply and other resources. Our next step in complexity is therefore to replace the constant growth rate r with one that depends on the population:

$$\frac{dN}{dt} = Nr(N).$$

What can we say about the properties of $r(N)$? When the population is small, resources are abundant, so we expect the initial growth to be similar to the unbounded case described in the previous section. As the population grows, competition for resources causes the death rate to slowly rise in comparison to the birth rate, so we expect $r(N)$ to be a strictly decreasing function (because it is the net growth rate). Furthermore, as N becomes very large, competition for resources becomes so stiff that the death rate eventually exceeds the birth rate, and the net growth becomes negative.

There are many ways to select a function $r(N)$ that satisfies the properties we want. We will choose one of the simplest to study,

$$\frac{dN}{dt} = rN\left(1 - \frac{N}{k}\right),$$

which is known as the *logistic equation*. How can we interpret the constants r and k? In the limit that N is very small, the right hand side of the equation is dominated by the rN term. In other words, when the population is small, the limiting behavior of the equation is essentially the same as unbounded exponential growth with a net growth rate r. As N increases, it eventually approaches the value k; at this point, the right hand side of the equation vanishes, and the population change is zero. We can therefore interpret k as the *carrying capacity*, or the maximum population that can be sustained given limited resources. Finally, if N happens to be larger than k, then the right hand side of the equation is negative, so the population decreases. In other words, the population is greater than the maximum sustainable number, so the net "growth" rate is dominated by deaths rather than births.

Now that we have an understanding of the logistic equation, let's solve it. We can combine terms on the right hand side so that the equation becomes $dN/dt = rN(k-N)/k$. Separating variables, we obtain

$$\int \frac{kdN}{N(k-N)} = \int rdt.$$

We can use partial fractions to rewrite the integrand on the left hand side as $1/N + 1/(k-N)$. Integrating then gives us

$$\ln N - \ln(k-N) = rt + C,$$

where C is an arbitrary constant. Cleaning up and solving for N gives us

$$N = \frac{k}{1 + Be^{-rt}},$$

where B is a constant (that can be expressed in terms of C, and that we've renamed for convenience). If we substitute in the initial condition $N(0) = N_0$, then we obtain $B = (k - N_0)/N_0$. Figure 1 illustrates the behavior of the logistic curve for two different initial populations, but with the same growth rates and carrying capacities. The behavior is as we expect: in the beginning, when t is small, the growth is roughly exponential. The maximum growth *rate* occurs when the population is at half of the carrying capacity. When t becomes very large, the population growth tapers off and asymptotes to the carrying capacity value.

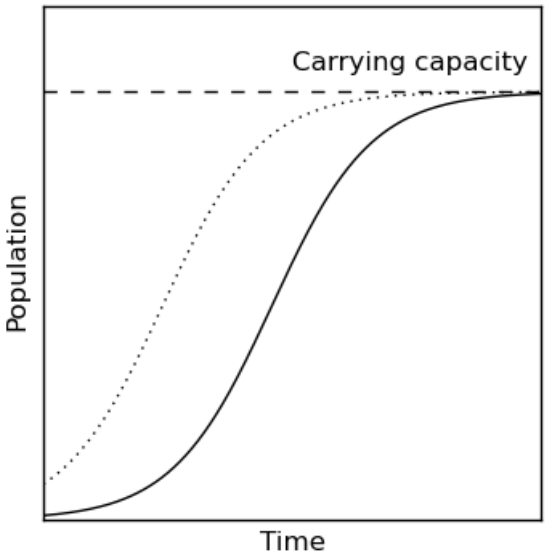


Figure 1: Two logistic curves (solid and dotted lines) with two different initial populations but same k and r values. The carrying capacity is indicated by the dashed line.

2.4 Beyond the logistic curve

The logistic curve is an improvement over our original unbounded exponential growth solution because the former more accurately captures the long-term behavior. However, every model has its failings, and we can ask ourselves: how could we improve the logistic equation to be even more realistic? The current logistic curve solution has a couple of outstanding problems:

- The population only moves asymptotically closer to k as time goes on. However, real populations often overshoot the steady-state size.

- Even steady-state population numbers tend to fluctuate slightly because of the random nature of births and deaths.

How can we capture those above two points in mathspeak? One possible solution is to separate the net growth rate into a death rate m and a birth rate b, and to introduce a *time lag* τ for the birth rate. That is, b changes from zero to a constant at age τ. (In contrast, the death rate does not depend on age.) Putting these statements together, we can write the following candidate differential equation:

$$\frac{dN}{dt} = -mN(t) + bN(t-\tau).$$

The above expression is an example of a *delay differential equation*—the solution at any given point in time depends on the value of $N(t)$ at time lag τ in the past. The solution is not easy, so we'll put this equation aside for now and return to some simpler examples for the time being. The most important lesson to learn here is how to translate desired model properties from English into math.

3 Lake pollution

In this example, we will consider the effects of pollution in lakes. Because lakes are, for the most part, isolated bodies of water, any pollution that enters can take a very long time to dissipate—the only way to really clean out all the pollution is to gradually replace all the water. There are many complex factors that we could worry about, including sedimentation and decay effects, how pollution moves in and out of the food chain, etc. As a first pass, let's ruthlessly simplify the picture: we will ignore all biological action, sedimentation, etc., and we will assume that all pollutants are simply dissolved in the water. In other words, we are treating a polluted lake as a perfect mixing problem. We will make the following assumptions:

- The lake level stays nearly constant, which means that the average rates of inflow and outflow are equal.
- The average inflow/outflow rates do not vary much seasonally.
- When water enters the lake, perfect mixing occurs so that the pollutants are uniformly distributed.
- Pollutants are removed from the lake *only* by outflow and not by decay, sedimentation, or any other mechanism.
- Pollutants flow freely out of the lake and are not retained in any way.

The first two of these assumptions are reasonable, and the last three are somewhat questionable. Nevertheless, let's see how far we get with extremely crude picture. We now need to introduce some variables. We're interested in the lake pollution levels as a function of time t. Let's call V the volume of the lake, $P(t)$ is the pollution concentration (moles or mass per unit volume) in the lake, P_i is the pollution concentration at the source of the pollution, which we'll assume is constant, and r is the rate of input flow from the pollution source (or equivalently, the rate of outgoing flow from the lake).

We're interested in how the total amount of lake pollution, PV, varies in a short time interval Δt. The small change in the lake pollution level can be written as

$$\Delta(PV) = P_i r \Delta t - P r \Delta t = (P_i - P) r \Delta t.$$

The first term, $P_i r \Delta t$, describes how much pollution flows *in* during the time interval Δt, and the second term, $P r \Delta t$, corresponds to the amount of pollution that flows *out* during that time. Our next step is to divide both sides of the equation by Δt and take the limit in which Δt goes to zero. Our governing differential equation then becomes

$$\frac{dP}{dt} = \frac{(P_i - P) r}{V}.$$

This is another example of a separable differential equation that we solve by taking the integrals

$$\int_{P_0}^{P} \frac{dP'}{P_i - P'} = \int_0^t \frac{r}{V} dt'.$$

Note that we are explicitly setting the limits of integration here so that we are evaluating the integrals from time of zero to t, during which the lake pollution level changes from its initial value P_0 to its present value P. Because our final variables of interest appear in the limits of integration, we must change the variables inside the integrands to "dummy" variables P' and t'. Evaluating the integrals gives us

$$\ln\left(\frac{P_i - P}{P_i - P_0}\right) = -\frac{rt}{V}$$

$$P(t) = P_i - (P_i - P_0)e^{-t/\tau},$$

where $\tau \equiv V/r$. The quantity τ is sometimes known as the *retention time* of the lake; that is, if all inflow were stopped, τ is the amount of time required for the lake to empty completely from outflow alone. We can now use this equation to study the behavior of lake pollution with time.

Let's consider the best-case scenario in which the incoming pollution stops entirely, and the lake cleaning rate is maximized. The first condition implies that $P_i = 0$, so our equation reduces to $P(t) = P_0 e^{-t/\tau}$. We can then solve for the time required to achieve a specified pollution fraction:

$$t = \tau \ln(P_0/P).$$

Let's plug in a few numbers to put this in perspective. South African dams and fresh water reservoirs have retention times that vary between 1 and 10 years, so let's take 3 years as our ballpark figure. If an average reservoir were polluted and we wanted to reduce the pollution level to 50% of its present level, we would need to wait $t = (3 \text{ years}) \times \ln(2) \approx 2$ years. If we wanted to reduce the pollution level to 25% of its present level, we would need to wait $t = (3 \text{ years}) \times \ln(4) \approx 4$ years. A reduction to 10% and 5% would take 7 and 9 years, respectively. Incidentally, a 2008 study[1] by the South African Water Research Commission found that out of 13 dams studied, 8 required pollution reduction in excess of 50%, and 5 of those required greater than 75%. The main pollutant in this particular study was phosphorus. I include the following quote from the report's executive summary as motivation for the utility of simple pollution models like the one we've studied:

"*The use of simple modelling approaches, at screening level, have enormous value in focusing management needs and actions at a fundamental level—this as opposed to the extremely costly and time-consuming expedient of more sophisticated models having intensive data requirements. The approach described here allows for relatively rapid ranking and prioritization of needs and actions based on limited and generally-available data.*"

As with all simple models, we need to perform the exercise of questioning the validity of the assumptions that we've made. If our assumptions are wrong, then how are our answers affected?

- Let's start with the perfect mixing assumption. If a lake has one source and one outlet, then water tends to preferentially follow a path connecting those two points, and mixing is reduced outside of that path. Therefore, although cleanup time will be reduced for water that sits along the flow path, there will be semistagnant regions elsewhere that require much longer cleaning times. Our conclusion is thus that although the perfect mixing assumption may be far from true, it will likely cause our cleaning time calculations to be *underestimates*.

- Now let's move on to the assumptions regarding pollution movement, and we'll discuss them in the context of two important pollutants, mercury and phosphorus. Mercury does not break down and persists for long times in the biosphere. If ingested by animals, mercury is carried through the food chain and reaches greater concentration in higher predators. The predators are unaffected by lake outflow, so they provide a mechanism for *retaining* mercury pollutants

[1] `http://www.wrc.org.za/Knowledge Hub Documents/Research Reports/1687 Final Report.pdf`

in the lake environment. Phosphorus is commonly used in fertilizers, detergents, etc, and is a common pollutant. The presence of phosphorus can cause algae blooms, which grow rapidly. When the algae dies, it sinks to the bottom of the lake, carrying the pollutants with it. However, the decay process eventually returns the phosphorus to the water. We therefore conclude that some pollutants like mercury cause cleanup times to be longer, and the effect is unclear for pollutants like phosphorus.

In summary, although there's a fair amount of uncertainty in the assumptions that we've made, we can guess that our typical cleanup times that we've calculated probably provide rough lower bounds for the actual cleanup times of persistent pollutants.

4 Long chain polymers

4.1 Chemistry 101

For this example, we're going to put on our chemistry hats and talk about polymerization reactions, which are widely used in industry. The particular reaction we're going to look at is illustrated in Figure 2.

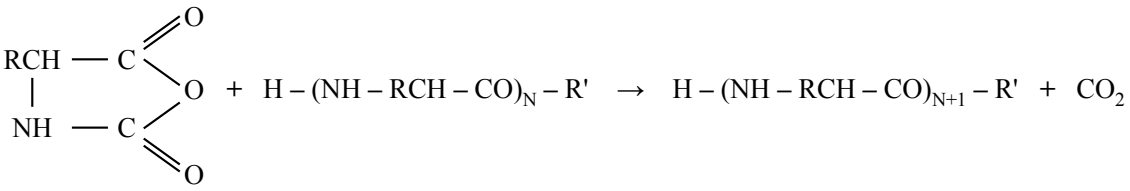


Figure 2: Example polymerization reaction.

In this reaction, R′ is a radical that breaks open the anhydride ring on the left, thus adding a link to the polymer chain ($N \to N+1$) and generating a CO_2 molecule in the process. We will write the reaction symbolically as

$$A + M_N \to M_{N+1} + CO_2,$$

where A is the anhydride ring, and M_N is a polymer of length $N \geq 0$. Our goal here is to understand the dynamics of this reaction: what are the various polymer lengths at any given time, and how quickly does the reaction proceed? For fixed temperature and pressure, the *rate* of our reaction depends on the probability of a collision between an A molecule and an M_N molecule. This probability is proportional to to the product of the concentrations, i.e. $k[A][M_N]$, where k is a rate constant that we will assume is the same for all N.

4.2 Governing differential equations for reaction rates

That's it for the chemistry refresher; let's translate the above into math now. We'll call our concentrations $a \equiv [A]$ and $m_N \equiv [M_N]$, and both of these are functions of time t. The reaction process begins with initial concentrations $a(0)$ and $m_0(0)$ (note that M_0 is simply R′H). Since the concentration of R′ does not change, we can write down the following conservation constraint:

$$\sum_{N=0}^{\infty} m_N(t) = m_0(0).$$

For each molecular species of interest, we can write down the relationship between the rate of change and the relevant concentrations:

$$\frac{dm_0}{dt} = -ka(t)m_0(t),$$

$$\frac{dm_N}{dt} = ka(t)[m_{N-1}(t) - m_N(t)],$$

$$\frac{da(t)}{dt} = -ka(t)\sum_{N=0}^{\infty} m_N(t) = -ka(t)m_0(0).$$

The first equation describes how the concentration of M_0 = R′H changes with time. Since M_0 is the starting point for polymerization, its concentration can only *decrease* at a rate proportional to the product of the concentrations of M_0 and A, which are the ingredients needed to form M_1. The second equation is valid for $N \geq 1$, and it describes the competing processes of creation and destruction for intermediate-length polymers. Any polymer of length $N \geq 1$ can be either *created* by a polymer of length $N-1$, or it can be *consumed* in the process of creating a polymer of length $N+1$. These processes are reflected in the $ka(t)m_{N-1}(t)$ and $-ka(t)m_N(t)$ terms, respectively. Finally, the third equation describes how the concentration of A decreases as it is continually consumed. One A molecule is used in each reaction—across *all* intermediate polymer lengths—so the total rate depends on the sum of all reaction rates with each M_N population.

We now have three differential equations that we'd like to solve. The equation for $da(t)/dt$ is simple to solve using separation of variables, and the answer is

$$a(t) = a(0)e^{-\lambda t} \quad , \quad \lambda \equiv km_0(0).$$

We will simplify the equations for m_0 and m_N by introducing a helper variable y such that $dy \equiv ka(t)dt$, and $y = 0$ at $t = 0$. Our equations then become

$$\frac{dm_0(y)}{dy} = -m_0(y),$$

$$\frac{dm_N(y)}{dy} = m_{N-1}(y) - m_N(y).$$

The solution to the first equation can be easily obtained by separating variables, and the answer is $m_0(y) = m_0(0)e^{-y}$. To tackle the second equation, let's first consider the $N = 1$ case. Now that we have the $N = 0$ solution in hand, the equation for m_1 becomes

$$\frac{dm_1(y)}{dy} = m_0(y) - m_1(y) = m_0(0)e^{-y} - m_1(y).$$

Although this equation now only explicitly involves m_1, note that we cannot solve it with separation of variables. To obtain a solution, we need to take an intermission and discuss another differential equation technique.

4.3 Intermission: solving differential equations with integrating factors

We're interested in solving generic first-order, linear differential equations of the form

$$\frac{dy}{dx} + p(x)y = q(x),$$

where $y(x)$ is our function of interest, and $p(x), q(x)$ are generic functions of x. The trick we'll pull here is finding a new function, $I(x)$, that we can multiply to both sides in order to turn the left hand side into a total derivative. Let's say that we have this function, and let's see what happens when we do the multiplication:

$$I\frac{dy}{dx} + Ipy = Iq.$$

(I've suppressed the x-dependence for clarity.) If we want to write the left hand side as a total derivative, then the product rule dictates that Ip must be equal to dI/dx:

$$I\frac{dy}{dx} + \frac{dI}{dx}y = \frac{d}{dx}(Iy) = Iq.$$

Because the right hand side, Iq, is expressed solely in terms of x, we simply need to integrate both sides of the equation once it's been put into this form. Solving for y is simple after that.

The only thing we need to figure out now is: what is the exact form of this magical function, $I(x)$? From our constraint that $Ip = dI/dx$, we can solve for I using separation of variables:

$$\int \frac{dI}{I} = \int p\, dx \quad \rightarrow \quad I(x) = e^{\int p dx}.$$

The function $I(x)$ is known as the *integrating factor*. Putting everything together, here's a summary of the steps for solving a first-order, linear differential equation:

1. Make sure that your differential equation fits the form $dy/dx + p(x)y = q(x)$, and identify $p(x)$.
2. Solve for the integrating factor $I(x) = e^{\int p(x)dx}$.
3. Multiply both sides of the differential equation by $I(x)$, and express the left hand side as $d(Iy)/dx$.
4. Integrate both sides of the differential equation to obtain $Iy = \int I(x)q(x)dx$, and then rejoice in your final solution $y = [\int I(x)q(x)dx]/I(x)$.

That's it for our intermission, and now let's return to the polymer problem.

4.4 Final solution for polymer reaction rates

The equation we were looking at was

$$\frac{dm_1(y)}{dy} + m_1(y) = m_0(0)e^{-y},$$

and we can now solve it with an integrating factor $I = e^{\int dy} = e^y$. Multiplying both sides by I gives us

$$e^y \frac{dm_1(y)}{dy} + e^y m_1(y) = \frac{d}{dy}[e^y m_1(y)] = m_0(0),$$

and integrating gives the final solution $m_1(y) = m_0(0)ye^{-y}$. Let's move on and see what the equation for m_2 looks like:

$$\frac{dm_2(y)}{dy} + m_2(y) = m_0(0)ye^{-y}.$$

The integrating factor here is the same, $I = e^y$, and applying it gives us

$$e^y \frac{dm_2(y)}{dy} + e^y m_2(y) = \frac{d}{dy}[e^y m_2(y)] = m_0(0)y.$$

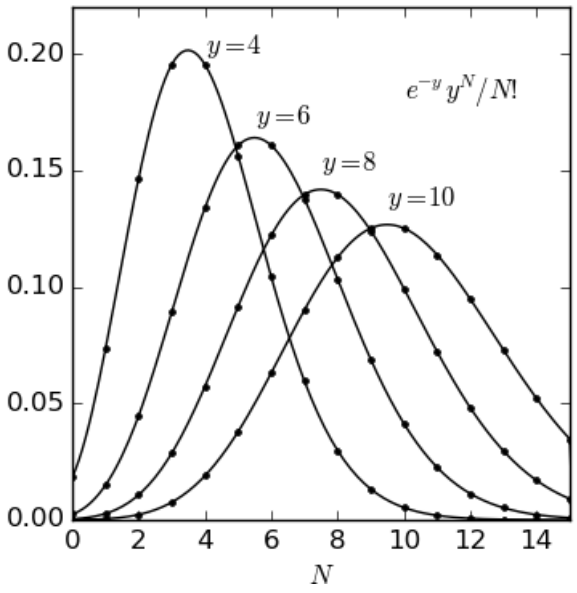


Figure 3: The distribution of polymers with length N is governed by a Poisson curve with parameter y, shown here for values of 4, 6, 8, and 10.

The solution is thus $m_2(y) = m_0(0)y^2e^{-y}/2$. The pattern continues so that the general solution is

$$\frac{m_N(y)}{m_0(0)} = \frac{y^N e^{-y}}{N!}.$$

Those of you who have taken statistics will recognize this expression as a *Poisson distribution*, as illustrated in Figure 3. In other words, *the distribution of polymer lengths follows a Poisson curve with parameter y, and the significance of y is that it corresponds to both the mean polymer chain length and the variance of that length.*

We now return to our previous solution $a(t) = a(0)e^{-\lambda t}$ and our "dummy variable" definition $dy = ka(t)dt$ to solve for y as a function of t:

$$y(t) = \int_0^t ka(t')dt' = \int_0^t ka(0)e^{-\lambda t'}dt' = \frac{ka(0)}{\lambda}(1 - e^{-\lambda t}) = \frac{a(0)}{m_0(0)}(1 - e^{-\lambda t}).$$

We can now finally address our original question: how long does it take to generate a desired polymer length L? Because there are many polymer lengths at any given time, the best we can do is specify the *mean length*, i.e. enforce $y = L$. If we make this substitution in the equation above, we obtain the following solution for time:

$$t = -\frac{\ln[1 - m_0(0)L/a(0)]}{km_0(0)}.$$

In other words, given initial concentrations of our starting reagents, A and M_0, as well as the associated rate constant, we can calculate the reaction time required so that the polymer length distribution peaks at our desired value L. What about the rest of the polymers? When y is very large, the Poisson distribution can be approximated by a Gaussian (normal) distribution with variance $y = L$, or standard deviation $\sqrt{L}$. In this limit, 95% of the polymer lengths lie between $L - \sqrt{L}$ and $L + \sqrt{L}$ (this is the "two-sigma" rule of Gaussians).

Let's revisit our assumption that the rate constant is the same for all polymer lengths, i.e. $k_N = k$. Suppose we relax this constraint and assume that longer polymers, on average, have a reduced reaction rate so that k_N is a decreasing function of N. The consequence is that as time goes on, the reaction proceeds more slowly than expected because the polymers increase in length. We expect the final distribution of chain lengths to be more peaked than a Poisson distribution because the shorter chains increase in length faster than the longer chains.

5 Beyond one variable

5.1 Introduction

We've worked through several example models that require differential equations involving a single dependent variable and a single independent variable. In our examples, that independent variable has been time. We'll now move on to the next step in complexity and consider models where the independent variable is still time, but there is *more than one dependent variable*. For simplicity, we'll deal with just two dependent variables here, each governed by a first-order differential equation, but the results can be generalized to systems with more variables and higher orders. Let's call our two dependent variables $x = x(t)$ and $y = y(t)$, and the general form of the differential equations that these variables satisfy is

$$x' = \frac{dx}{dt} = f(x, y)$$

$$y' = \frac{dy}{dt} = g(x, y).$$

Here, f and g are functions of x, y that describe the time derivatives x' and y', but we can't say much about the functions themselves at this point.

If we divide the above equations, we can eliminate the time variable and obtain

$$\frac{dy}{dx} = \frac{g(x, y)}{f(x, y)},$$

which depends only on x and y. Why bother doing this? The first reason is that the expression for dy/dx is often more analytically tractable than dealing with x' and y' directly. The second reason is that by eliminating time, we can plot the results in two-dimensional xy space—this is referred to as the *phase plane*. Because x and y are a collection of points that evolve with time, we can draw arrows on the curves in the phase plane to denote the direction of motion as time increases. As we've seen before, graphical representations are extremely useful for building intuition, and phase diagrams for coupled differential equations are another example of a valuable graphical technique. The phase diagram contains all of the information about the time evolution of x and y with the exception of the rate of movement along the curves. There are a couple of points to note about the expression for dy/dx:

- If $f(x, y)$ vanishes at any values of x, y, then dy/dx is undefined at those points. Assuming $g(x, y) \neq 0$, the curve in the phase plane approaches a vertical line near the points where $f(x, y) = 0$.
- If both $f(x, y)$ and $g(x, y)$ vanish at some value(s) of x, y, then $x' = y' = 0$, and there is no motion. In other words, the points where the derivatives vanish are *equilibrium points*. A solution that starts as an equilibrium point will never evolve away from that location, and it is plotted as a single point in the phase plane.

Once we have a phase diagram in hand, we can ask the following kinds of questions regarding the evolution of x and y:

- If a solution starts near an equilibrium point, will it move toward or away from that point? In other words, what is the *local stability* of that equilibrium point?
- If a solution starts at any generic location in the xy plane, how does it evolve? In other words, what is the *global stability*?

5.2 Warmup exercises

Example 1: To get a feel for phase diagrams, let's start with the following simple example of a system of two differential equations:

$$x' = -x$$

$$y' = -y.$$

The conditions $x' = 0$ and $y' = 0$ give us the (single) equilibrium point $(0, 0)$ in the phase plane. To determine the trajectories elsewhere in the plane, we look at the ratio of the differential equations:

$$\frac{dy}{dx} = \frac{y}{x}.$$

This equation is separable, and the solution is $y = Cx$, where C is an arbitrary constant. The trajectories in the phase plane therefore consist of lines radiating from the origin. The direction of time evolution is *inward* because the original differential equations state that as time increases, both the x and y coordinates go toward zero. Note that in this simple example, we could have solved the original system directly (both x and y have decaying exponential solutions). In general, our systems will not be so simple.

Example 2: Consider the following system of differential equations:

$$x' = y$$

$$y' = -x.$$

Unlike the previous example, it's not obvious how to solve each equation by itself. To find the equilibrium point(s), we set $x' = 0$ and $y' = 0$, which gives us the result $(0, 0)$ as the only equilibrium point in the phase plane. To analyze the behavior elsewhere in the plane, we once again divide the differential equations:

$$\frac{dy}{dx} = -\frac{x}{y}.$$

Separating variables and integrating gives us the result $x^2 + y^2 = C$, where C is an arbitrary constant. The trajectories in the phase plane are therefore *circles* of arbitrary radii that are centered on the origin. The direction of time evolution is in the *clockwise* direction: from the original system of equations, we can read off the fact that when x is positive (negative), the change in y is downward (upward). Similarly, when y is positive (negative), the change in x is toward the right (left).

5.3 Local stability theory

In many systems of differential equations, the behavior near an equilibrium point can be *approximated* by linear, first-order differential equations. As mentioned earlier, we will consider systems of only two equations to keep things simple. Suppose that (x_0, y_0) is an equilibrium point; that is,

$$f(x_0, y_0) = g(x_0, y_0) = 0.$$

To approximate the values of f and g near (x_0, y_0), we can express small changes in the functions in terms of their partial derivatives. To first order, we have

$$f(x, y) \approx f(x_0, y_0) + (x - x_0)\frac{\partial f}{\partial x}\bigg|_{(x_0,y_0)} + (y - y_0)\frac{\partial f}{\partial y}\bigg|_{(x_0,y_0)},$$

and a similar approximation for $g(x, y)$. To make these expressions more compact, we'll introduce the shorthand notation $f_x \equiv \partial f/\partial x$ and $f_y \equiv \partial f/\partial y$ for partial derivatives (and similar expressions for g_x, g_y), and we'll also define the variables $u \equiv x - x_0$ and $v \equiv y - y_0$. Using this new notation, our system of two equations becomes

$$u' \approx f_x u + f_y v$$
$$v' \approx g_x u + g_y v.$$

If we assume that the approximate equalities are exact, then we can solve this system of equations. The most convenient way to solve this system is to write it in matrix form:

$$\begin{bmatrix} u' \\ v' \end{bmatrix} = \begin{bmatrix} f_x & f_y \\ g_x & g_y \end{bmatrix} \begin{bmatrix} u \\ v \end{bmatrix}.$$

This system is analogous to the equation $dN/dt = rN$, which we encountered in our population growth example and which has the solution $N = N_0 e^{rt}$. For our system of two equations, we will therefore guess an exponential solution of the form

$$\begin{bmatrix} u \\ v \end{bmatrix} = \begin{bmatrix} a \\ b \end{bmatrix} e^{\alpha t},$$

where a, b, α are quantities that we have yet to determine. Plugging this candidate solution into our original system gives us the result

$$\alpha \begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} f_x & f_y \\ g_x & g_y \end{bmatrix} \begin{bmatrix} a \\ b \end{bmatrix}.$$

In other words, α corresponds to the *eigenvalues* of the two-by-two matrix containing f_x, f_y, g_x, g_y, and a, b are the elements of the *eigenvectors*.

We can solve for the eigenvalues by inserting an identity matrix on the left hand side so that our equation becomes

$$\alpha \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} f_x & f_y \\ g_x & g_y \end{bmatrix} \begin{bmatrix} a \\ b \end{bmatrix} \quad \rightarrow \quad \begin{bmatrix} f_x - \alpha & f_y \\ g_x & g_y - \alpha \end{bmatrix} \begin{bmatrix} a \\ b \end{bmatrix} = 0.$$

The determinant $(f_x - \alpha)(g_y - \alpha) - f_y g_x$ must vanish, which gives us the following solution for α:

$$\alpha = \frac{f_x + g_y}{2} \pm \sqrt{\left(\frac{f_x + g_y}{2}\right)^2 - (f_x g_y - f_y g_x)}.$$

How can we make sense of this result? Without knowing the details of f and g, the solution for α could either be real or complex depending on the terms under the square root. For now, let's loosely say that $\alpha = c + di$, a generic complex number. Note that α is real only when $d = 0$, which occurs when $[(f_x + g_y)/2]^2 - (f_x g_y - f_y g_x) \geq 0$.

If we focus on u for the moment (v is similar), our solution is then $u = ae^{\alpha t} = ae^{(c+di)t}$. We haven't bothered to solve for a, but it's just an overall scale factor; the interesting behavior is determined by α. Let's consider the following three cases for c and d:

- $c < 0$, $d = 0$. In other words, α is *real* and *negative*. In this case, $u = ae^{-|c|t}$ and *decreases* as a function of time. Because $u \equiv x - x_0$, this means that the solution moves closer to the equilibrium point as it evolves (with a rate given by c). The equilibrium point is therefore *stable*.

- $c > 0$, $d = 0$. In other words, α is *real* and *positive*. In this case, $u = ae^{ct}$ and *increases* as a function of time. The solution therefore moves away from the equilibrium point as it evolves, and once again, the rate is given by c. The equilibrium point is *unstable*.

- $c \neq 0$, $d \neq 0$. In other words, α is *complex*. In this case, $u = ae^{(c+di)t} = ae^{ct}[\cos(dt) + i\sin(dt)]$, so the solution *oscillates* as a function of time. Because v follows a similar oscillatory pattern, the overall motion in the phase plane is roughly elliptical. The sign of c determines whether the oscillation increases or decreases as time goes on.

Figure 4 illustrates a few typical phase plane diagrams for each of the cases described above, when α is both real and complex.

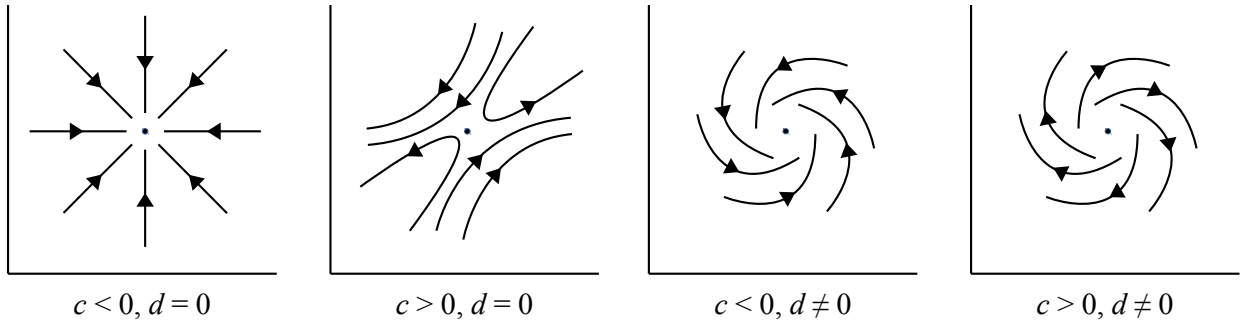


Figure 4: Example phase plane diagrams.

5.4 Example: damping of a pendulum

Now that we've talked about systems of equations and local stability in the abstract sense, let's apply the theory to a practical example. We'll consider a simple pendulum that swings back and forth in the presence of air resistance. What is the solution for the pendulum's motion, and how is it slowed because of air resistance?

We'll make the simplifying assumption that the pendulum's mass m is concentrated at a point that is located at a fixed distance ℓ from the pivot. Let's define $\theta = \theta(t)$ as the angle of the pendulum, with $\theta = 0$ corresponding to the rest position. The motion of the pendulum is governed by Newton's second law, which states that the net force is equal to the product of the mass and acceleration. There are two forces at play. First, the pendulum moves under the influence of the gravitational force mg, and we are interested in the *tangential* component of this force, $-mg\sin\theta$, that lies along the direction of motion. Our second force arises from air resistance, and we've seen earlier in this class that air resistance depends on velocity. We'll therefore call the air resistance force $-r(\omega)$, where r is some differentiable function, and ω is the angular velocity. Note that we need the minus sign because air resistance always opposes the direction of motion, and $r(0) = 0$. We won't say anything else about the properties of r for now, but we'll come back to it later.

Combining gravity with air resistance and applying Newton's second law gives us the following governing differential equation:

$$m\ell\theta'' = -mg\sin\theta - r(\omega).$$

Since $\omega = \theta'$, we can rewrite our differential equation as a pair of linear, first-order equations:

$$\theta' = \omega$$

$$\omega' = -\frac{g}{\ell}\sin\theta - \frac{r(\omega)}{\ell m}.$$

Note that this pair of equations now matches the generic form that we discussed earlier [$x' = f(x, y)$ and $y' = g(x, y)$], with $x = \theta$ and $y = \omega$. In order to find the equilibrium points, we must set $\theta' = 0$ and $\omega' = 0$. The first condition gives us $\omega = 0$, and the second condition gives us $\sin\theta = 0$, or $\theta = 0, \pi$. Of these two possibilities, we will show that only $\theta = 0$ is a stable equilibrium (I'll leave it to you to convince yourself that $\theta = \pi$ is unstable).

To analyze the local stability of the system around the equilibrium point $\theta_0 = 0, \omega_0 = 0$, we need to calculate the partial derivatives:

$$(\theta')_\theta = 0, \qquad (\theta')_\omega = 1,$$

$$(\omega')_\theta = -\frac{g}{\ell}, \qquad (\omega')_\omega = -\frac{r'(0)}{\ell m}.$$

Our expression for α is then

$$\alpha = -\frac{r'(0)}{2\ell m} \pm \sqrt{\left(\frac{r'(0)}{2\ell m}\right)^2 - \frac{g}{\ell}} = c + di.$$

At this point, we need to make an assumption about $r(\omega)$: we know that it's an increasing function near $\omega = 0$, and we'll make the further assumption that $r'(0) > 0$. (Previously in class, we've encountered air resistance that scales as velocity squared, so this assumption isn't crazy.) Now let's examine the implications for α:

- Case 1: If $r'(0) > 2m\sqrt{g\ell}$, then the square root term is real and positive, but less than $r'(0)/(2\ell m)$. We can therefore conclude that in this particular case, $c < 0$ and $d = 0$. The equilibrium point is stable, and there are no oscillations. In physical terms, this case corresponds to very large air resistance forces (with a steep derivative $r'(0)$), i.e. the pendulum is moving through a highly viscous medium.
- Case 2: If $r'(0) < 2m\sqrt{g\ell}$, then the square root term is imaginary. In this particular case, $c < 0$ and $d \neq 0$. The equilibrium point is stable because $c < 0$, but there are oscillations because $d \neq 0$.

In case 1, the solutions to the differential equations near the equilibrium point are $\theta \propto e^{-|c|t}$ and $\omega \propto e^{-|c|t}$. The phase diagram therefore looks similar to the leftmost panel in Figure 4, but centered on the origin in the ω versus θ plane. In other words, solutions in the neighborhood of the equilibrium point move inward toward that equilibrium with an exponentially decreasing rate. In case 2, the solutions to the differential equations are $\theta \propto e^{-|c|t}[\cos(dt) + i\sin(dt)]$ and $\omega \propto e^{-|c|t}[\cos(dt) + i\sin(dt)]$. The phase diagram therefore looks similar to the third panel in Figure 4, but centered on the origin. Both θ and ω oscillate with a decreasing amplitude, so the phase diagram consists of spiral paths that asymptotically converge on the equilibrium point.

As a final note, the pendulum example is a bit contrived because there are other methods for solving the governing differential equation directly, without having to split it into two separate equations. Nevertheless, it's a useful illustration of the phase plane analysis machinery. We'll now discuss another example that is (hopefully) a bit less contrived.

5.5 Example: predator–prey interactions

Interaction between species lies at the heart of ecology. Some claim that these interactions cause the nearly cyclic fluctuations observed in some populations. Can we construct a simple mathematical model to help shed light on this debate? Since our mathematical machinery currently involves two dependent variables, we will consider interactions between only two species. There are several kinds of interactions we can think about:

- The first species preys on the second.
- Both species compete for the same limited resources, e.g. plants competing for sunlight.
- Two species live in a symbiotic relationship with each other, e.g. nitrogen-fixing bacteria on the roots of peas and beans.

We will discuss only predation in this example, but I encourage you to think about the other two scenarios as well. (This list is not exhaustive, by the way; recall the example of the "biological arms race" between the garter snake and Oregon newt from the fourth set of lecture notes.)

Let's define x as the number of predators and y as the number of prey, both of which vary with time t. We can then express the fractional change in each population as generic functions that depend on both x and y:

$$\frac{x'}{x} = r(x, y), \qquad \frac{y'}{y} = s(x, y).$$

The equilibrium condition is then set by $r = s = 0$. If we want the system to be stable, then we need to make some assumptions about the form of r and s. Let's pause and think about what $r(x, y)$ actually means. If r does not depend on x, i.e. $r_x = 0$, then the implication is that the fractional growth of the predator population does not depend on its own population size. If r does not depend on y, i.e. $r_y = 0$, then the fractional growth of the predator population does not depend on the population of prey. Similar arguments can be made for $s(x, y)$. Of course, the predator and prey populations do affect each other in practice, so let's go through this exercise again and think about how the partial derivatives *should* behave:

- How is the prey net growth rate affected by a change in predator population, i.e. how should s depend on x? Since predators consume prey, we can conclude that $s_x < 0$. That is, as the population of predators increases, the growth rate of prey decreases.
- How is the prey net growth rate affected by a change in its own population, i.e. how should s depend on y? If the population of prey grows (while the predator population stays the same), then there are two competing effects. First, there is reduced predator pressure per member of prey, so one would expect the growth rate to increase. However, increased competition for resources amongst the prey tends to make the growth rate decrease. These effects tend to cancel, so we'll guess that overall, $s_y \approx 0$.
- How is the predator net growth rate affected by a change in prey population, i.e. how should r depend on y? If the prey population increases, there will be more food per predator, so we expect $r_y > 0$.
- How is the predator net growth rate affected by a change in its own population, i.e. how should r depend on x? If the predator population increases (while the prey population stays the same), there will be less food per predator, so we expect $r_x < 0$. However, let's assume that the food chain extends higher so that the predators are consumed by even higher-level predators. Therefore, if the predator population increases, there is reduced probability of being eaten by the higher-level predators, and the growth rate should tend to increase ($r_x > 0$). We'll therefore guess that $r_x \approx 0$ overall.

Putting the above arguments together, we obtain the following general conditions on r and s:

$$r_x \approx 0, \qquad r_y > 0,$$

$$s_x < 0, \qquad s_y \approx 0.$$

There are infinitely many functions that can satisfy the above conditions. For our purposes, we will choose one of the simplest possibilities, and we will express both r and s as *linear* functions:

$r(x,y) = a + by$ and $s(x,y) = c + dx$. This special case is known as the *Lotka–Volterra equations* and is historically significant in predator–prey population studies, although it isn't entirely realistic because of its simplicity. Note that r has no x-dependence, and s has no y-dependence, so the conditions $r_x \approx 0$ and $s_y \approx 0$ are satisfied. Since $s_x < 0$, we will rewrite our equation as $s(x,y) = c - dx$ and take d to be a positive number. Similarly, b is positive because $r_y > 0$. What about the signs of a and c? In the complete *absence* of prey ($y = 0$), the predators die of starvation, so we expect $a < 0$. If there are no predators ($x = 0$), then the prey are able to flourish and increase their population, i.e. $c > 0$. Putting all of this together, we obtain the following equations:

$$x' = x(-a + by) \equiv f(x,y)$$

$$y' = y(c - dx) \equiv g(x,y),$$

where a, b, c, d are all *positive* constants.

Now let's analyze the behavior in the phase plane. By setting $x' = y' = 0$, we obtain equilibrium points of $(0,0)$ and $(c/d, a/b)$. The first is the trivial solution where there are no creatures and nothing interesting happens, so let's focus our attention on the second point. To assess the stability of the equilibrium, we'll calculate the following partial derivatives:

$$f_x = 0, \qquad f_y = bc/d,$$

$$g_x = -ad/b, \qquad g_y = 0.$$

The expression for α is then

$$\alpha = 0 \pm \sqrt{(0 - (0 + abcd/bd)} = \pm i\sqrt{ac}.$$

In other words, the solutions for x and y oscillate around the equilibrium point with a rate determined by $\sqrt{ac}$. There is no growth or decay in the oscillation amplitude because α is purely imaginary.

To complete our picture, we will now explicitly solve the differential equations by taking the ratio and separating variables:

$$\frac{dy}{dx} = \frac{y(c - dx)}{x(-a + by)}$$

$$\int \frac{-a + by}{y} dy = \int \frac{c - dx}{x} dx.$$

Integrating both sides gives us

$$-a \ln y + by = c \ln x - dx + k,$$

where k is an arbitrary constant. This equation is plotted in Figure 5 for multiple values of k. Note that the plot shows predator versus prey, or x plotted against y. The closed trajectories are centered on the equilibrium point $(x,y) = (c/d, a/b)$. We can show that the direction of motion is in the counterclockwise direction by considering the following four quadrants relative to the equilibrium position:

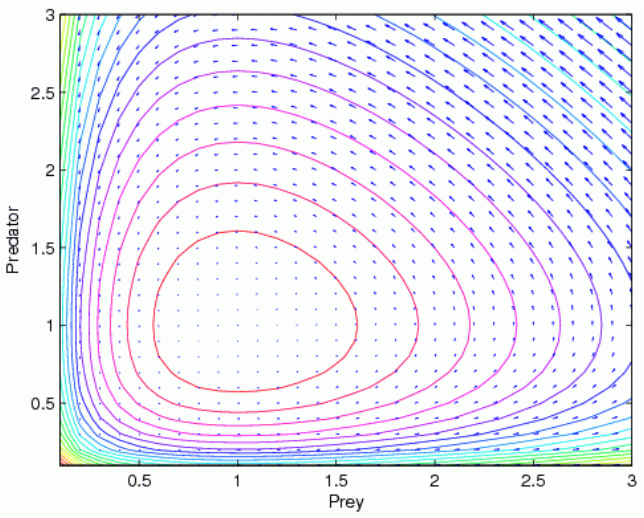


Figure 5: Cyclical solutions to the Lotka–Volterra equations.

- Upper right quadrant, $x > c/d, y > a/b$: in this case, $x' > 0$ and $y' < 0$, so the motion along the trajectory moves up and to the left (don't forget x and y are reversed on the plot).
- Upper left quadrant, $x > c/d, y < a/b$: in this case, $x' < 0$ and $y' < 0$, so the motion is down and to the left.

- Lower left quadrant, $x < c/d, y < a/b$: in this case, $x' < 0$ and $y' > 0$, so the motion is down and to the right.
- Lower right quadrant, $x < c/d, y > a/b$: in this case, $x' > 0$ and $y' > 0$, so the motion is up and to the right.

In practice, the solution properties mean that the populations of predator and prey fluctuate up and down, balancing each other according to the above constraints. If we start in the lower left corner of one of the solution curves, the population of prey increases since the predator population is near its minimum, i.e. the solution evolves to the right. As the number of prey increases, the population of predators also increases because food becomes more abundant. i.e. the solution moves upward. As the number of predators continues to increase, more and more prey are consumed, and the population decreases, i.e. the solution moves to the left. Eventually, the supply of prey is depleted to a point where the predators begin to starve, and their population drops, i.e. the solution moves down and back to the lower left corner, where we started.

Although the Lotka–Volterra equations are a useful starting point for understanding interactions between predator and prey, there are a number of limitations. Some of the underlying assumptions are overly simplistic; for example, we assume that the predators' food supply depends entirely on the size of the prey population, that predators have boundless appetite, that the prey has ample food at all times, etc. There are many extensions of the Lotka–Volterra equations that account for additional effects, but we'll end our discussion here and move on to another example.

5.6 Example: Keynesian economics

J. M. Keynes's theory described in *The General Theory of Employment, Interest and Money* has had a profound impact on the study of economics and its practice. Here we will consider a bare bones model of Keynesian economics. In addition to being minimal, our model will also be abstract: we will identify a list of relevant quantities, but rather than write down specific functional forms relating these quantities, we will restrict our discussion to the qualitative behavior of the derivatives. With the signs of derivatives in hand, we will hopefully be able to make some broad predictions about stability in a national economy.

Let's begin by writing down a list of variables that are relevant when describing a national economy:

- C = desired level of consumption
- I = desired level of capital investment, i.e. total amount of money people want to spend on building large production infrastructure such as factories
- D = total demand for goods
- Y = national income
- L = desired amount of money to be held as cash on hand
- M = total amount of money available, i.e. the country's money supply
- R = cost of money (interest rate)

We will keep our model simple by considering only short time periods, since long-term trends (e.g. increased labor efficiency from capital investment) complicate the picture. Normally we'd go through the list of variables and would attempt to identify inputs and outputs. However, in this case, the list is long and complicated, and it's easier to begin by identifying relationships between the variables, one step at a time.

If we assume that there's an equilibrium in the economy, then we can say that $D = Y$ and $L = M$. In other words, what we want is what we have. We'd like to assess the stability, which means that

we need to consider *departures from equilibrium*, i.e. the quantities $D-Y$ and $L-M$. The first quantity describes the excessive demand for goods, which can be satisfied if income increases. The second quantity describes excessive demand for money (as cash in hand), which tends to drive up interest rates. (In other words, if the desire for liquid cash exceeds the available money supply, then it costs more to obtain a loan.) We can therefore guess the very basic, generic relations

$$Y' = y(D-Y), \qquad y'(0) > 0, \qquad y(0) = 0,$$

$$R' = r(L-M), \qquad r'(0) > 0, \qquad r(0) = 0,$$

where y and r are some increasing functions that we won't specify for now. This pair of equations looks like a promising starting point, so let's now try to find a way to reduce the right-hand-side variables (L, M, D, Y) to the left-hand-side variables (Y, R). We get $Y = Y$ for free, so let's focus on the other variables.

Since the money supply M is determined by the government, this variable is an input. We already stated that we are dealing with short time periods for our model, so we will assume that M is roughly *constant* for now. We'll come back to this later.

Let's move on to L. In general, if interest rates rise, then people prefer to invest more money and to have less liquid cash on hand, so we expect $L_R < 0$. If income rates rise, then there is more available cash on hand, so we can also guess $L_Y > 0$.

To study the partial derivatives of D, it is convenient to break it into two parts, $D = C + I$. That is, the total demand for goods comes from the sum of raw consumption C and materials required for capital investment I. How do each of these factors depend on R and Y?

- C_R should be zero or negative because if interest rates rise, people have an incentive to reduce consumption and save interest-bearing money in order to maximize financial return. However, reduced consumption is merely an option, and people are not compelled to do this.
- C_Y should be positive because if income increases, people will tend to also increase their consumption.
- I_R should be negative because if interest rates rise, people will tend to save interest-bearing money rather than purchase capital.
- I_Y should be zero or positive. If income increases, people have the flexibility and means to spend more money on capital investments, although they are not compelled to do so.

Putting the above arguments together, we arrive at the conclusion that $D_R < 0$ and $D_Y > 0$.

In summary,

$$L_R < 0, \qquad D_R < 0, \qquad I_R < 0, \qquad C_R \leq 0,$$

$$L_Y > 0, \qquad D_Y > 0, \qquad I_Y \geq 0, \qquad C_Y > 0.$$

We can now write down the partial derivatives of r and y that are needed to evaluate the stability of the equilibrium:

$$r_R = r'(0)L_R < 0, \qquad r_Y = r'(0)L_Y > 0,$$

$$y_R = y'(0)D_R < 0, \qquad y_Y = y'(0)(D_Y - 1).$$

Note that the sign of y_Y cannot be determined from our above list of partial derivatives. The α for this system is then

$$\alpha = \frac{r_R + y_Y}{2} \pm \sqrt{\left(\frac{r_R + y_Y}{2}\right)^2 - (r_R y_Y - r_Y y_R)}.$$

In order for the solutions to be stable, we must have the conditions

$$r_R + y_Y < 0, \qquad r_R y_Y > r_Y y_R.$$

The first condition arises because the leading term in α must be negative in order to produce solutions that exponentially decay toward equilibrium. The second condition arises because the magnitude of the square root term cannot exceed $(r_R + y_Y)/2$; otherwise, α can become positive and the equilibrium is unstable. The combination $r_R y_Y - r_Y y_R$ must therefore be a positive quantity that is subtracted from $[(r_R + y_Y)/2]^2$, i.e. $r_R y_Y$ must be larger than $r_Y y_R$.

Now let's combine the stability conditions with our knowledge of the signs of the partial derivatives $r_R < 0, r_Y > 0, y_R < 0$. We can be guaranteed to satisfy the first stability condition $r_R + y_Y < 0$ by enforcing $y_Y < 0$. In the second stability condition $r_R y_Y > r_Y y_R$, the right hand side is negative; once again, we can guarantee the condition by enforcing $y_Y < 0$, which makes the left hand side positive. Therefore, $y_Y = y'(0)(D_Y - 1) < 0$ is a *sufficient but not necessary* condition to ensure stability. Because $y'(0) > 0$, the condition implies that $D_Y < 1$, which we can translate into words:

If the sensitivity of total demand to changes in the national income is less than unity, our Keynesian model is locally stable.

(In economics "sensitivity" is a term for a partial derivative; the sensitivity of A to B is the amount A changes when B changes by one unit.) We can unpack the above statement a bit further by noting that we must have $C_Y < 1$ in order to ensure $D_Y < 1$. (Recall that $D = C + I$, and $I_Y \geq 0$.) To interpret $C_Y < 1$, it's easier to think about the opposite scenario when $C_Y > 1$. Under these circumstances, as national income increases, the desired consumption increases even faster. This scenario is an unlikely possibility except in underdeveloped countries, where it can cause severe problems.

As a final exercise, let's revisit our assumption that the money supply M is fixed. This parameter is one that can be tweaked by the government, so we can ask the question: how does the equilibrium shift in response to the government adjusting the money supply? At equilibrium, we have $L = M$ and $D - Y = 0$. Using the chain rule to differentiate these expressions with respect to M (using R, Y as intermediate variables), we obtain

$$L_R R_M + L_Y Y_M = 1, \qquad D_R R_M + D_Y Y_M - Y_M = 0.$$

Solving for R_M and Y_M gives us

$$R_M = \frac{D_Y - 1}{\Delta}, \qquad Y_M = -\frac{D_R}{\Delta},$$

where $\Delta \equiv (D_Y - 1)L_R - D_R L_Y$. Recall that one of our stability criteria was $r_R y_Y > r_Y y_R$; this expression can be written as $L_R(D_Y - 1) > L_Y D_R$, which is equivalent to $\Delta > 0$. What does this imply for the signs of R_M and Y_M?

- Y_M: We identified previously that $D_R < 0$, and putting this together with $\Delta > 0$ gives us $Y_M > 0$.
- R_M: If we have stability, then we identified earlier that $D_Y < 1$. We can therefore conclude that $R_M < 0$.

What does all of this mean in practice? Let's go back to the original premise: the money supply M is a quantity that the government can adjust in an attempt to improve the economic situation. When unemployment is high, the government sometimes tries to influence the national income by adjusting M and increasing the money supply. Because $Y_M > 0$, increasing M should also increase the national income, which means an increase in labor in the short term, thus alleviating the unemployment problem. The amount of change depends on the change in M and the size of Y_M. If D_R is small, we can see from the expression for Y_M that changing M may not be a very effective way to fight large scale unemployment.

This model can be extended to address many other ways in which the government may try to influence the economy. For example, the government may try to control interest rates, influence D through deficit spending, influence C and I through taxation, etc.

Math 246 notes #7: Testing models with data

1 Motivation

- By now, we've discussed a lot of examples of mathematical models, many of which are accompanied by supporting data. As an example, when we discussed scaling relations, we derived the relation that trunk thickness scales as trunk length to the 3/2 power for four-legged animals. Data of vertebrae diameter versus length suggest that this scaling relation is correct; that is, the data points "look like" the model curve.

- What does it actually mean for data to "look like" a predicted model? This statement is fundamentally imprecise. Our goal in this section is develop basic mathematical machinery that will allow us to *quantitatively* compare models with data. With this machinery in place, we can then answer questions such as "do the data prefer model A or model B, i.e. which model is better?"

- Being able to intelligently compare data and models is a skill that is important in many aspects of daily life. The media, for example, is often a minefield of incorrect conclusions that are based on faulty data/model comparisons. Figure 1 illustrates one such example from the British tabloid, Mail on Sunday. The graph shows data of global average temperatures (black line) compared to model predictions (red areas), and the newspaper's conclusion was that global warming had stopped and that the world average temperature was "about to crash out." For other (infuriating) examples of misleading graphs in the media, I recommend checking out `http://www.statisticshowto.com/misleading-graphs/`.

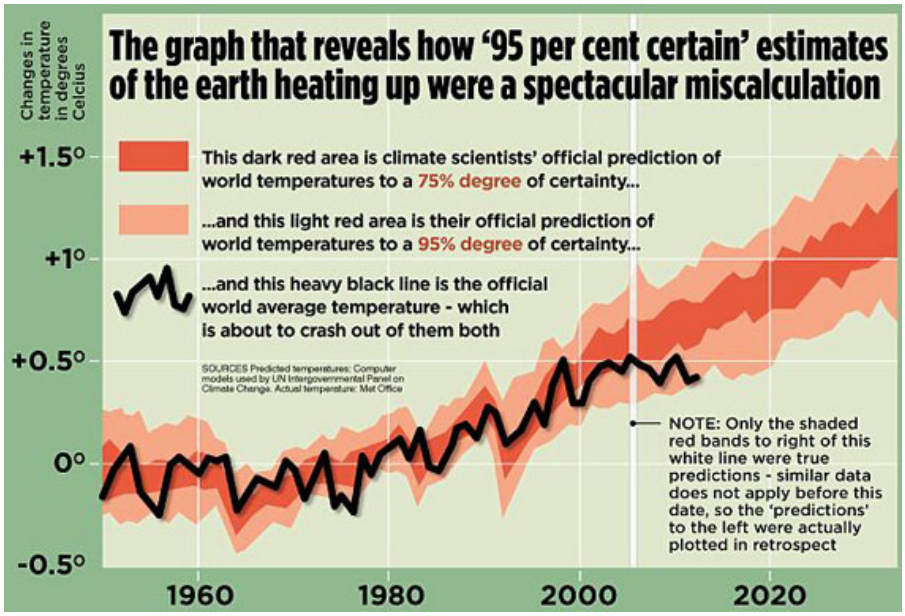


Figure 1: The Mail on Sunday used this graph to claim that global warming had stopped. Image credit: Mail on Sunday.

2 Fitting models to data graphically

When we formulate a model to be tested with data, that model generally contains one or more parameters. One of the main goals of comparing a model to data is to determine the values of those parameters. *At minimum, there need to be as many data points as there are parameters in the model.* There are many other factors to consider when collecting data, e.g. proper spacing of data points, accuracy of the data collection process (systematic errors), and identification of spurious data points. We'll

gloss over these issues for now, and we'll say that as a starting point for model comparison, we're given a collection of data points, each of which can be thought of as an *interval of relative confidence* rather than a single point. The sizes of these intervals should be commensurate with the appraisal of the errors present in the data collection process.

Most of the models we've discussed in this class involve functions of two variables (one independent and one dependent). We'll restrict our discussion to this simple case, although the concepts can be extended to multivariable functions. Suppose we have a very simple model, $y = ax + b$, where x is the independent variable, y is the dependent variable, and a, b are the parameters that we wish to determine. Now let's say that we have a collection of data points; Figure 2 illustrates one example. Our goal is to find the values of a, b that gives us the best match between the resulting line and the data points. Not all of the points lie perfectly on that line, but we'd like to choose the line in such a way that minimizes the vertical discrepancies, i.e. the minimizes the differences of data-minus-model.

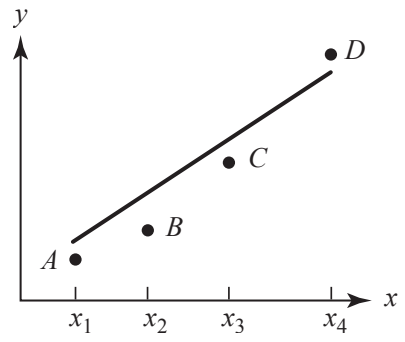


Figure 2: An example of four data points that could be used to determine the parameters in our dummy model $y = ax + b$.

Before we discuss the mathematical details of minimizing the discrepancies between data and models, it's worth noting that some more complicated models, such as exponentials, can be *transformed* so that they take the same form as the simple linear case we discussed above. For example, if we had an exponential model of the form $y = Ce^x$, we could transform the equation by taking the natural logarithm of both sides: $\ln y = \ln C + x$. Plotting $\ln y$ versus x now gives us a line, so the problem reduces a linear fit.

3 Basic least squares fitting

One of the most frequently used methods for fitting model curves to data is the method of *least squares*. Suppose our model is governed by a function $y = f(x)$, which has some parameters that we won't write out explicitly for now. The best-fit parameters are the ones that minimize the sum

$$\sum_{i=1}^{N} |y_i - f(x_i)|^2,$$

where y_i are the measured data points (N in total), and $f(x_i)$ are the calculated model values at the corresponding x_i positions. The above expression depends on $|y_i - f(x_i)|$, which represents the deviations between the data and the model. The reason why we're interested in the sum of the squares can be explained using a geometric interpretation. Suppose that we have three data points, and $R_i \equiv |y_i - f(x_i)|$ is defined as the absolute deviation between the observed and predicted values for $i = 1, 2, 3$. We can think of R_i as the scalar components of a *deviation or residual vector*, $\vec{R} = R_1\hat{i} + R_2\hat{j} + R_3\hat{k}$. The magnitude of this vector is given by $|\vec{R}| = \sqrt{R_1^2 + R_2^2 + R_3^2}$, and we can minimize $|\vec{R}|$ by minimizing $|\vec{R}|^2$. Thus, the least-squares problem is to determine the parameters of the function $y = f(x)$ such that

$$|\vec{R}|^2 = \sum_{i=1}^{3} R_i^2 = \sum_{i=1}^{3} |y_i - f(x_i)|^2$$

is minimized. That is, we may interpret the least-squares criterion as minimizing the magnitude of the vector whose coordinates represent the absolute deviation between the observed and predicted values. This concept extends beyond three dimensions and applies to data sets that contain any number of points.

Now let's look at how to apply the least-squares criterion in practice, using our simple linear model $y = Ax + B$ as a starting point. We'll distinguish between the true model parameters A, B and our least-squares *estimate*, which we'll denote with lowercase letters $y = ax + b$. Let's say that

we're given a set of data (x_i, y_i) with $i = 1, 2, ..., N$. Applying the least-squares criterion to this situation requires the minimization of

$$S = \sum_{i=1}^{N} |y_i - f(x_i)|^2 = \sum_{i=1}^{N} (y_i - ax_i - b)^2.$$

Specifically, our goal is to find the values of a and b that minimize S. We must therefore set the following two partial derivatives equal to zero:

$$\frac{\partial S}{\partial a} = -2 \sum_{i=1}^{N} (y_i - ax_i - b) x_i = 0$$

$$\frac{\partial S}{\partial b} = -2 \sum_{i=1}^{N} (y_i - ax_i - b) = 0.$$

We can rewrite these equations as

$$a \sum_{i=1}^{N} x_i^2 + b \sum_{i=1}^{N} x_i = \sum_{i=1}^{N} x_i y_i$$

$$a \sum_{i=1}^{N} x_i + Nb = \sum_{i=1}^{N} y_i.$$

Note that since we're given the data points (x_i, y_i), the coefficients of a and b are just numbers that can be computed. We can thus solve for a and b, and our best estimates are

$$a = \frac{N \sum x_i y_i - \sum x_i \sum y_i}{N \sum x_i^2 - (\sum x_i)^2}$$

$$b = \frac{\sum x_i^2 \sum y_i - \sum x_i y_i \sum x_i}{N \sum x_i^2 - (\sum x_i)^2}.$$

It's easy plug these expressions and data points into a computer in order to obtain exact values for a, b.

4 Weighted least squares

The previous section gives the basic picture of least squares fitting, but we've glossed over one important point: we started off by saying that the measured data points can be thought of as intervals of relative confidence, and yet we haven't accounted for this uncertainty at all. If we knew A and B exactly, then for every value of x_i, we could compute the true value of the corresponding y_i. However, the actual *measurement* of y_i has some associated uncertainty or error: mathematically speaking, the measured y_i is governed by a normal distribution centered on the true value, with width parameter σ_y. The probability of obtaining the observed value y_i is

$$Prob_{a,b}(y_i) \propto \frac{1}{\sigma_y} e^{-(y_i - ax_i - b)/2\sigma_y^2},$$

where the subscripts a, b indicate that this probability depends on the (unknown) values of a and b. Therefore, the total probability of obtaining our *complete* set of measurements $y_1, ..., y_N$ is the product

$$Prob_{a,b}(y_1, ..., y_N) = Prob_{a,b}(y_1) \cdot \cdot \propto Prob_{a,b}(y_N) \propto \frac{1}{\sigma_y^N} e^{-\chi^2/2},$$

where the exponent is given by

$$\chi^2 = \sum_{i=1}^{N} \frac{(y_i - ax_i - b)^2}{\sigma_y^2}.$$

We assume that the best estimates for the unknown constants a and b, based on the given measurements, are those values of a and b for which the probability $Prob_{a,b}(y_1, ..., y_N)$ is maximum, i.e. for which the sum of squares χ^2 is a *minimum*. Note the similarity between χ^2 and our expression for S in the previous section: χ^2 is essentially the same as S, but *weighted* by the uncertainty in the y_i measurements. To minimize χ^2, we set the following derivatives equal to zero:

$$\frac{\partial \chi^2}{\partial a} = -\frac{2}{\sigma_y^2} \sum_{i=1}^{N} x_i(y_i - ax_i - b) = 0$$

$$\frac{\partial \chi^2}{\partial B} = -\frac{2}{\sigma_y^2} \sum_{i=1}^{N} (y_i - ax_i - b) = 0.$$

Apart from the leading coefficients, the above expressions are identical to those in the previous section. The weighted least-squares estimates for a and b are therefore the same as what we previously derived.

You might be wondering: why did we go through all that trouble just to arrive at the same result? The answer is that now that we have statement of the uncertainties in the y_i measurements, we can also say something about the errors in our estimates of a and b. For a general function q that depends on several variables $x, ..., z$, the uncertainty σ_q can be related to the uncertainties of the independent variables $\sigma_x, ..., \sigma_z$ through the expression

$$\sigma_q = \sqrt{\left(\frac{\partial q}{\partial x}\sigma_x\right)^2 + ... + \left(\frac{\partial q}{\partial z}\sigma_z\right)^2},$$

assuming the errors are independent and random. If we start with the errors in y_i and propagate them through to errors in our estimates of a and b, we obtain

$$\sigma_a = \sqrt{\sum_{i=1}^{N} \left(\frac{\partial a}{\partial y_i}\sigma_y\right)^2} = \frac{\sigma_y}{\Delta}\sqrt{\sum (Nx_i - \sum x_i)^2} = \sigma_y\sqrt{\frac{N}{\Delta}},$$

$$\sigma_b = \sqrt{\sum_{i=1}^{N} \left(\frac{\partial b}{\partial y_i}\sigma_y\right)^2} = \frac{\sigma_y}{\Delta}\sqrt{\sum (\sum x_i^2 - x_i \sum x_i)^2} = \sigma_y\sqrt{\frac{\sum x_i^2}{\Delta}},$$

where $\Delta \equiv N\sum x_i^2 - (\sum x_i)^2$ is defined for convenience. The above expressions are a bit messy and tedious to calculate (which is why we normally leave that task to a computer), but they capture an important concept: *when estimating model parameters from data, you not only need to find the best-fit values, but you also need to calculate the associated uncertainties*. The parameter estimates themselves are of limited value unless you know how confident you are in the answers.